

Problem solutions of Many-Body Theory of Condensed Matter Systems by Michael G. Cottam and Zahra Haghshenasfard

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1 Introduction to Second Quantization

1.1 Commutator of $\exp(-i\mathbf{k} \cdot \mathbf{r})$ with $\mathbf{p}$ and $\mathbf{p}^2$

Introducing a mute wave function ψ at the right of the commutator we obtain

$$\begin{aligned} [e^{-i\mathbf{k} \cdot \mathbf{r}}, \mathbf{p}] \psi &= (e^{-i\mathbf{k} \cdot \mathbf{r}}(-i\nabla) - (-i\nabla)e^{-i\mathbf{k} \cdot \mathbf{r}}) \psi \\ &= \cancel{-ie^{-i\mathbf{k} \cdot \mathbf{r}}\nabla\psi} + i(-i\mathbf{k}e^{-i\mathbf{k} \cdot \mathbf{r}}\psi + \cancel{e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla\psi}) \\ &= +\mathbf{k}e^{-i\mathbf{k} \cdot \mathbf{r}}\psi \end{aligned}$$

Because ψ can be any wave function this relation is in general true

$$\boxed{[e^{-i\mathbf{k} \cdot \mathbf{r}}, \mathbf{p}] = \mathbf{k}e^{-i\mathbf{k} \cdot \mathbf{r}}}$$

Analogously for the second commutator we get

$$\begin{aligned} [e^{-i\mathbf{k} \cdot \mathbf{r}}, \mathbf{p}^2] \psi &= (e^{-i\mathbf{k} \cdot \mathbf{r}}(-\nabla^2) - (-\nabla^2)e^{-i\mathbf{k} \cdot \mathbf{r}}) \psi \\ &= -e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla^2\psi + \nabla((-i\mathbf{k})e^{-i\mathbf{k} \cdot \mathbf{r}}\psi + e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla\psi) \end{aligned}$$

As stated, $\mathbf{k}$ is a constant vector so $\nabla\mathbf{k} = 0$,

$$\begin{aligned} &= \cancel{-e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla^2\psi} + (-i\mathbf{k}) \cdot (-i\mathbf{k})e^{-i\mathbf{k} \cdot \mathbf{r}}\psi + (-i\mathbf{k})e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla\psi + (-i\mathbf{k})e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla\psi + \cancel{e^{-i\mathbf{k} \cdot \mathbf{r}}\nabla^2\psi} \\ &= -k^2e^{-i\mathbf{k} \cdot \mathbf{r}}\psi + 2\mathbf{k}e^{-i\mathbf{k} \cdot \mathbf{r}}(-i\nabla)\psi \\ &= -e^{-i\mathbf{k} \cdot \mathbf{r}}k^2\psi + e^{-i\mathbf{k} \cdot \mathbf{r}}2\mathbf{k} \cdot \mathbf{p}\psi \\ &= e^{-i\mathbf{k} \cdot \mathbf{r}}(2\mathbf{k} \cdot \mathbf{p} - k^2)\psi \end{aligned}$$

Again, because ψ can be any wave function this relation is in general true

$$\boxed{[e^{-i\mathbf{k} \cdot \mathbf{r}}, \mathbf{p}^2] = e^{-i\mathbf{k} \cdot \mathbf{r}}(2\mathbf{k} \cdot \mathbf{p} - k^2)}$$

1.2 EM field operators commutators at the same time (1)

As seen in section 1.1.2 Eletromagnetic Field in Free Space, we can write the vector potential $\mathbf{A}(\mathbf{r}, t)$ in the Coulomb gauge and in with periodic condition of a cubical box of volume $V = L^3$ in second quantization as

$$\mathbf{A}(\mathbf{r}, t) = \sum_j \sum_{\lambda=1,2} \sqrt{\frac{1}{2\omega_j \epsilon_0 V}} \hat{\mathbf{e}}_{j,\lambda} \left[a_{j,\lambda} e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} + a_{j,\lambda}^\dagger e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} \right],$$

where ω_j are the frequency modes, $k_{j\alpha} = 2\pi/Ln_{j\alpha}$ with $n_{j\alpha} = 0, \pm 1, \pm 2, \dots$ (where $\alpha = x, y, z$) are the wave vectors modes of modulus $|\mathbf{k}_j| = \omega_j/c$, $\hat{\mathbf{e}}_{j,\lambda}$ are the polarization vectors with $\lambda = 1, 2$ (corresponding to the polarization s and p) and $a_{j,\lambda}/a_{j,\lambda}^\dagger$ the ladder operators for the annihilations or creation of a photon in the mode j and polarization λ .

From the potencial representation of the eletric field $\mathbf{E}(\mathbf{r}, t)$ and the magnetic field $\mathbf{B}(\mathbf{r}, t)$ in the Coulomb gauge we have that

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = \frac{-\partial \mathbf{A}}{\partial t}$$

Substituting the vector potential from the above expression we obtain

$$\begin{aligned} \mathbf{H}(\mathbf{r}, t) &= \frac{-i}{c\mu_0} \sum_j \sum_{\lambda=1,2} \sqrt{\frac{\omega_j}{2\epsilon_0 V}} (\hat{\mathbf{e}}_{j,\lambda} \times \mathbf{k}_j) \left[a_{j,\lambda} e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} - a_{j,\lambda}^\dagger e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} \right] \\ \mathbf{E}(\mathbf{r}, t) &= i \sum_j \sum_{\lambda=1,2} \sqrt{\frac{\omega_j}{2\epsilon_0 V}} \hat{\mathbf{e}}_{j,\lambda} \left[a_{j,\lambda} e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} - a_{j,\lambda}^\dagger e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} \right] \end{aligned}$$

1.2.1 Commutator $[\mathbf{A}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)]$

Of the four terms only two of them survive given the commutators proprieties of the ladder operators. We are left with

$$\begin{aligned} [\mathbf{A}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)] &= \sum_{j,\lambda,j',\lambda'} \frac{i}{2V\epsilon_0} e_{j,\lambda}^n e_{j',\lambda'}^m \left(-e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} e^{-i(\mathbf{k}_{j'} \cdot \mathbf{r}' - \omega_{j'} t)} \left[a_{j,\lambda}, a_{j',\lambda'}^\dagger \right] \right. \\ &\quad \left. + e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} e^{i(\mathbf{k}_{j'} \cdot \mathbf{r}' - \omega_{j'} t)} \left[a_{j,\lambda}^\dagger, a_{j',\lambda'} \right] \right) \\ &= \sum_{j,\lambda,j',\lambda'} \frac{i}{2V\epsilon_0} e_{j,\lambda}^n e_{j',\lambda'}^m \left(-e^{i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} e^{-i(\mathbf{k}_{j'} \cdot \mathbf{r}' - \omega_{j'} t)} \delta_{j,j'} \delta_{\lambda,\lambda'} - e^{-i(\mathbf{k}_j \cdot \mathbf{r} - \omega_j t)} e^{i(\mathbf{k}_{j'} \cdot \mathbf{r}' - \omega_{j'} t)} \delta_{j,j'} \delta_{\lambda,\lambda'} \right) \\ &= \sum_{j,\lambda} \frac{i}{2V\epsilon_0} e_{j,\lambda}^n e_{j,\lambda}^m \left(-e^{i\mathbf{k}_j \cdot (\mathbf{r} - \mathbf{r}')} - e^{-i\mathbf{k}_j \cdot (\mathbf{r} - \mathbf{r}')} \right) \end{aligned}$$

For the next step we define the projector on the plane perpendicular to the vector $\mathbf{k}_j$

$$P_{n,m}^T(\mathbf{k}_j) = \sum_{\lambda} e_{j,\lambda}^n e_{j,\lambda}^m = \delta_{nm} - \frac{k_n k_m}{k^2}$$

Notice that, because a plane perpendicular to $\mathbf{k}_j$ is also perpendicular to $-\mathbf{k}_j$ we have the propriety $P_{n,m}^T(\mathbf{k}_j) = P_{n,m}^T(-\mathbf{k}_j)$. Using this, we can make the substitution in the second term $-\mathbf{k}_j \rightarrow \mathbf{k}_j$ to obtain

$$\begin{aligned} [\mathbf{A}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)] &= \sum_j \frac{i}{2V\epsilon_0} \left(-P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - P_{n,m}^T(\mathbf{k}_j) e^{-i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\ &= \sum_j \frac{i}{2V\epsilon_0} \left(-P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - P_{n,m}^T(-\mathbf{k}_j) e^{+i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\ &= \sum_j \frac{i}{2V\epsilon_0} \left(-P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\ &= \sum_j \frac{-i}{V\epsilon_0} P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \end{aligned}$$

In the limit where $V \rightarrow +\infty$, the summation over the j modes gets extended to an integral over all $\mathbf{k}_{j,\alpha}$

$$\frac{1}{V} \sum_j [\dots] = \frac{1}{2\pi^3} \int d^3\mathbf{k} [\dots]$$

In this limit, if we now substituting the expression for $P_{n,m}^T(\mathbf{k}_j)$ we obtain

$$[\mathbf{A}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)] = \frac{-i}{\epsilon_0} \left[\frac{1}{2\pi^3} \int d^3\mathbf{k} \left(\delta_{nm} - \frac{k_n k_m}{k^2} \right) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right]$$

Recognising the term inside the straight brackets as the given defining of the transverse delta δ_{nm}^T we get

$$[\mathbf{A}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)] = -\frac{i}{\epsilon_0} \delta_{nm}^T(\mathbf{r} - \mathbf{r}')$$

1.2.2 Commutator $[\mathbf{A}_n(\mathbf{r}, t), \mathbf{A}_m(\mathbf{r}', t)]$

Because $\mathbf{A}_n(\mathbf{r}, t)$ and $\mathbf{E}_m(\mathbf{r}', t)$ only differ by a minus sign on the second term and on the prefactor, the procedure is analogous to what we did we before. We have immediatly

$$[\mathbf{A}_n(\mathbf{r}, t), \mathbf{A}_m(\mathbf{r}', t)] = \sum_j \frac{1}{2\omega_j \epsilon_0 V} \left(P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - P_{n,m}^T(\mathbf{k}_j) e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right)$$

This time the terms end up canceling out and so

$$[\mathbf{A}_n(\mathbf{r}, t), \mathbf{A}_m(\mathbf{r}', t)] = 0$$

1.2.3 Commutator $[\mathbf{E}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)]$

It goes the same as the commutator above,

$$[\mathbf{E}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)] = 0$$

1.2.4 Commutator $[\mathbf{H}_n(\mathbf{r}, t), \mathbf{A}_m(\mathbf{r}', t)]$

$$\begin{aligned}
[\mathbf{H}_n(\mathbf{r}, t), \mathbf{A}_m(\mathbf{r}', t)] &= \sum_{j,\lambda} \frac{-1}{\mu_0 c} \frac{i}{2V\epsilon_0} (\hat{\mathbf{e}}_{j,\lambda} \times \mathbf{k}_j)^n e_{j,\lambda}^m \left(-e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - e^{-i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\
&= \sum_{j,\lambda} \frac{-1}{\mu_0 c} \frac{i}{2V\epsilon_0} \epsilon_{abn} e_{j,\lambda}^a k_j^b e_{j,\lambda}^m \left(-e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - e^{-i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\
&= \sum_{\lambda} \frac{-1}{\mu_0 c} \frac{i}{2V\epsilon_0} \epsilon_{abn} k^b P_{a,m}^T \left(-e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} - e^{-i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')} \right) \\
&= \sum_{\lambda} \frac{1}{\mu_0 c} \frac{i}{V\epsilon_0} \epsilon_{abn} k^b P_{a,m}^T e^{i\mathbf{k}_j \cdot (\mathbf{r}-\mathbf{r}')}
\end{aligned}$$

1.2.5 Commutator $[\mathbf{H}_n(\mathbf{r}, t), \mathbf{H}_m(\mathbf{r}', t)]$

1.2.6 (Extra) Commutator $[\mathbf{H}_n(\mathbf{r}, t), \mathbf{E}_m(\mathbf{r}', t)]$

1.3 Boson operators commutation relations

Let a and $a^\dagger$ be boson operators and f a differentiable function of the operators. Let's try

$$\frac{\partial}{\partial a} \left(e^{-\alpha a^\dagger a} f(a, a^\dagger) e^{\alpha a^\dagger a} \right) = -\alpha a^\dagger e^{-\alpha a^\dagger a} f e^{\alpha a^\dagger a} + \left(e^{-\alpha a^\dagger a} \frac{\partial f}{\partial a} e^{\alpha a^\dagger a} \right) + \left(e^{-\alpha a^\dagger a} f \alpha a^\dagger e^{-\alpha a^\dagger a} \right)$$

1.4 Commutator Identities

In the following exercises we assume that the commutator of any two operators A and B is a scalar quantity i.e $[A, B] = \text{const} = c$

1.4.1 (a)

As usual in this type of proves, we start by differentiating the first expression in order to a real scalar parameter λ

$$\begin{aligned}
\frac{d}{d\lambda} f(\lambda) &= \frac{d}{d\lambda} (e^{\lambda A} B e^{-\lambda A}) = A e^{\lambda A} B e^{-\lambda A} + e^{\lambda A} \frac{dB}{d\lambda} e^{-\lambda A} + e^{\lambda A} B (-A) e^{-\lambda A} \\
&= e^{\lambda A} (AB - BA) e^{-\lambda A} = e^{\lambda A} [A, B] e^{-\lambda A} \\
&= e^{\lambda A} c e^{-\lambda A} = c
\end{aligned}$$

The solution of this differential equation is simply

$$f(\lambda) = f(0) + \lambda c = e^0 A e^{-0} + \lambda c = A + \lambda c$$

Which is exactly what we wanted

$$\boxed{e^{\lambda A} B e^{-\lambda A} = A + \lambda [A, B]}$$

1.4.2 (b)

$$\begin{aligned}
\frac{d}{d\lambda} (e^{\lambda A} e^{\lambda B}) &= A e^{\lambda A} e^{\lambda B} + e^{\lambda A} B e^{\lambda B} \\
&= A e^{\lambda A} e^{\lambda B} + ([e^{\lambda A}, B] + B e^{\lambda A}) e^{\lambda B} \\
&= (A + B + [e^{\lambda A}, B] e^{-\lambda A}) e^{\lambda A} e^{\lambda B}
\end{aligned}$$

Notice that because $[A, B] = c$ we get

$$[e^{\lambda A}, B] = [\sum_n \frac{\lambda^n A^n}{n!}, B] = \sum_n \frac{\lambda^n}{n!} [A^n, B]$$

If we use the relation

$$[AB, C] = A[B, C] + [A, C]B$$

First, for example, for $n = 2$, we get

$$[A^2, B] = A[A, B] + [A, B]A = 2cA$$

Now, for $n = 3$, we get

$$[A^3, B] = [AA^2, B] = A[A^2, B] + [A, B]A^2 = A(2cA) + cA^2 = 3cA^2$$

In general, we obtain

$$\boxed{[A^n, B] = ncA^{n-1}}$$

Which can be proved true by induction. For $n = n + 1$,

$$\begin{aligned}
[A^{n+1}, B] &= [A^n A, B] = [A^n, B]A + A^n[A, B] = ncA^{n-1}A + A^n c \\
&= ncA^n + cA^n = (n+1)cA^n
\end{aligned}$$

This way, the above commutator takes the form

$$[e^{\lambda A}, B] = \sum_n \frac{\lambda^n}{n!} ncA^{n-1} = c\lambda \sum_n \frac{\lambda^{n-1}}{(n-1)!} A^{n-1} = c\lambda e^{\lambda A}$$

Now, we just have to substitute above

$$\frac{d}{d\lambda} (e^{\lambda A} e^{\lambda B}) = (A + B + c\lambda e^{\lambda A} e^{-\lambda A}) e^{\lambda A} e^{\lambda B} = (A + B + c\lambda) e^{\lambda A} e^{\lambda B}$$

1.4.3 (c)

Recurring to Baker-Campbell-Hausdorff

$$\boxed{e^A e^B = e^{A+B} e^{[A,B]/2} e^{[A,[A,B]]/12}}$$

the result is immediate because a scalar always commutes with an operator and so, the term following the commutator of first order cancel all out.

$$e^{\lambda A} e^{\lambda B} = e^{(\lambda A + \lambda B + c\lambda^2/2)}$$

1.5 Properties of coherent states

A coherent state is defined as

$$a|\alpha\rangle = \alpha|\alpha\rangle$$

where the α eigenvalues are a complex number, $\alpha = |\alpha|e^{i\theta}$, and the eigenstate are found to be (as derived in

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha a^\dagger)^n}{n!} |0\rangle = e^{\alpha a^\dagger - |\alpha|^2/2} |0\rangle$$

with normalization such that $\langle\alpha|\alpha\rangle = 1$. We want to prove that

$$a^\dagger |\alpha\rangle \langle\alpha| = \left(\alpha^* + \frac{\partial}{\partial\alpha} \right) |\alpha\rangle \langle\alpha|$$

Multiplying both sides by the ket $|\alpha\rangle$, and given the normalization, we get

$$a^\dagger |\alpha\rangle = \left(\alpha^* + \frac{\partial}{\partial\alpha} \right) |\alpha\rangle$$

The partial derivative is immediate given the second form presented above. We have

$$\cancel{a^\dagger |\alpha\rangle} = \left(\alpha^* + (\cancel{a^\dagger} - \alpha^*/2) \right) |\alpha\rangle$$

The terms with $a^\dagger$ cancel out and we are left with

$$\alpha^* |\alpha\rangle = 0$$

1.6 Validity of the exponential representation of the Kronecker delta

Consider two discrete wave vectors, $\mathbf{k}$ and $\mathbf{k}'$, for plane waves states $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ in a cubic system of volume $L^3 = V$ with periodic boundary conditions over the length L . Take, for example, the x coordinate. The periodic boundary condition imply that

$$e^{i(k_x x - \omega t)} = e^{i(k_x (x+L) - \omega t)} = e^{i(k_x x - \omega t)} e^{i(k_x L)}$$

$$\Rightarrow e^{ik_x L} = 1 \Rightarrow k_x L = 2\pi n_x \Leftrightarrow k_x = 2\pi n_x / L$$

with $n_x = 0, \pm 1, \pm 2, \dots$. This is perfect analogous to the y and z coordinate. ONDE É QUE ENTRA ESTAS CENAS? EU PERCEBO QUE QUANDO ESTIVER A INTEGRAL POSSO FAZER UMA MUDANÇA DE VARIÁVEL DE x PARA $2\pi x L$ E VAI APARECER O FACTOR DE $V/(2\pi)^3$. TAMBÉM VAI APARECER OS N SEREM TODOS IGUAIS. ESTOU É MEIO CONFUSO, QUERIA ISTO DIREITINHO. Because the integral

$$\int e^{ik_x} dk$$

does not converge we need to add a regularization factor $e^{-x\epsilon}$ so that it does converge. Once we evaluate the integral we can make $\epsilon \rightarrow 0$. The procedure is as follows

$$\begin{aligned}
\int e^{ikx} dx &= \int_{-\infty}^0 e^{ikx} dx + \int_0^{+\infty} e^{ikx} dx \\
&= \int_0^{\infty} e^{-ikx} dx + \int_0^{+\infty} e^{ikx} dx \\
&= \int_0^{+\infty} (e^{-ikx} + e^{ikx}) dx \\
&= \lim_{\epsilon \rightarrow 0} \int_0^{+\infty} (e^{-ikx} + e^{ikx}) e^{-\epsilon x} dx \\
&= \lim_{\epsilon \rightarrow 0} \int_0^{+\infty} (e^{-(ik+\epsilon)x} + e^{(ik-\epsilon)x}) dx \\
&= \lim_{\epsilon \rightarrow 0} \left(\frac{1}{-(ik+\epsilon)} e^{-(ik+\epsilon)x} + \frac{1}{(ik-\epsilon)} e^{(ik-\epsilon)x} \right)_0^{\infty} dx \\
&= \lim_{\epsilon \rightarrow 0} \left((0+0) + \left(\frac{1}{-(ik+\epsilon)} + \frac{1}{(ik-\epsilon)} \right) \right) \\
&= \lim_{\epsilon \rightarrow 0} \left(\frac{(ik-\epsilon) - (ik+\epsilon)}{-(ik+\epsilon)(ik-\epsilon)} \right) \\
&= \lim_{\epsilon \rightarrow 0} \left(\frac{2\epsilon}{k^2 + \epsilon^2} \right) = \begin{cases} 0 & k \neq 0 \\ \infty & k = 0 \end{cases}
\end{aligned}$$

1.7 Interaction term for a 3D and 2D interacting electron gas

The Coulom potential energy with screening is given, in terms of the spatial variables, by

$$w(|\mathbf{r} - \mathbf{r}'|) = \frac{e^2 e^{-\lambda|\mathbf{r} - \mathbf{r}'|}}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|}$$

where λ is a positive screening parameter. To represent it in the wave-vector space we take it's Fourier transform

$$v(\mathbf{q}) = \int d^3\mathbf{R} e^{-i\mathbf{q} \cdot \mathbf{R}} w(R)$$

$$\begin{aligned}
&= \frac{e^2}{4\pi\epsilon_0} \int_0^\infty \int_0^{2\pi} \int_0^\pi R^2 \sin(\theta) dR d\theta d\phi e^{-i\mathbf{q}\cdot\mathbf{R}} \frac{e^{-\lambda R}}{R} \\
&= \frac{e^2}{2\epsilon_0} \int_0^\infty \int_{-1}^1 R dR d(\cos(\theta)) e^{-iqR \cos(\theta)} e^{-\lambda R} \\
&= \frac{e^2}{2\epsilon_0} \int_0^\infty \frac{1}{-iqR} (e^{-iqR} - e^{+iqR}) R dR e^{-\lambda R} \\
&= \frac{e^2}{-iq2\epsilon_0} \int_0^\infty (e^{-(iq+\lambda)R} - e^{+(iq-\lambda)R}) dR \\
&= \frac{e^2}{-iq2\epsilon_0} \left(\frac{e^{-(iq+\lambda)R}}{-(iq+\lambda)} - \frac{e^{+(iq-\lambda)R}}{+(iq-\lambda)} \right)_0^\infty \\
&= \frac{e^2}{iq2\epsilon_0} \left(\frac{e^{-(iq+\lambda)R}(iq-\lambda) + e^{+(iq-\lambda)R}(iq+\lambda)}{-q^2 - \lambda^2} \right)_0^\infty \\
&= \frac{-e^2}{iq2\epsilon_0} \frac{1}{q^2 + \lambda^2} (-((iq-\lambda) + (iq+\lambda))) \\
&= \frac{e^2}{iq2\epsilon_0} \frac{1}{q^2 + \lambda^2} \cancel{(2iq)} \\
&= \frac{e^2}{\epsilon_0} \frac{1}{q^2 + \lambda^2}
\end{aligned}$$

1.8 Commutators with the Hamiltonian of a free gas of boson

The Hamiltonian for a free gas of bosons is given by

$$H = \sum_{\mathbf{k}} E_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$

To facilitate on the notation we drop the bold font on the vector wave $\mathbf{k}$. We start by evaluating the commutator

$$[a_k^\dagger, H] = [a_k^\dagger, \sum_k E_k a_k^\dagger a_k] = \sum_{k'} E_{k'} [a_k^\dagger, a_{k'}^\dagger a_{k'}]$$

Using the relation

$$[AB, C] = [A, C]B + A[B, C]$$

we get

$$\begin{aligned}
[a_k^\dagger, H] &= - \sum_{k'} E_{k'} \left([a_{k'}^\dagger, a_k^\dagger] a_{k'} + a_{k'}^\dagger [a_{k'}, a_k^\dagger] \right) \\
&= - \sum_{k'} E_{k'} \left(0 + a_{k'}^\dagger \delta_{k, k'} \right) \\
&= -E_k a_k^\dagger
\end{aligned}$$

Analogously for the commutator with a_k we get

$$\begin{aligned}[a_k^\dagger, H] &= - \sum_{k'} E_{k'} \left([a_{k'}^\dagger, a_k] a_{k'} + a_{k'}^\dagger [a_{k'}, a_k] \right) \\ &= - \sum_{k'} E_{k'} (-\delta_{k,k'} a_{k'} + 0) \\ &= E_k a_k\end{aligned}$$

1.9 Commutator of pair of electrons (2)

1.9.1 Proof of a commutation of 4 fermionic operators

Because the operators are fermionic we use instead this straightforward relation

$$[AB, C] = A\{B, C\} - \{A, C\}B$$

Using this relation to expand for the 4 operators we get

$$\begin{aligned}[AB, CD] &= A[B, CD] + [A, CD]B \\ &= -A[CD, B] - [CD, A]B \\ &= -A(C\{D, B\} - \{C, B\}D) - (C\{D, A\} - \{C, A\}D)B\end{aligned}$$

We then obtain

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

1.9.2 First Commutator

$$[b_{\mathbf{k}}, b_{\mathbf{k}'}^\dagger] = [c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow}, c_{\mathbf{k}'\uparrow}^\dagger c_{-\mathbf{k}'\downarrow}^\dagger]$$

Because c and $c^\dagger$ are fermionic operators it's commutation relations are

$$\begin{aligned}\{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}^\dagger\} &= \delta_{\mathbf{k},\mathbf{k}'} \delta_{\sigma,\sigma'} \\ \{c_{\mathbf{k}\sigma}, c_{\mathbf{k}'\sigma'}\} &= \{c_{\mathbf{k}\sigma}^\dagger, c_{\mathbf{k}'\sigma'}^\dagger\} = 0\end{aligned}$$

Using the expression derived above we obtain by substitution

$$[b_{\mathbf{k}}, b_{\mathbf{k}'}^\dagger] = -c_{-\mathbf{k}\downarrow} c_{\mathbf{k}'\uparrow}^\dagger \delta_{-\mathbf{k}',\mathbf{k}} \delta_{\downarrow,\uparrow} + c_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow}^\dagger \delta_{\mathbf{k}',\mathbf{k}} \delta_{\uparrow,\uparrow} - c_{\mathbf{k}'\uparrow}^\dagger c_{\mathbf{k}\uparrow} \delta_{-\mathbf{k}',-\mathbf{k}} \delta_{\downarrow,\downarrow} + c_{-\mathbf{k}'\downarrow}^\dagger c_{\mathbf{k}\uparrow} \delta_{\mathbf{k}',-\mathbf{k}} \delta_{\uparrow,\downarrow}$$

By definition we have that

$$\delta_{\sigma,\sigma'} = \begin{cases} 1 & , \sigma = \sigma' \\ 0 & , \sigma \neq \sigma' \end{cases}$$

and so $\delta_{\downarrow,\uparrow} = \delta_{\uparrow,\downarrow} = 0$ and $\delta_{\uparrow,\uparrow} = \delta_{\downarrow,\downarrow} = 1$. Because of this the first and last term are zero. We are left with

$$\begin{aligned} [b_{\mathbf{k}}, b_{\mathbf{k}'}^\dagger] &= (c_{-\mathbf{k}\downarrow} c_{-\mathbf{k}'\downarrow}^\dagger - c_{\mathbf{k}'\uparrow}^\dagger c_{\mathbf{k}\uparrow}) \delta_{\mathbf{k}',\mathbf{k}} \\ &= (c_{-\mathbf{k}\downarrow} c_{-\mathbf{k}\downarrow}^\dagger - c_{\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}\uparrow}) \delta_{\mathbf{k}',\mathbf{k}} \\ &= ([1 - n_{-\mathbf{k}\downarrow}] - n_{\mathbf{k}\uparrow}) \delta_{\mathbf{k}',\mathbf{k}} \\ &= (1 - n_{\mathbf{k}\uparrow} - n_{-\mathbf{k}\downarrow}) \delta_{\mathbf{k}',\mathbf{k}} \end{aligned}$$

1.9.3 Second Commutator

$$[b_{\mathbf{k}}, b_{\mathbf{k}'}] = [c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow}, c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}] = 0$$

1.9.4 Third Commutator

Analogous to what we did in 1.9.1 it can be demonstrated that

$$\{AB, CD\} = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

1.10 Bogoliubov transformation for an interacting gas of bosons

One can try to find a new set of creation and annihilation such that the Hamiltonian of an interacting gas of bosons is diagonal

$$H = \sum_{\mathbf{k}} E_{\mathbf{k}} \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$$

in the limit where we assume the ground state of the system $|\phi_0\rangle$ to be when all the particles reside in the lowest level such that $\alpha_{\mathbf{k}} |\phi_0\rangle = 0$ for all $\mathbf{k}$. “Physically” this means we are looking for a new quasiparticle which is referred as Bogoliouvons. As found at the end of page 24 this operators are found to be

$$\Leftrightarrow \begin{cases} a_{\mathbf{k}} = \cosh(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}} + \sinh(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}}^\dagger \\ a_{\mathbf{k}}^\dagger = \cosh(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}}^\dagger + \sinh(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \end{cases}$$

The number of particles $F(\mathbf{k})$ with wave vector $\mathbf{k}$ in the quasiparticle ground state is given by

$$\begin{aligned} F(\mathbf{k}) &= \langle \phi_0 | a_{\mathbf{k}}^\dagger a_{\mathbf{k}} | \phi_0 \rangle \\ &= \langle \phi_0 | \left(\cosh(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}}^\dagger + \sinh(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \right) \left(\cosh(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}} + \sinh(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}}^\dagger \right) | \phi_0 \rangle \\ &= \langle \phi_0 | \cosh^2(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \sinh^2(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger + \cosh(\theta_{\mathbf{k}}) \sinh(\theta_{\mathbf{k}}) \alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger + \sinh(\theta_{\mathbf{k}}) \cosh(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \alpha_{\mathbf{k}} | \phi_0 \rangle \end{aligned}$$

Given the terminal condition of the ladder operator the first and last term are zero. We are left with

$$F(\mathbf{k}) = \langle \phi_0 | \sinh^2(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger + 2 \sinh(2\theta_{\mathbf{k}}) \alpha_{\mathbf{k}}^\dagger \alpha_{-\mathbf{k}}^\dagger | \phi_0 \rangle$$

Because the occupation of the ground state is $N_0 \approx N \rightarrow +\infty \Rightarrow N_0 - 1 \approx N_0$ the coefficient of the creation operator is $\approx N$. Looking at the last term, because we go up twice and never come down the bracket of $\langle \phi_0 |$ with this excited state gives zero. We are left with

$$F(\mathbf{k}) = \langle \phi_0 | \sinh^2(\theta_{\mathbf{k}}) \alpha_{-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger | \phi_0 \rangle = \sinh^2(\theta_{\mathbf{k}}) \sqrt{N}^2 \langle \phi_0 | \phi_0 \rangle = N \sinh^2(\theta_{\mathbf{k}})$$

As derived in the section it can be found the expression for $\sinh^2(\theta_{\mathbf{k}})$ by comparing the original hamiltonean in Eq.(1.70) and the one one gets by substituting the bogolibouv operators expression. By this method we get

$$\sinh^2(\theta_{\mathbf{k}}) = \frac{v(\mathbf{k})N^2}{E_{\mathbf{k}}} \quad , \text{with} \quad E_{\mathbf{k}} = \frac{k^2}{2m}$$

Which means that the number of particles $F(\mathbf{k})$ is

$$F(\mathbf{k}) = 2mN^3 \frac{v(\mathbf{k})}{k^2}$$

1.11 Commutator of Holstein-Primakoff Spin operators

One can rewrite the set of spin operators (S^+ , S^- and S^z) in terms of boson or fermion operators while still satisfying the spin commutation relations

$$\begin{cases} [S_i^+, S_j^-] = 2S_i^z \delta_{i,j} \\ [S_i^\pm, S_j^z] = \mp S_i^\pm \delta_{i,j} \end{cases}$$

This transformation is known and called Holstein-Primakoff (H-P) transformations. They are as follows

$$\begin{cases} S_i^+ = \sqrt{2S} \left(1 - \frac{a_i^\dagger a_i}{2S}\right)^{1/2} a_i \\ S_i^- = \sqrt{2S} a_i^\dagger \left(1 - \frac{a_i^\dagger a_i}{2S}\right)^{1/2} \\ S_i^z = S - a_i^\dagger a_i \end{cases}$$

Let's verify if after the HP transformation the spin commutators hold for bosonic operators.

1.11.1 Commutator S^+ with S^-

As indicated we will use that $S = 1/2$.

$$[S_i^+, S_j^-] = \left[\left(1 - a_i^\dagger a_i\right)^{1/2} a_i, a_j^\dagger \left(1 - a_j^\dagger a_j\right)^{1/2} \right]$$

Now we use the recurrent properties for $[AB, C]$

$$= \left(1 - a_i^\dagger a_i\right)^{1/2} \left[a_i, a_j^\dagger \left(1 - a_j^\dagger a_j\right)^{1/2} \right] + \left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_j^\dagger \left(1 - a_j^\dagger a_j\right)^{1/2} \right] a_i$$

And again

$$\begin{aligned}
&= - \left(1 - a_i^\dagger a_i\right)^{1/2} \left(a_j^\dagger \left[\left(1 - a_j^\dagger a_j\right)^{1/2}, a_i \right] + \left[a_j^\dagger, a_i \right] \left(1 - a_j^\dagger a_j\right)^{1/2} \right) \\
&- \left(a_j^\dagger \left[\left(1 - a_j^\dagger a_j\right)^{1/2}, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] + \left[a_j^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} \right) a_i
\end{aligned}$$

Using the commutator relation of the bosonic operators we have

$$\begin{aligned}
&= - \left(1 - a_i^\dagger a_i\right)^{1/2} \left(a_j^\dagger \left[\left(1 - a_j^\dagger a_j\right)^{1/2}, a_i \right] - \delta_{i,j} \left(1 - a_j^\dagger a_j\right)^{1/2} \right) \\
&- \left(a_j^\dagger \delta_{i,j} + \left[a_j^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} \right) a_i
\end{aligned}$$

Putting $i = j$ because of the kronecker delta

$$= - \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_i \right] + \delta_{i,i} \left(1 - a_i^\dagger a_i\right)^{1/2} - \left[a_i^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} a_i$$

Reagrouping terms to make obvious the appearance of S_z

$$\begin{aligned}
&= 2 \left(1/2 - a_i^\dagger a_i\right) \delta_{i,i} - \left(\left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_i \right] + \left[a_i^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} a_i \right) \\
&= 2 S_i^z \delta_{i,i} - \left(\left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_i \right] + \left[a_i^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} a_i \right)
\end{aligned}$$

To make the result hold the last term has to be zero. Let's then check if it is.

$$\begin{aligned}
&\left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_i \right] + \left[a_i^\dagger, \left(1 - a_i^\dagger a_i\right)^{1/2} \right] \left(1 - a_i^\dagger a_i\right)^{1/2} a_i \\
&= \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left(\left(1 - a_i^\dagger a_i\right)^{1/2} a_i - a_i \left(1 - a_i^\dagger a_i\right)^{1/2} \right) \\
&+ \left(a_i^\dagger \left(1 - a_i^\dagger a_i\right)^{1/2} - \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \right) \left(1 - a_i^\dagger a_i\right)^{1/2} a_i \\
&= \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left(1 - a_i^\dagger a_i\right)^{1/2} a_i - \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger a_i \left(1 - a_i^\dagger a_i\right)^{1/2} \\
&+ a_i^\dagger \left(1 - a_i^\dagger a_i\right)^{1/2} \left(1 - a_i^\dagger a_i\right)^{1/2} a_i - \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left(1 - a_i^\dagger a_i\right)^{1/2} a_i \\
&= \left(1 - a_i^\dagger a_i\right)^{1/2} a_i^\dagger \left(\left(1 - a_i^\dagger a_i\right)^{1/2} - \left(1 - a_i^\dagger a_i\right)^{1/2} \right) a_i +
\end{aligned}$$

1.11.2 Commutator S^+ with S^z

$$\begin{aligned} [S_i^+, S_j^z] &= \left[\left(1 - a_i^\dagger a_i\right)^{1/2} a_i, \frac{1}{2} - a_j^\dagger a_j \right] \\ &= - \left[\left(1 - a_i^\dagger a_i\right)^{1/2} a_i, a_j^\dagger a_j \right] \end{aligned}$$

Now we use the recurrent propertie for $[AB, C]$ and because a function of an operator always commutes with that operator we are left with

$$\begin{aligned} &= - \left(\left[\left(1 - a_i^\dagger a_i\right)^{1/2}, a_j^\dagger a_j \right] a_i + \left(1 - a_i^\dagger a_i\right)^{1/2} [a_i, a_j^\dagger a_j] \right) \\ &= - \left(1 - a_i^\dagger a_i\right)^{1/2} [a_i, a_j^\dagger a_j] \\ &= \left(1 - a_i^\dagger a_i\right)^{1/2} \left(a_j^\dagger [a_i, a_j] + [a_j^\dagger, a_i] a_j \right) \\ &= \left(1 - a_i^\dagger a_i\right)^{1/2} a_i (-\delta_{i,j}) \\ &= -S_i^+ \delta_{i,j} \end{aligned}$$

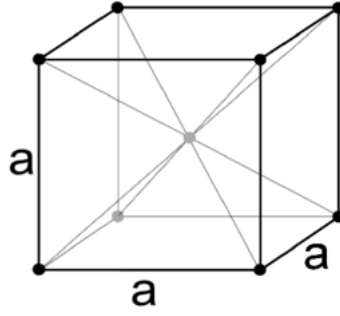
1.12 Exchange term $J(\mathbf{k})$ for a b.c.c ferromagnetic

For a body-centered cubic (b.c.c) ferromagnetic lattice with nearest-neighbour exchange J and a lattice constant a , the exchange term is given by

$$J(\mathbf{k}) = \sum_{\mathbf{r}_i - \mathbf{r}_j} J(\mathbf{r}_i - \mathbf{r}_j) e^{+i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)} = J \sum_{\boldsymbol{\delta}} e^{+i\mathbf{k} \cdot \boldsymbol{\delta}}$$

If we consider one atom to be the origin at the position of its 7 nearest-neighbours are

$$\begin{aligned} J(\mathbf{k}) &= J \left[e^{-ik_z a/2} \left\{ e^{+ik_y a/2} \left(e^{-ik_x a/2} + e^{+ik_x a/2} \right) + e^{-ik_y a/2} \left(e^{-ik_x a/2} + e^{+ik_x a/2} \right) \right\} \right. \\ &\quad \left. + J \left[e^{+ik_z a/2} \left\{ e^{+ik_y a/2} \left(e^{-ik_x a/2} + e^{+ik_x a/2} \right) + e^{-ik_y a/2} \left(e^{-ik_x a/2} + e^{+ik_x a/2} \right) \right\} \right] \right] \\ &= J \left[e^{-ik_z a/2} \left\{ e^{+ik_y a/2} + e^{-ik_y a/2} \right\} \times 2 \cos(k_x a/2) \right] \\ &\quad + J \left[e^{+ik_z a/2} \left\{ e^{+ik_y a/2} + e^{-ik_y a/2} \right\} \times 2 \cos(k_x a/2) \right] \\ &= J \left[\left(e^{-ik_z a/2} + e^{+ik_z a/2} \right) \times 2 \cos(k_y a/2) \times 2 \cos(k_x a/2) \right] \\ &= J [2 \cos(k_z a/2) \times 2 \cos(k_y a/2) \times 2 \cos(k_x a/2)] \\ &= 8J \cos(k_z a/2) \cos(k_y a/2) \cos(k_x a/2) \end{aligned}$$



Body-centered cubic (I)

1.13 Coupling between Spin-Waves and Phonons (3)

The coupling between spin-waves and phonons may be described by the Hamiltonian

$$H = \sum_{\mathbf{k}} \left(\omega_k a_k^\dagger a_k + \Omega_k b_k^\dagger b_k + c_k (a_k b_k^\dagger + a_k^\dagger b_k) \right)$$

where the $a/a^\dagger$ are the spin-waves operators while $b/b^\dagger$ are the phonons operators. Also c_k is a real, positive coupling coefficient. We choose a transformation to a new bosons operators given by

$$\begin{cases} a_k = A_k \cos(\phi_k) + B_k \sin(\phi_k) \\ b_k = B_k \cos(\phi_k) - A_k \sin(\phi_k) \end{cases}$$

with ϕ_k representing a real quantity, to reexpress the Hamiltonian in a quasi-particle/diagonalized form as

$$H = \sum_k \left(\epsilon_k A_k^\dagger A_k + E_k B_k^\dagger B_k \right)$$

By direct substitution of the new operators in the original hamiltonean we obtain

$$\begin{aligned} H = \sum_{\mathbf{k}} & \omega \left(A^\dagger \cos(\phi) + B^\dagger \sin(\phi) \right) \left(A \cos(\phi) + B \sin(\phi) \right) + \Omega \left(B^\dagger \cos(\phi) - A^\dagger \sin(\phi) \right) \left(B \cos(\phi) - A \sin(\phi) \right) \\ & + c \left((A \cos(\phi) + B \sin(\phi)) (B^\dagger \cos(\phi) - A^\dagger \sin(\phi)) + (A^\dagger \cos(\phi) + B^\dagger \sin(\phi)) (B \cos(\phi) - A \sin(\phi)) \right) \end{aligned}$$

Reagrouping common terms and noting that the new set of operators commute with each other i.e $[A, B] = [A, B^\dagger] = [A^\dagger, B] = [A^\dagger, B^\dagger] = 0$ one can get we have

$$\begin{aligned} &= \sum_{\mathbf{k}} \omega \left\{ A^\dagger A \cos^2(\phi) + B^\dagger B \sin^2(\phi_k) + (A^\dagger B + B^\dagger A) \cos(\phi) \sin(\phi) \right\} \\ &+ \Omega \left\{ B^\dagger B \cos^2(\phi) + A^\dagger A \sin^2(\phi) - (A^\dagger B + B^\dagger A) \cos(\phi) \sin(\phi) \right\} \\ &+ c \left\{ (AB^\dagger + A^\dagger B) \cos^2(\phi) - (A^\dagger B + AB^\dagger) \sin^2(\phi) \right. \\ &\left. + \left(-(\cancel{A} - A^\dagger \cancel{A}) + (\cancel{B} - B^\dagger \cancel{B}) - \cancel{A^\dagger A} + \cancel{B^\dagger B} \right) 2 \cos(\phi) \sin(\phi) \right\} \end{aligned}$$

Regrouping in terms of the operators

$$\begin{aligned}
&= \sum_k A^\dagger A (\omega \cos^2(\phi) + \Omega \sin^2(\phi)) + B^\dagger B (\omega \sin^2(\phi) + \Omega \cos^2(\phi)) \\
&+ (A^\dagger B + B^\dagger A) \left((\omega - \Omega) \frac{1}{2} \sin(2\phi) + c \cos(2\phi) \right)
\end{aligned}$$

We obtain a system of 3 equations for 3 unknowns ϵ_k, E_k and ϕ_k , being

$$\begin{cases} \epsilon = \omega \cos^2(\phi) + \Omega \sin^2(\phi) \\ E = \omega \sin^2(\phi) + \Omega \cos^2(\phi) \\ (\omega - \Omega) \frac{1}{2} \sin(2\phi) + c \cos(2\phi) = 0 \end{cases}$$

Adding and subtracting the first equation to the second we obtain a different set of more usefull equations

$$\begin{cases} \epsilon + E = \omega + \Omega \\ \epsilon - E = (\omega - \Omega) \cos(2\phi) \\ \omega - \Omega = -2c \arctan(2\phi) = 0 \end{cases}$$

2 Time Evolution and Equation of Motion

2.1 Time evolution of the wave function in the interaction picture

The wave function in the interaction picture is defined as

$$|\phi_{int}(t)\rangle = U_0^\dagger(t) |\phi_S(t)\rangle$$

This relation can be arranged using the unity property to give

$$|\phi_S(t)\rangle = U_0(t) |\phi_{int}(t)\rangle$$

where U_0 is the time-evolution operator in the absence of the interacting part of the total Hamiltonian $H = H_0 + H_{int}(t)$. As in equation (2.9) it's solution if of the form

$$U_0(t) = e^{-i \int_0^t H_0(\tau) d\tau} \xrightarrow{\text{if } H_0(t) = H_0} U_0(t) = e^{-i H_0 t}$$

To study the time evolution of the wave function in the interacting in the interaction picture we start from the Schrodinger equation

$$\begin{aligned}
&i \frac{d}{dt} |\phi_S(t)\rangle = H(t) |\phi_S(t)\rangle \\
&\Leftrightarrow i \frac{d}{dt} (U_0(t) |\phi_{int}(t)\rangle) = H(t) (U_0(t) |\phi_{int}(t)\rangle) \\
&\Leftrightarrow i \left(\frac{d}{dt} (e^{-i H_0 t}) |\phi_{int}(t)\rangle + U_0(t) \frac{d}{dt} |\phi_{int}(t)\rangle \right) = H(t) e^{-i H_0 t} |\phi_{int}(t)\rangle \\
&\Leftrightarrow i U_0(t) \left(-i H_0 |\phi_{int}(t)\rangle + \frac{d}{dt} |\phi_{int}(t)\rangle \right) = H(t) U_0(t) |\phi_{int}(t)\rangle
\end{aligned}$$

If we multiply both sides by $U_0^\dagger(t)$ at the right, and because U_0 is unitary i.e $U_0^\dagger(t)U_0(t) = \mathbb{1}$ we obtain

$$\begin{aligned} i \frac{d}{dt} |\phi_{int}(t)\rangle &= U_0^\dagger(t)(H - H_0)U_0(t) |\phi_{int}(t)\rangle \\ &= U_0^\dagger(t)H_{int}(t)U_0(t) |\phi_{int}(t)\rangle \end{aligned}$$

If we define a interaction picture Hamiltonian $V_{int}(t)$ as

$$V_{int}(t) = U_0^\dagger(t)H_{int}(t)U_0(t)$$

we can rewrite the equation as a Schrodinger-like equation but in the interaction picture

$$i \frac{d}{dt} |\phi_{int}(t)\rangle = V_{int}(t) |\phi_{int}(t)\rangle$$

2.2 Forced Quantum Harmonic Oscillator (F1.6)

The Hamiltonian of a forced harmonic oscillator in 1D can be written as

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - x f(t)$$

where $f(t)$ is a time-dependent external force that is switched on at time $t = 0$, so that $f(t) \rightarrow 0$ as $t \rightarrow 0$. As usual, if we define the ladder operators as

$$\begin{cases} a^\dagger = (2m\omega)^{-1/2}(m\omega x + ip) \\ a = (2m\omega)^{-1/2}(m\omega x - ip) \end{cases}$$

we obtain an undisturbed hamiltonian of the form

$$H_0 = \omega(a^\dagger a + 1/2).$$

The perturbation term given to us as

$$\begin{aligned} H_1 &= -h(t)(a^\dagger + a) \\ &= -h(t)(2m\omega)^{-1/2}((m\omega x + ip) + (m\omega x - ip)) \\ &= -h(t)(2m\omega)^{-1/2}(2m\omega)x \\ &= -h(t)(2m\omega)^{+1/2}x \end{aligned}$$

By comparison we find that

$$\begin{aligned} -h(t)(2m\omega)^{+1/2}x &= -x f(t) \\ \Leftrightarrow h(t) &= (2m\omega)^{-1/2}f(t) \end{aligned}$$

2.3 Time-dependent potential FHO

2.3.1 Verifying the expression

The time-dependent potential energy of the FHO in the interaction picture is given by

$$V_{int}(t) = F(t) [ae^{-i\omega t} + a^\dagger e^{+i\omega t}]$$

2.3.2 Commutator

$$\begin{aligned} [V_{int}(t_1), V_{int}(t_2)] &= [F(t_1) (ae^{-i\omega t_1} + a^\dagger e^{+i\omega t_1}), F(t_2) (ae^{-i\omega t_2} + a^\dagger e^{+i\omega t_2})] \\ &= F(t_1)F(t_2) \left(e^{-i\omega(t_1-t_2)} [a, a^\dagger] + e^{+i\omega(t_1-t_2)} [a^\dagger, a] \right) \\ &= F(t_1)F(t_2) \left(e^{-i\omega(t_1-t_2)} - e^{+i\omega(t_1-t_2)} \right) \\ &= -2iF(t_1)F(t_2) \sin(\omega(t_1 - t_2)) \end{aligned}$$

2.4 Coupling between two sets of bosons (4)

The following Hamiltonian H describes the coupling between two sets of bosons

$$H = \lambda a^\dagger a + \epsilon(ab^\dagger + a^\dagger b)$$

The equation of motion for the operator a is given by

$$\begin{aligned} \partial_t a &= i[H, a] \\ &= i [\lambda a^\dagger a + \epsilon(ab^\dagger + ba^\dagger), a] \\ &= i\lambda [a^\dagger a, a] + i\epsilon [ba^\dagger, a] \\ &= -i\lambda a - i\epsilon b \end{aligned}$$

In a similar way, for the operator b , we get

$$\begin{aligned} \partial_t b &= i[H, b] \\ &= i [\lambda a^\dagger a + \epsilon(ab^\dagger + ba^\dagger), b] \\ &= i\epsilon [ab^\dagger, b] \\ &= -i\epsilon a. \end{aligned}$$

To deduce the values for the frequencies of the coupled modes we need to solve the system of E.D

$$\frac{d}{dt} \begin{bmatrix} a(t) \\ b(t) \end{bmatrix} = -i \begin{bmatrix} \lambda & \epsilon \\ \epsilon & 0 \end{bmatrix} \begin{bmatrix} a(t) \\ b(t) \end{bmatrix}.$$

We look for solution of the time

$$\begin{bmatrix} a(t) \\ b(t) \end{bmatrix} = \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} c_1 e^{-i\Omega_1 t} \\ c_2 e^{-i\Omega_2 t} \end{bmatrix}$$

$$\begin{aligned}\Rightarrow \frac{d}{dt} \begin{bmatrix} a(t) \\ b(t) \end{bmatrix} &= \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} -i\Omega_1 c_1 e^{-i\Omega_1 t} \\ -i\Omega_2 c_2 e^{-i\Omega_2 t} \end{bmatrix} \\ &= -i \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{bmatrix} \begin{bmatrix} c_1 e^{-i\Omega_1 t} \\ c_2 e^{-i\Omega_2 t} \end{bmatrix}\end{aligned}$$

Comparing the two form we get

$$\begin{aligned}\begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{bmatrix} \begin{bmatrix} c_1 e^{-i\Omega_1 t} \\ c_2 e^{-i\Omega_2 t} \end{bmatrix} &= \begin{bmatrix} \lambda & \epsilon \\ \epsilon & 0 \end{bmatrix} \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} c_1 e^{-i\Omega_1 t} \\ c_2 e^{-i\Omega_2 t} \end{bmatrix} \\ \Leftrightarrow \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} \Omega_1 & 0 \\ 0 & \Omega_2 \end{bmatrix} &= \begin{bmatrix} \lambda & \epsilon \\ \epsilon & 0 \end{bmatrix} \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix}\end{aligned}$$

We can solve this equations solving for the eigenvalues

$$\begin{aligned}\begin{bmatrix} \lambda & \epsilon \\ \epsilon & 0 \end{bmatrix} \begin{bmatrix} u_{1,n} \\ u_{2,n} \end{bmatrix} &= \Omega_n \begin{bmatrix} u_{1,n} \\ u_{2,n} \end{bmatrix}, n = 1, 2 \\ \Rightarrow 0 &= \det \begin{bmatrix} \lambda - \Omega_n & \epsilon \\ \epsilon & -\Omega_n \end{bmatrix} = -(\lambda - \Omega_n) \Omega_n - \epsilon^2 \\ &\Rightarrow \Omega_n^2 - \lambda \Omega_n - \epsilon^2 = 0 \\ \Rightarrow \Omega_n &= \frac{\lambda \pm \sqrt{\lambda^2 + 4\epsilon^2}}{2} = \frac{\lambda}{2} \pm \sqrt{\left(\frac{\lambda}{2}\right)^2 + \epsilon^2}\end{aligned}$$

2.5 Diatomic vibration chain

We found in section 2.4.2 that the two modes of a diatomic chain of vibration atoms are

$$\omega_k = C \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \pm C \left[\left(\frac{1}{m_1} + \frac{1}{m_2} \right)^2 - \frac{4 \sin^2(ka)}{m_1 m_2} \right]^{1/2}$$

Taking the special case of $m_1 = m_2 = m$ we obtain

$$\omega_k^\pm = \frac{2C}{m} (1 \pm \cos(ka))$$

2.6 System of Boson

Consider the Hamiltonian

$$H = \sum_{\mathbf{k}} \left(F(\mathbf{k}) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} G(\mathbf{k}) (a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}) \right)$$

Here $F(\mathbf{k})$ and $G(\mathbf{k})$ are real function satisfying $F(\mathbf{k}) > G(\mathbf{k}) > 0$. The equation of motion for the operator $a_{\mathbf{k}}^\dagger$ is given by

$$\begin{aligned}
-i\partial_t a_{\mathbf{k}'}^\dagger &= [H, a_{\mathbf{k}'}^\dagger] \\
&= \sum_{\mathbf{k}} F(\mathbf{k}) [a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] + \sum_{\mathbf{k}} \frac{1}{2} G(\mathbf{k}) \left(\cancel{[a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}^\dagger]} + [a_{\mathbf{k}} a_{-\mathbf{k}}, a_{\mathbf{k}'}^\dagger] \right) \\
&= \sum_{\mathbf{k}} F(\mathbf{k}) \left(a_{\mathbf{k}}^\dagger \delta_{\mathbf{k}, \mathbf{k}'} \right) + \sum_{\mathbf{k}} \frac{1}{2} G(\mathbf{k}) (a_{\mathbf{k}} (\delta_{-\mathbf{k}, \mathbf{k}'} + \delta_{\mathbf{k}, \mathbf{k}'} a_{-\mathbf{k}})) \\
&= F(\mathbf{k}') a_{\mathbf{k}'}^\dagger + \frac{1}{2} (G(-\mathbf{k}') + G(\mathbf{k}')) a_{-\mathbf{k}'}
\end{aligned}$$

We now need to calculate the equation of motion for the operator $a_{-\mathbf{k}}$. Similarly

$$\begin{aligned}
-i\partial_t a_{-\mathbf{k}'} &= [H, a_{-\mathbf{k}'}] \\
&= \sum_{\mathbf{k}} F(\mathbf{k}) [a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{-\mathbf{k}'}] + \sum_{\mathbf{k}} \frac{1}{2} G(\mathbf{k}) \left([a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger, a_{-\mathbf{k}'}] + \cancel{[a_{\mathbf{k}} a_{-\mathbf{k}}, a_{-\mathbf{k}'}]} \right) \\
&= \sum_{\mathbf{k}} F(\mathbf{k}) (-\delta_{\mathbf{k}, -\mathbf{k}'} a_{\mathbf{k}}) + \sum_{\mathbf{k}} \frac{1}{2} G(\mathbf{k}) \left(a_{\mathbf{k}}^\dagger (-\delta_{-\mathbf{k}, -\mathbf{k}'} + (-\delta_{\mathbf{k}, -\mathbf{k}'} a_{-\mathbf{k}}^\dagger) \right) \\
&= -F(-\mathbf{k}') a_{-\mathbf{k}'} - \frac{1}{2} (G(-\mathbf{k}') + G(\mathbf{k}')) a_{\mathbf{k}'}^\dagger
\end{aligned}$$

We need to solve the system of E.D

$$\begin{aligned}
\frac{d}{dt} \begin{bmatrix} a_{\mathbf{k}}^\dagger(t) \\ a_{-\mathbf{k}}(t) \end{bmatrix} &= \frac{1}{-i} \begin{bmatrix} F(\mathbf{k}) & +\frac{1}{2}(G(-\mathbf{k}) + G(\mathbf{k})) \\ -\frac{1}{2}(G(-\mathbf{k}) + G(\mathbf{k})) & -F(-\mathbf{k}) \end{bmatrix} \begin{bmatrix} a_{\mathbf{k}}^\dagger(t) \\ a_{-\mathbf{k}}(t) \end{bmatrix} \\
&= i \begin{bmatrix} F(\mathbf{k}) & +g(\mathbf{k}) \\ -g(\mathbf{k}) & -F(-\mathbf{k}) \end{bmatrix} \begin{bmatrix} a_{\mathbf{k}}^\dagger(t) \\ a_{-\mathbf{k}}(t) \end{bmatrix}
\end{aligned}$$

Where we define $g(\mathbf{k}) = \frac{1}{2} (G(-\mathbf{k}) + G(\mathbf{k}))$ to facilitate on the notation. We look for solution like

$$\begin{aligned}
\begin{bmatrix} a_{\mathbf{k}}^\dagger(t) \\ a_{-\mathbf{k}}(t) \end{bmatrix} &= \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} e^{+i\Omega_{\mathbf{k}} t} \\ e^{-i\Omega_{-\mathbf{k}} t} \end{bmatrix} \\
\Rightarrow \frac{d}{dt} \begin{bmatrix} a_{\mathbf{k}}^\dagger(t) \\ a_{-\mathbf{k}}(t) \end{bmatrix} &= \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} +i\Omega_{\mathbf{k}} e^{+i\Omega_{\mathbf{k}} t} \\ -i\Omega_{-\mathbf{k}} e^{-i\Omega_{-\mathbf{k}} t} \end{bmatrix} \\
&= -i \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} -\Omega_{\mathbf{k}} & 0 \\ 0 & \Omega_{-\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{-i\Omega_{\mathbf{k}} t} \\ e^{-i\Omega_{-\mathbf{k}} t} \end{bmatrix}
\end{aligned}$$

Comparing the two equations we get

$$\begin{aligned}
-\not{t} \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} -\Omega_{\mathbf{k}} & 0 \\ 0 & \Omega_{-\mathbf{k}} \end{bmatrix} \begin{bmatrix} e^{+i\Omega_{\mathbf{k}}t} \\ e^{-i\Omega_{-\mathbf{k}}t} \end{bmatrix} &= \not{t} \begin{bmatrix} F(\mathbf{k}) & +g(\mathbf{k}) \\ -g(\mathbf{k}) & -F(-\mathbf{k}) \end{bmatrix} \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} e^{+i\Omega_{\mathbf{k}}t} \\ e^{-i\Omega_{-\mathbf{k}}t} \end{bmatrix} \\
\Rightarrow \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix} \begin{bmatrix} -\Omega_{\mathbf{k}} & 0 \\ 0 & \Omega_{-\mathbf{k}} \end{bmatrix} &= \begin{bmatrix} -F(\mathbf{k}) & -g(\mathbf{k}) \\ +g(\mathbf{k}) & +F(-\mathbf{k}) \end{bmatrix} \begin{bmatrix} u_{1,1} & u_{1,2} \\ u_{2,1} & u_{2,2} \end{bmatrix}
\end{aligned}$$

We can solve this equations solving for the eigenvalues

$$\begin{bmatrix} -F(\mathbf{k}) & -g(\mathbf{k}) \\ +g(\mathbf{k}) & +F(-\mathbf{k}) \end{bmatrix} \begin{bmatrix} u_{1,n} \\ u_{2,n} \end{bmatrix} = \Omega_n \begin{bmatrix} u_{1,n} \\ u_{2,n} \end{bmatrix}, n = 1, 2$$

$$\Rightarrow 0 = \det \begin{bmatrix} -F(\mathbf{k}) - \Omega_n & -g(\mathbf{k}) \\ +g(\mathbf{k}) & +F(-\mathbf{k}) - \Omega_n \end{bmatrix} = -(F(\mathbf{k}) + \Omega_n)(F(-\mathbf{k}) - \Omega_n) + g^2(\mathbf{k})$$

$$\Rightarrow \Omega_n^2 - (F(-\mathbf{k}) - F(+\mathbf{k}))\Omega_n + (g^2(\mathbf{k}) - F(-\mathbf{k})F(\mathbf{k})) = 0$$

$$\Rightarrow \Omega_n = \frac{(F(-\mathbf{k}) - F(+\mathbf{k})) \pm \sqrt{(F(-\mathbf{k}) - F(+\mathbf{k}))^2 - 4(g^2(\mathbf{k}) - F(-\mathbf{k})F(\mathbf{k}))}}{2}$$

If we consider $F(\mathbf{k})$ and $G(\mathbf{k})$ to be

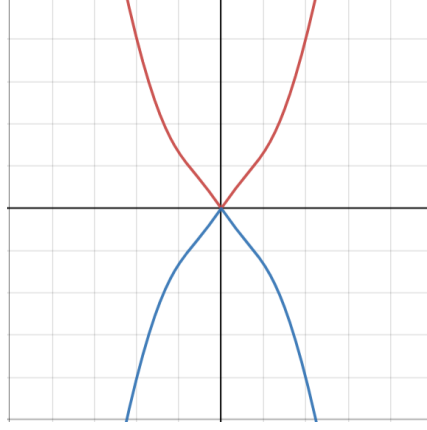
$$F(\mathbf{k}) = \alpha k^2 + e^{-k^2} \quad \text{and} \quad G(\mathbf{k}) = e^{-k^2}$$

assuming α to be a positive constant as indicated. This way the functions are simetric and so $F(\mathbf{k}) = F(-\mathbf{k}) \equiv F(|k|)$ and $G(\mathbf{k}) = G(-\mathbf{k}) \equiv G(|k|)$ which implies $g(\mathbf{k}) = \frac{1}{2}(G(-\mathbf{k}) + G(\mathbf{k})) = G(\mathbf{k})$. Our solutions reduces to

$$\begin{aligned}
\Omega_n &= \frac{(\cancel{F(-\mathbf{k})} - \cancel{F(+\mathbf{k})}) \pm \sqrt{(\cancel{F(-\mathbf{k})} - \cancel{F(+\mathbf{k})})^2 - 4(g^2(\mathbf{k}) - F(-\mathbf{k})F(\mathbf{k}))}}{\cancel{2}} \\
&= \pm \sqrt{-(G^2(k) - F^2(k))} \\
&= \pm \sqrt{-\left((e^{-k^2})^2 - (\alpha k^2 + e^{-k^2})^2\right)} \\
&= \pm \sqrt{-\left(\cancel{(e^{-k^2})^2} + (\alpha k^2)^2 + \cancel{(e^{-k^2})^2} + 2\alpha k^2 e^{-k^2}\right)} \\
&= \pm |k| \sqrt{\alpha^2 k^2 + 2\alpha e^{-k^2}}
\end{aligned}$$

Choosing $\alpha = 1$ for the plot we obtain

$$\Omega_n^\pm = \pm |k| \sqrt{k^2 + 2e^{-k^2}}$$



2.6.1 Checking the answer

We obtain the equations of motion

$$\begin{aligned}
& \begin{cases} -i\partial_t a_{\mathbf{k}}^\dagger(t) = F(\mathbf{k})a_{\mathbf{k}}^\dagger + \frac{1}{2}(G(-\mathbf{k}) + G(\mathbf{k}))a_{-\mathbf{k}} \\ -i\partial_t a_{-\mathbf{k}}(t) = -F(-\mathbf{k})a_{-\mathbf{k}} - \frac{1}{2}(G(-\mathbf{k}) + G(\mathbf{k}))a_{\mathbf{k}}^\dagger \end{cases} \\
& \Rightarrow \begin{cases} -ia_{\mathbf{k}}^\dagger \partial_t e^{+i\Omega t} = Fa_{\mathbf{k}}^\dagger e^{+i\Omega t} + Ga_{-\mathbf{k}} e^{+i\Omega t} \\ -ia_{-\mathbf{k}} \partial_t e^{+i\Omega t} = -Fa_{-\mathbf{k}} e^{+i\Omega t} - Ga_{\mathbf{k}}^\dagger e^{+i\Omega t} \end{cases} \\
& \Rightarrow \begin{cases} -\Omega a_{\mathbf{k}}^\dagger = Fa_{\mathbf{k}}^\dagger + Ga_{-\mathbf{k}} \\ -\Omega a_{-\mathbf{k}} = -Fa_{-\mathbf{k}} - Ga_{\mathbf{k}}^\dagger \end{cases}
\end{aligned}$$

Adding one to another

$$\begin{aligned}
& \Rightarrow \Omega^2 (a_{\mathbf{k}}^\dagger)^2 - (Fa_{\mathbf{k}}^\dagger + Ga_{-\mathbf{k}})^2 + \Omega^2 (a_{-\mathbf{k}})^2 + (Fa_{-\mathbf{k}} + Ga_{\mathbf{k}}^\dagger)^2 = 0 \\
& \Leftrightarrow (F^2 - G^2) (a_{\mathbf{k}}^\dagger)^2 + (F^2 - G^2) (a_{-\mathbf{k}})^2 - (Fa_{\mathbf{k}}^\dagger)^2 - (Ga_{-\mathbf{k}})^2 + (Fa_{-\mathbf{k}})^2 + (Ga_{\mathbf{k}}^\dagger)^2 = 0
\end{aligned}$$

All the terms cancel out. QED

2.7 Dipole-exchange Hamiltonian in second-quantization

The total spin Hamiltonian can now be written as $\mathcal{H} = \mathcal{H}_{ex} + \mathcal{H}_{dip}$, where $\mathcal{H}_{ex}$ includes the exchange interactions plus an applied magnetic field term

$$\mathcal{H}_{ex} = -\frac{1}{2} \sum_{i,j} J_{i,j} \mathbf{S}_i \cdot \mathbf{S}_j - g\mu_B B_0 \sum_i S_i^z$$

whereas $\mathcal{H}_{dip}$ is the extra term describing the dipolar interactions between all sites

$$\mathcal{H}_{dip} = \frac{1}{2} \left(\frac{\mu_0 g^2 \mu_B^2}{4\pi} \right) \sum_{i,j} \left\{ \frac{\mathbf{S}_i \cdot \mathbf{S}_j}{r_{ij}^3} - \frac{3(\mathbf{r}_{ij} \cdot \mathbf{S}_i)(\mathbf{r}_{ij} \cdot \mathbf{S}_j)}{r_{ij}^5} \right\}.$$

Here $\mathbf{r}_{ij}$ is the vector joining the sites i and j . By contrast with the short-range exchange interactions between nearest neighbors, the dipolar interactions falls off more slowly with distance apart of the atomic sites. This Hamiltonians can be written in second quantization if we employ the Holstein-Primakoff transformations

$$\begin{cases} S_i^+ = \sqrt{2S} \left(1 - \frac{a_i^\dagger a_i}{2S}\right)^{1/2} a_i \\ S_i^- = \sqrt{2S} a_i^\dagger \left(1 - \frac{a_i^\dagger a_i}{2S}\right)^{1/2} \\ S_i^z = S - a_i^\dagger a_i \end{cases}$$

at low temperature which means that the spins align along the direction of the applied magnetic field (taken as defining the z direction), and so $S_i^z \approx S$ for each spin, implying $a_i^\dagger a_i \ll S$ on average. Our job is to find that the Hamiltonian takes the form

$$\mathcal{H} = \sum_{\mathbf{k}} \left\{ P(\mathbf{k}) a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + Q(\mathbf{k}) a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + Q^*(\mathbf{k}) a_{\mathbf{k}} a_{-\mathbf{k}} \right\},$$

where we have dropped an unimportant constant term to focus on the dynamics. Here the coefficients $P(\mathbf{k})$ and $Q(\mathbf{k})$ are given by

$$\begin{aligned} P(\mathbf{k}) &= g\mu_B B_0 + S[J(0) - J(\mathbf{k})] - S \left[D^{zz}(0) + \frac{1}{2} D^{zz}(\mathbf{k}) \right], \\ Q(\mathbf{k}) &= \frac{1}{4} S [D^{xx}(\mathbf{k}) - D^{yy}(\mathbf{k}) + 2iD^{xy}(\mathbf{k})]. \end{aligned}$$

The Fourier-transformed exchange terms $J(\mathbf{k})$ were defined previously in Subsection 1.5.2, and the $D^{\alpha\beta}(\mathbf{k})$ are corresponding terms for the dipole-dipole interactions (with α and β denoting Cartesian components):

$$D^{\alpha\beta}(\mathbf{k}) = \frac{\mu_0 g^2 \mu_B^2}{4\pi} \sum_{\mathbf{r}} \frac{3r^\alpha r^\beta - |\mathbf{r}|^2 \delta_{\alpha,\beta}}{|\mathbf{r}|^5} \exp(-i\mathbf{k} \cdot \mathbf{r}).$$

Here the sum over $\mathbf{r}$ is over all lattice translation vectors (but excluding the $\mathbf{r} = 0$ term).

2.8 Hamiltonian of a Static Magnetic Field

The hamiltonian for a single spin with quantum number $S = 1/2$ placed in a static magnetic field B_0 along the z is given by

$$H = -\gamma B_0 S^z \equiv -\omega_L S^z$$

where ω_L is known as the Larmor frequency. Assuming an initial state

$$|\phi(t)\rangle = \alpha(t) |\uparrow\rangle + \beta(t) |\downarrow\rangle$$

2.8.1 (a)

The time evolution of the system is given by

$$\begin{aligned} |\phi(t)\rangle &= e^{-iHt} |\phi(0)\rangle \\ &= e^{-i\omega_L S^z t} (\alpha_0 |\uparrow\rangle + \beta_0 |\downarrow\rangle) \end{aligned}$$

The spin operator in term of the pauli matrices is given by

$$S^z = \frac{1}{2} \sigma^z = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

Because the matrix is diagonal the result is direct

$$\begin{aligned} |\phi(t)\rangle &= e^{-i\omega_L S^z t} (\alpha_0 |\uparrow\rangle + \beta_0 |\downarrow\rangle) = \begin{pmatrix} e^{-i\omega_L t/2} & 0 \\ 0 & e^{+i\omega_L t/2} \end{pmatrix} \left[\alpha_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta_0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] \\ &= \alpha_0 e^{-i\omega_L t/2} |\uparrow\rangle + \beta_0 e^{+i\omega_L t/2} |\downarrow\rangle \end{aligned}$$

Giving

$$\begin{cases} \alpha(t) &= \alpha_0 e^{-i\omega_L t/2} \\ \beta(t) &= \beta_0 e^{+i\omega_L t/2} \end{cases}$$

2.8.2 (b)

The time evolution of the longitudinal spin average $\langle S^z \rangle$ is given by

$$\begin{aligned} \langle S^z \rangle &= \langle \phi(t) | S^z | \phi(t) \rangle \\ &= (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} (\alpha(t) |\uparrow\rangle + \beta(t) |\downarrow\rangle) \\ &= \frac{1}{2} (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) (\alpha(t) |\uparrow\rangle - \beta(t) |\downarrow\rangle) \\ &= \frac{1}{2} (|\alpha(t)|^2 - |\beta(t)|^2) \\ &= \frac{1}{2} (\alpha_0^2 - \beta_0^2) \end{aligned}$$

Similarly, for the other coordinates we get

$$\begin{aligned} \langle S^y(t) \rangle &= \langle \phi(t) | S^y | \phi(t) \rangle \\ &= (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) i \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} (\alpha(t) |\uparrow\rangle + \beta(t) |\downarrow\rangle) \\ &= \frac{1}{2} i (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) (\alpha(t) |\downarrow\rangle - \beta(t) |\uparrow\rangle) \\ &= \frac{1}{2} i (-\alpha^*(t) \beta(t) + \beta^*(t) \alpha(t)) \\ &= -\frac{1}{2} i (\alpha_0^* \beta_0 e^{i\omega_L t} - \beta_0^* \alpha_0 e^{-i\omega_L t}) \end{aligned}$$

and

$$\begin{aligned}
\langle S^x(t) \rangle &= \langle \phi(t) | S^x | \phi(t) \rangle \\
&= (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (\alpha(t) | \uparrow \rangle + \beta(t) | \downarrow \rangle) \\
&= \frac{1}{2} (\alpha^*(t) \langle \uparrow | + \beta^*(t) \langle \downarrow |) (\alpha(t) | \downarrow \rangle + \beta(t) | \uparrow \rangle) \\
&= \frac{1}{2} (\alpha^*(t) \beta(t) + \beta^*(t) \alpha(t)) \\
&= \frac{1}{2} (\alpha_0^* \beta_0 e^{i\omega_L t} + \beta_0^* \alpha_0 e^{-i\omega_L t})
\end{aligned}$$

2.9 Graphene reciprocal lattice vectors

2.10 Graphene Eletronic Dispersion

2.11 Heisenberg Hamiltonian Spins as Classical Vectors

2.12 Free Particle Hamiltonian

Consider a free particle with mass m and charge e placed in a uniform magnetic field B_0 along the z direction. Its Hamiltonian is

$$H = \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2$$

where $\mathbf{p}$ is the particle momentum and $\mathbf{A}$ is the magnetic vector potential corresponding to the applied field in the EM gauge

$$\mathbf{A} = \frac{B_0}{2} (-y, x, 0)$$

Necessary to the equations of motion we calculate first the commutator

$$\begin{aligned}
[A_i, p_j] \psi &= (A_i p_j - p_j A_i) \psi \\
&= i A_i \partial_j \psi - i \partial_j (A_i \psi) \\
&= \cancel{i A_i \partial_j \psi} - i ((\partial_j A_i) \psi + \cancel{A_i (\partial_j \psi)}) \\
&= \begin{cases} +\frac{B_0}{2} & , j = x \text{ e } i = y \\ -\frac{B_0}{2} & , j = y \text{ e } i = x \\ 0 & , \text{outros} \end{cases}
\end{aligned}$$

while $[\mathbf{A}, \mathbf{r}] = 0$. The equation of motion for the position operator is then given by

$$\begin{aligned}\partial_t r_j &= i[H, x_j] \\ &= \frac{i}{2m} \left[\mathbf{p}^2 - \frac{e}{c} \mathbf{p} \cdot \mathbf{A} + \left(\frac{e}{c} \right)^2 \mathbf{A}^2, r_j \right] \\ &= \frac{i}{2m} (2p_i \delta_{i,j} + 0) \\ &= \frac{i}{m} p_j\end{aligned}$$

and now for the momentum operator

$$\begin{aligned}\partial_t p_j &= i[H, p_j] \\ &= \frac{i}{2m} \left[\mathbf{p}^2 - \frac{e}{c} \mathbf{p} \cdot \mathbf{A} + \left(\frac{e}{c} \right)^2 \mathbf{A}^2, p_j \right] \\ &= \frac{i}{2m} \left(\frac{e}{c} \right)^2 (0 + A_i [A_i, p_j] + [A_i, p_j] A_i) \\ \Rightarrow \partial_t p_j &= \begin{cases} \frac{i}{m} \left(\frac{e}{c} \right)^2 \left(\frac{B_0}{2} x \right) \left(+\frac{B_0}{2} \right) & j = x \text{ e } i = y \\ \frac{i}{m} \left(\frac{e}{c} \right)^2 \left(-\frac{B_0}{2} y \right) \left(-\frac{B_0}{2} \right) & j = y \text{ e } i = x \\ 0 & , \text{outros} \end{cases}\end{aligned}$$

Substituting in the equation of motion of the position

$$\begin{cases} \partial_t x = \int dt \left(\frac{i}{m} \right) \left(\frac{e}{c} \right)^2 \left(\frac{B_0}{2} \right)^2 x \\ \partial_t y = \int dt \left(\frac{i}{m} \right) \left(\frac{e}{c} \right)^2 \left(\frac{B_0}{2} \right)^2 y \end{cases} \Rightarrow \partial_t^2 r_i = -\frac{1}{4} \left(\frac{eB_0}{mc} \right)^2 r_i, i = 1, 2$$

Identificando a frequência ciclotrônica como $\omega_c = (eB_0)/(mc)$ obtemos

$$\begin{aligned}\frac{\partial^2}{\partial t^2} r_i &= -\left(\frac{\omega_c}{2} \right)^2 r_i \\ \Rightarrow r_i &= e^{\pm i \frac{\omega_c}{2} t}\end{aligned}$$

2.13 Plasmon Dispersion Relation

The plasmon dispersion relation derived in section 2.8.2 and as shown is Eq. (2.91) is

$$\sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 - \left(\frac{q^2}{2m} \right)^2 \right]^{-1} = \frac{m}{q^2 v(\mathbf{q})}$$

For the case of a Coulomb interaction in 3D we absence of screening we have $v(\mathbf{q}) \propto 1/q^2$ and so we obtain

$$\sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 - \left(\frac{q^2}{2m} \right)^2 \right]^{-1} = \frac{\epsilon_0 m}{e^2}$$

For small nonzero $\mathbf{q}$, we start to insert a power-series expansion for $\Omega(\mathbf{q})$ ignoring terms of fourth order and behind. We obtain

$$\begin{aligned} \sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 - \left(\frac{q^2}{2m} \right)^2 \right]^{-1} &= \sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 \right]^{-1} \\ &= \sum_{\mathbf{k} < \mathbf{k}_F} \left[\Omega^2(\mathbf{q}) \left(1 - \frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 \right]^{-1} \\ &= \frac{1}{\Omega^2(\mathbf{q})} \sum_{\mathbf{k} < \mathbf{k}_F} \frac{1}{\left(1 - \frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2} \end{aligned}$$

If we use the geometric series formula

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$$

Notice that the square² is not over the summation so we need to do

$$\begin{aligned} \frac{1}{(1-x)^2} &= \frac{1}{1-x} \frac{1}{1-x} \\ &= (1 + x + x^2 + x^3 + \dots) (1 + x + x^2 + x^3 + \dots) \\ &= (1 + x + x^2 + x^3 + \dots) + x(1 + x + x^2 + x^3 + \dots) + x^2(1 + x + x^2 + x^3 + \dots) + x^3(1 + x + x^2 + x^3 + \dots) \\ &= (1 + x + x^2 + x^3 + \dots) + x(1 + x + x^2) + x^2(1 + x) + x^3 \\ &= 1 + 2x + 3x^2 + 4x^3 \end{aligned}$$

where our x is

$$x = \frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m}$$

Notice however that because we are adding over all $\mathbf{k}$ from $-\mathbf{k}_F$ to $\mathbf{k}_F$, the odd terms on $\mathbf{k}$ end up canceling each other. We need only to work with the terms

$$\frac{1}{(1-x)^2} = 1 + 3 \left(\frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2$$

which means that

$$\sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 - \left(\frac{q^2}{2m} \right)^2 \right]^{-1} \approx \frac{1}{\Omega^2(\mathbf{q})} \sum_{\mathbf{k} < \mathbf{k}_F} \left(1 + 3 \left(\frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 \right)$$

Now, we need to promote the summation to an integral. We do this because the expression given depends on k_F .

$$\sum_{\mathbf{k} < \mathbf{k}_F} \rightarrow 2 \times \frac{V}{(2\pi)^3} \int d^3\mathbf{k}$$

where the factor of 2 is because of the spin degeneracy we obtain

$$\begin{aligned} & 2 \frac{1}{(2\pi)^3} \frac{V}{\Omega^2(\mathbf{q})} \int d^3\mathbf{k} \left(1 + 3 \left(\frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 \right) \\ &= 2 \frac{1}{(2\pi)^3} \frac{V}{\Omega^2(\mathbf{q})} \left(\int d^3\mathbf{k} + 3 \int d^3\mathbf{k} \left(\frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 \right) \end{aligned}$$

Because the first term does not depend on $\mathbf{k}$ the integral value is just the total number of states

$$n_0 = \frac{N}{V} = 2 \frac{1}{(2\pi)^3} \frac{4}{3} \pi k_F^3$$

So we just need to solve the integral

$$\begin{aligned} \int (\mathbf{k} \cdot \mathbf{q})^2 d^3\mathbf{k} &= \int_0^{2\pi} d\phi \int_0^\pi \sin(\theta) d\theta \int_0^{k_F} k^2 dk (k^2 q^2 \cos^2 \theta) \\ &= q^2 (2\pi) \int_1^{-1} (\cos^2 \theta) d(-\cos \theta) \int_0^{k_F} k^4 dk \\ &= q^2 (2\pi) \frac{1}{3} (+2) \frac{1}{5} k_F^5 \\ &= +q^2 \frac{4\pi}{5} \frac{1}{3} k_F^5 \end{aligned}$$

With the factors we let out we can rewrite it making N explicit giving

$$\begin{aligned} 2 \frac{V}{(2\pi)^3} \frac{1}{\Omega^2(\mathbf{q})} 3 \int d^3\mathbf{k} \left(\frac{1}{\Omega(\mathbf{q})} \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 &= 2 \frac{V}{(2\pi)^3} \frac{1}{\Omega^2(\mathbf{q})} 3 \left(q^2 \frac{4\pi}{5} \frac{1}{3} k_F^5 \right) \frac{1}{\Omega^2(\mathbf{q})} \frac{1}{m^2} \left\{ \times \frac{4/3\pi k_F^3}{4/3\pi k_F^3} \right\} \\ &= n_0 \frac{1}{\Omega^4(\mathbf{q})} \left(\frac{3 \times 4\pi}{5} \frac{1}{3} k_F^5 \frac{1}{m^2} \frac{1}{\pi} k_F^{-3} \right) q^2 \\ &= n_0 \frac{1}{\Omega^4(\mathbf{q})} \left(\frac{3k_F^2}{5m^2} \right) q^2 \end{aligned}$$

The final result is then

$$\sum_{\mathbf{k} < \mathbf{k}_F} \left[\left(\Omega(\mathbf{q}) - \frac{\mathbf{k} \cdot \mathbf{q}}{m} \right)^2 - \left(\frac{q^2}{2m} \right)^2 \right]^{-1} = \frac{n_0}{\Omega^2(\mathbf{q})} \left(1 + \frac{1}{\Omega^2(\mathbf{q})} \left(\frac{3k_F^2}{5m^2} \right) q^2 \right) \stackrel{!}{=} \frac{\epsilon_0 m}{e^2}$$

Multiplying both sides by $\Omega^4(\mathbf{q})$ we obtain

$$\begin{aligned} &\Leftrightarrow n_0 \Omega^2(\mathbf{q}) + \left(\frac{3n_0 k_F^2}{5m^2} \right) q^2 - \frac{\epsilon_0 m}{e^2} \Omega^4(\mathbf{q}) = 0 \left\{ \times - \frac{e^2}{\epsilon_0 m} \right\} \\ &\Leftrightarrow \Omega^4(\mathbf{q}) - \frac{n_0 e^2}{\epsilon_0 m} \Omega^2(\mathbf{q}) - \left(\frac{3\mu_\sigma k_F^2}{5m^2} \frac{e^2}{\epsilon_0 m} \times \frac{n_0}{\mu_\sigma} \right) q^2 = 0 \end{aligned}$$

We can substitute the definition of the plasma frequency

$$\Omega_0 \equiv \sqrt{\frac{n_0 e^2}{\epsilon_0 m}}$$

and obtain

$$\begin{aligned} (\Omega^2(\mathbf{q}))^2 - \Omega_0^2 \Omega^2(\mathbf{q}) - \left(\frac{3k_F^2}{5m^2} \Omega_0^2 \right) q^2 &= 0 \\ \Leftrightarrow \Omega_{\pm}^2(\mathbf{q}) &= \frac{\Omega_0^2 \pm \sqrt{\Omega_0^4 + 4 \left(\frac{3k_F^2}{5m^2} \Omega_0^2 \right) q^2}}{2} \\ &= \frac{\Omega_0^2}{2} \left(1 \pm \sqrt{1 + 4 \left(\frac{3k_F^2}{5m^2} \frac{1}{\Omega_0^2} \right) q^2} \right) \end{aligned}$$

We can now taylor expand the square root as $(1+x)^{1/2} = (1+0)^{1/2} + \frac{1}{2}(+1)(1-0)^{1/2} = 1 + \frac{1}{2}x$. We get

$$\Omega_{\pm}^2(\mathbf{q}) = \frac{\Omega_0^2}{2} \left(1 \pm \left(1 + 2 \left(\frac{3k_F^2}{5m^2} \frac{1}{\Omega_0^2} \right) q^2 \right) \right)$$

Because our solution has to be stable we can't have the solution with an inverted parabola because it wouldn't have an energy minimum. The desired solution is the one with the plus sign so we will only work with that one.

$$\begin{aligned} \Omega_+^2(\mathbf{q}) &= \Omega_0^2 \left(1 + \left(\frac{3k_F^2}{5m^2} \frac{1}{\Omega_0^2} \right) q^2 \right) \\ &= \Omega_0^2 \left(1 + \left(\frac{3k_F^2}{5m^2} \frac{\epsilon_0 \cancel{m}}{n_0 e^2} \right) q^2 \right) \\ &= \Omega_0^2 \left(1 + \left(\frac{3\epsilon_0 k_F^2}{5m n_0 e^2} \right) q^2 \right) \end{aligned}$$

There is no contribution from the terms proportional to q and q^3 as we were told. The extra $1/2$ factor on the q^2 term comes from including the spin contribution.

3 Formal Properties of Green's Functions

3.1 Thermal Average of the Energy of a Canonical Ensemble of particle

The hamiltonean of a spinless particle of mass m confined in cubic box is

$$H = -\frac{\hbar^2}{2m} \nabla^2$$

From the time independent Schrodinger equation

$$H\psi = E\psi$$

assuming a wave function like a plane wave

$$\psi = e^{-i\mathbf{k}\cdot\mathbf{r}}$$

The energy eigenvalues are

$$E = \frac{\hbar^2 k^2}{2m}$$

The grand partition function Q is defined as

$$Q = \sum_i e^{-\beta(E_i - \mu N_i)}$$

Assuming that the summation over states can be replaced by an integral

$$\sum_i \rightarrow \frac{V}{(2\pi)^3} \int d^3\mathbf{k}$$

We obtain

$$\begin{aligned} Q &= \sum_i \exp\left(-\beta\left(\frac{\hbar^2 k_i^2}{2m} - \mu N_i\right)\right) \\ &= \frac{V}{(2\pi)^3} \int_0^{2\pi} \int_0^\pi \int_0^\infty k^2 \sin(\theta) d\theta d\phi dk \exp\left(-\beta\left(\frac{\hbar^2 k^2}{2m} - \mu N\right)\right) \end{aligned}$$

The number of particles N inside of the box is constant so we can take it out of the integral. Next we do the change of variable $\beta \frac{\hbar^2 k^2}{2m} \rightarrow k^2$

$$\begin{aligned} Q &= \frac{4\pi V}{(2\pi)^3} e^{\beta\mu N} \left(\frac{2m}{\hbar^2\beta}\right)^{3/2} \int_0^\infty k^2 e^{-k^2} dk \\ &= \frac{4\pi V}{(2\pi)^3} e^{\beta\mu N} \left(\frac{2m}{\hbar^2\beta}\right)^{3/2} \frac{\pi^{1/2}}{4} \\ &= \frac{V}{(2\pi)^3} \left(\frac{2\pi m}{\hbar^2\beta}\right)^{3/2} e^{\beta\mu N} \end{aligned}$$

The thermal average of the energy of the particle is given by

$$\begin{aligned} \langle H \rangle &= \frac{1}{Q} \sum_i \langle i | H | i \rangle e^{-\beta(E_i - \mu N_i)} \\ &= \frac{(2\pi)^3}{V} \left(\frac{2\pi m}{\hbar^2\beta}\right)^{-3/2} \cancel{e^{-\beta\mu N}} \sum_k e^{-i\mathbf{k}\cdot\mathbf{r}} \left(\frac{-\hbar^2}{2m} \nabla^2\right) e^{-i\mathbf{k}\cdot\mathbf{r}} e^{-\beta(E_k - \mu N)} \\ &= \frac{(2\pi)^3}{V} \left(\frac{2\pi m}{\hbar^2\beta}\right)^{-3/2} \frac{-\hbar^2}{2m} \sum_k \cancel{e^{-i\mathbf{k}\cdot\mathbf{r}}} (-i\mathbf{k})^2 \cancel{e^{-i\mathbf{k}\cdot\mathbf{r}}} e^{-\beta E_k} \\ &= \frac{(2\pi)^3}{V} \left(\frac{2\pi m}{\hbar^2\beta}\right)^{-3/2} \frac{\hbar^2}{2m} \sum_k k^2 e^{-\beta E_k} \end{aligned}$$

If we promote the summation to an integral as we did before we obtain a similar integral but with k^4 instead of k^2 evaluating in $3/8\pi^{1/2}$. And so

$$\begin{aligned}
\langle H \rangle &= \frac{(2\pi)^3}{V} \left(\frac{2m\pi}{\hbar^2\beta} \right)^{-3/2} \frac{\hbar^2}{2m} \left(\frac{4\pi V}{(2\pi)^3} \left(\frac{2m}{\hbar^2\beta} \right)^{5/2} \int_0^\infty k^4 e^{-k^2} dk \right) \\
&= \frac{1}{\beta^{-1}} \pi^{-3/2} \frac{3}{4\pi} \pi^{1/2} \\
&= \frac{1}{V} \frac{3}{2} k_B T
\end{aligned}$$

in accordance with the quipartition theorem

3.2 Causal Green's Function in Frequency Domain

$$G_c(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} J(\omega') \left\{ \frac{e^{\beta\omega'}}{\omega - \omega' + i\eta} - \frac{\epsilon}{\omega - \omega' - i\eta} \right\} d\omega'$$

3.3 Complex Conjugate of a Green's Function (5)

By the definition

$$g_r(A; B | t - t') = -i\theta(t - t') \langle [A(t), B(t')]_\epsilon \rangle$$

where $\epsilon = \pm 1$. We obtain that

$$\begin{aligned}
g_r^*(A; B | t - t') &= i\theta(t - t') \langle [A(t), B(t')]_\epsilon \rangle^* \\
&= i\theta(t - t') \langle A(t)B(t') - \epsilon B(t')A(t) \rangle^* \\
&= i\theta(t - t') \langle B^\dagger(t')A^\dagger(t) - \epsilon A^\dagger(t)B^\dagger(t') \rangle \\
&= -\epsilon i\theta(t - t') \left\langle \frac{1}{-\epsilon} B^\dagger(t')A^\dagger(t) + A^\dagger(t)B^\dagger(t') \right\rangle \\
&= -\epsilon i\theta(t - t') \langle A^\dagger(t)B^\dagger(t') - \epsilon B^\dagger(t')A^\dagger(t) \rangle \\
&= \epsilon \times -i\theta(t - t') \langle [A^\dagger(t), B^\dagger(t')]_\epsilon \rangle \\
&= \epsilon g_r(A^\dagger; B^\dagger | t - t')
\end{aligned}$$

Given that

$$G_r(\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} g_r(t) e^{i\omega t} dt$$

we can also deduce the relation

$$\begin{aligned}
G_r^*(A; B | -\omega) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} g_r^*(A; B | t) e^{-i(-\omega)t} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \epsilon g_r(A^\dagger; B^\dagger | t) e^{i\omega t} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \epsilon (-i\theta(t) \langle A^\dagger(t) B^\dagger(t') - \epsilon B^\dagger(t') A^\dagger(t) \rangle) e^{i\omega t} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} i\theta(t) \langle B^\dagger(t') A^\dagger(t) - \epsilon A^\dagger(t) B^\dagger(t') \rangle e^{i\omega t} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} i\theta(t) \langle [B^\dagger(t'), A^\dagger(t),]_\epsilon \rangle e^{i\omega t} dt \\
&= G_a(B^\dagger; A^\dagger | \omega)
\end{aligned}$$

3.4 Inverse Kramers-Kronig relation

3.5 Spectral Intensity Calculation

Given the following analytically continued GF at the complex frequency

$$G(A; B | \tilde{\omega}) = \frac{2\tilde{\omega}}{(\omega_1 - \tilde{\omega})(\omega_2 - \tilde{\omega})}$$

the corresponding retarded GF at any real frequency label is

$$G_r(A; B | \tilde{\omega}) = \frac{2(\omega + i\eta)}{(\omega_1 - (\omega + i\eta))(\omega_2 - (\omega + i\eta))}$$

The spectral intensity $J(\omega)$ is given by

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \Im(G_r(\omega))$$

In order to apply the spectral theorem more easily its useful to use partial fractions in reexpressing the GF

$$\begin{aligned}
G(A; B | \tilde{\omega}) &= \frac{2\tilde{\omega}}{(\omega_1 - \tilde{\omega})(\omega_2 - \tilde{\omega})} = \frac{A}{(\omega_1 - \tilde{\omega})} + \frac{B}{(\omega_2 - \tilde{\omega})} \\
&= \frac{A(\omega_2 - \tilde{\omega}) + B(\omega_1 - \tilde{\omega})}{(\omega_1 - \tilde{\omega})(\omega_2 - \tilde{\omega})} = \frac{(A\omega_2 + B\omega_1) - (A+B)\tilde{\omega}}{(\omega_1 - \tilde{\omega})(\omega_2 - \tilde{\omega})} \\
\Rightarrow \begin{cases} A\omega_2 + B\omega_1 &= 0 \\ -(A+B) &= 2 \end{cases} &= \begin{cases} -(B+2)\omega_2 + B\omega_1 &= 0 \\ A &= -(B+2) \end{cases} \\
\Rightarrow \begin{cases} B &= \frac{-2\omega_2}{\omega_2 - \omega_1} \\ A &= \frac{2\omega_2}{\omega_2 - \omega_1} - 2 \end{cases} &= \frac{\cancel{2\omega_2} - \cancel{2\omega_2} + 2\omega_1}{\omega_2 - \omega_1} = \frac{2\omega_1}{\omega_2 - \omega_1}
\end{aligned}$$

This way we can rewrite our retarded GF as

$$G_r(A; B | \tilde{\omega}) = \frac{\left(\frac{2\omega_1}{\omega_2 - \omega_1}\right)}{\omega_1 - \omega - i\eta} + \frac{\frac{-2\omega_2}{\omega_2 - \omega_1}}{\omega_2 - \omega - i\eta}$$

Using the Sokhotski–Plemelj identity

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x \pm i\eta} = \mp i\pi\delta(x) + \mathcal{P}\left(\frac{1}{x}\right)$$

we can extract the imaginart part easily obtaining

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \left[\left(\frac{2\omega_1}{\omega_2 - \omega_1}\right) \pi\delta(\omega_1 - \omega) + \left(\frac{-2\omega_2}{\omega_2 - \omega_1}\right) \pi\delta(\omega_2 - \omega) \right]$$

3.6 GF Fourier Transform

Starting from

$$\mathcal{G}(i\omega_m) = \frac{1}{2} \int_{-\beta}^0 d\tau e^{i\omega_m \tau} g_M(\tau) + \frac{1}{2} \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau)$$

We now introduce a change of variable $\tilde{\tau} = \tau + \beta \Leftrightarrow \tau = \tilde{\tau} - \beta \Rightarrow d\tau = d\tilde{\tau}$ for the first integral

$$= \frac{1}{2} \int_0^{\beta} d\tilde{\tau} e^{i\omega_m(\tilde{\tau} - \beta)} g_M(\tilde{\tau} - \beta) + \frac{1}{2} \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau)$$

Given the periocity of the Matsubara GF $g_M(\tau) = \epsilon g_M(\tau + \beta)$ we have that $g_M(\tilde{\tau} - \beta) = \epsilon g_M(\tilde{\tau})$. Not only that we have that $e^{-i\omega_m \beta} = \epsilon$. Lets also remeber that $\epsilon = \pm 1$ such that $\epsilon^2 = 1$. This way we are left with

$$\begin{aligned} &= \frac{1}{2} \int_0^{\beta} d\tilde{\tau} e^{i\omega_m \tilde{\tau}} \epsilon^2 g_M(\tilde{\tau}) + \frac{1}{2} \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau) \\ &= \frac{1}{2} \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau) + \frac{1}{2} \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau) \\ &= \int_0^{\beta} d\tau e^{i\omega_m \tau} g_M(\tau) \end{aligned}$$

3.7 GF for more than 3 operators

The n -point Matsubara Green's function is defined as

$$g_M(\tau_1, \tau_2, \dots, \tau_n) = (-i)^n \langle T \{ A_1(\tau_1) A_2(\tau_2) \dots A_n(\tau_n) \} \rangle$$

Assume that all the times $0 < \tau_j \leq \beta$. Let us consider

$$g_M(\tau_1, \dots, \tau_i = \beta, \dots, \tau_n) = (-i)^n \langle T \{ A_1(\tau_1) \dots A_i(\beta) \dots A_n(\tau_n) \} \rangle$$

with all the times $0 < \tau_j < \beta$ (except for τ_i). We can change the order of the operators under the time ordering operator (at the cost of gaining a possible ϵ sign). We can write

$$T\{A_1(\tau_1) \dots A_i(\beta) \dots A_n(\tau_n)\} = \epsilon^{i-1} T\{A_i(\beta) A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\}$$

Assuming that $\beta > \tau_j$ for all $j \neq i$ we can take the operator $A_i(\beta)$ outside the time ordering operator (since it is already in the correct position):

$$T\{A_i(\beta) A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} = A_i(\beta) T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\}$$

We these considerations we can write

$$g_M(\tau_1, \dots, \tau_i = \beta, \dots, \tau_n) = (-i)^n \epsilon^{i-1} \langle A_i(\beta) T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} \rangle$$

Since we have that

$$A_i(\beta) = e^{\beta H} A_i(0) e^{-\beta H}$$

and

$$\langle \dots \rangle = \frac{1}{Z} \text{Tr} \{ e^{-\beta H} \dots \}$$

we can write

$$\begin{aligned} g_M(\tau_1, \dots, \tau_i = \beta, \dots, \tau_n) &= (-i)^n \epsilon^{i-1} \langle A_i(\beta) T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} \rangle \\ &= (-i)^n \epsilon^{i-1} \frac{1}{Z} \text{Tr} \left\{ \cancel{e^{-\beta H}} e^{\beta H} A_i(0) e^{-\beta H} T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} \right\} \\ &= (-i)^n \epsilon^{i-1} \frac{1}{Z} \text{Tr} \{ A_i(0) e^{-\beta H} T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} \} \\ &= (-i)^n \epsilon^{i-1} \frac{1}{Z} \text{Tr} \{ e^{-\beta H} T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} A_i(0) \} \\ &= (-i)^n \epsilon^{i-1} \left\langle T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} A_i(0) \right\rangle \end{aligned}$$

Since for all $j \neq i$ we have $\tau_j > 0$ we can write

$$T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n)\} A_i(0) = T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n) A_i(0)\}$$

Now we want to bring the operator $A_i(0)$ from the final position in the string of operators, to its original place at position i . We have to make $n - i$ moves. Therefore, we obtain

$$\begin{aligned} g_M(\tau_1, \dots, \tau_i = \beta, \dots, \tau_n) &= (-i)^n \epsilon^{i-1} \langle T\{A_1(\tau_1) \dots \cancel{A_i(\beta)} \dots A_n(\tau_n) A_i(0)\} \rangle \\ &= (-i)^n \epsilon^{i-1} \epsilon^{n-i} \langle T\{A_1(\tau_1) \dots A_i(0) \dots A_n(\tau_n)\} \rangle \\ &= (-i)^n \epsilon^{n-1} \langle T\{A_1(\tau_1) \dots A_i(0) \dots A_n(\tau_n)\} \rangle \end{aligned}$$

If n is an even number (in a Green's function we usually have an equal number of creation and annihilation operators), we have that $\epsilon^n = 1$. Also noticing that $\epsilon^{-1} = (\pm 1)^{-1} = (\pm 1)^1 = \epsilon$, we obtain

$$\begin{aligned} g_M(\tau_1, \dots, \tau_i = \beta, \dots, \tau_n) &= (-i)^n \epsilon \langle T\{A_1(\tau_1) \dots A_i(0) \dots A_n(\tau_n)\} \rangle \\ &= \epsilon g_M(\tau_1, \dots, \tau_i = 0, \dots, \tau_n) \end{aligned}$$

3.8 Lehmann Representation (6)

By definition the retarded GF is given by

$$\begin{aligned} g_r(t) &= -i\theta(t) \langle [A(t)B(0)]_\epsilon \rangle \\ &= -i\theta(t) \langle A(t)B(0) - \epsilon B(0)A(t) \rangle \end{aligned}$$

We will be using the definition of thermal average as

$$\langle A \rangle = Q^{-1} \sum_i \langle i|A|i \rangle e^{-\beta E_i}$$

noting that we can also write as $\sum_i A_{ii} = \sum_i \langle i|A|i \rangle$. We also remind that, in the Heisenberg picture

$$A(t) = e^{+iHt} A e^{-iHt}$$

Substituting this definitions we obtain

$$\begin{aligned} g_r(t) &= -i\theta(t) \langle e^{+iHt} A e^{-iHt} B - \epsilon B e^{+iHt} A e^{-iHt} \rangle \\ &= -i\theta(t) Q^{-1} \sum_j \{ e^{-\beta E_j} \langle j|e^{+iHt} A e^{-iHt} B|j \rangle - e^{-\beta E_j} \langle j|\epsilon B e^{+iHt} A e^{-iHt}|j \rangle \} \end{aligned}$$

Given that the energies E_j are the eigenvalues of the diagonalized Hamiltonian H we know for a fact that $H|j\rangle = E_j|j\rangle$ and so, given that the exponential of a diagonal matrix is also a diagonal matrix with its eigenvalues as exponential exponents we have that $e^{-iHt}|j\rangle = e^{-iE_j t}|j\rangle$. However, notice that we can't use this on the right side of the first term and on the left side of the second term because B does not necessarily commute with e^{+iHt} . To bypass this fact we employ the completeness property of the states

$$\sum_{j'} |j'\rangle \langle j'| = \mathbf{1}$$

and multiply it between $B(0)$ and the exponential term as follows

$$\begin{aligned} g_r(t) &= -i\theta(t) Q^{-1} \sum_j \{ e^{-\beta E_j} \langle j|e^{+iHt} A e^{-iHt} \mathbf{1} B|j \rangle - \epsilon e^{-\beta E_j} \langle j|B \mathbf{1} e^{+iHt} A e^{-iHt}|j \rangle \} \\ &= -i\theta(t) Q^{-1} \sum_{j,j'} \{ e^{-\beta E_j} \langle j|e^{+iHt} A e^{-iHt} |j'\rangle \langle j'| B|j \rangle - \epsilon e^{-\beta E_j} \langle j|B|j'\rangle \langle j'| e^{+iHt} A e^{-iHt}|j \rangle \} \\ &= -i\theta(t) Q^{-1} \sum_{j,j'} \{ e^{-\beta E_j} e^{iE_j t} \langle j|A|j'\rangle e^{-iE_{j'} t} \langle j'| B|j \rangle - \epsilon e^{-\beta E_j} \langle j|B|j'\rangle e^{+iE_{j'} t} \langle j'| A|j \rangle e^{-iE_j t} \} \\ &= -i\theta(t) Q^{-1} \sum_{j,j'} \left\{ e^{-\beta E_j} e^{i(E_j - E_{j'})t} A_{j,j'} B_{j',j} - \epsilon e^{-\beta E_j} e^{+i(E_{j'} - E_j)t} B_{j,j'} A_{j',j} \right\} \end{aligned}$$

We can always rename the mute variables on the second term to match the first one i.e $j \rightarrow j'$ and $j' \rightarrow j$. Notice also that A_{ii} are matrix elements, we have no

problem commuting this terms. We then have

$$\begin{aligned}
&= -i\theta(t)Q^{-1} \sum_{j,j'} \left\{ e^{-\beta E_j} e^{i(E_j - E_{j'})t} A_{j,j'} B_{j',j} - \epsilon e^{-\beta E_{j'}} e^{+i(E_j - E_{j'})t} A_{j,j'} B_{j',j} \right\} \\
&= -i\theta(t)Q^{-1} \sum_{j,j'} e^{i(E_j - E_{j'})t} \left\{ e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}} e^{+i(E_j - E_{j'})t} \right\} A_{j,j'} B_{j',j}
\end{aligned}$$

And so we arrive at the pretended Eq.(3.58). Now we just need to employ the Fourier transform

$$\begin{aligned}
G_r(\omega) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} g_r(t) e^{i\omega t} dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \left(i\theta(t)Q^{-1} \sum_{j,j'} e^{i(E_j - E_{j'})t} \left\{ e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}} e^{+i(E_j - E_{j'})t} \right\} A_{j,j'} B_{j',j} \right) e^{i\omega t} dt \\
&= \frac{-i}{2\pi} Q^{-1} \sum_{j,j'} \left\{ e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}} \right\} A_{j,j'} B_{j',j} \int_0^{+\infty} e^{i(\omega + (E_j - E_{j'}))t} dt
\end{aligned}$$

To make the integral converge we do analitically continue $\omega \rightarrow \omega + i\eta$ with $\eta \rightarrow 0^+$. We are left with

$$\begin{aligned}
&= \frac{-i}{2\pi Q} \sum_{j,j'} \left\{ e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}} \right\} A_{j,j'} B_{j',j} \frac{e^{i(\omega + (E_j - E_{j'}))t} e^{-\eta t}}{i((\omega + i\eta) + (E_j - E_{j'}))} \Big|_0^{+\infty} \\
&= \frac{-i}{2\pi Q} \sum_{j,j'} \left\{ e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}} \right\} A_{j,j'} B_{j',j} \frac{(0 \cancel{-} 1)}{\omega + E_j - E_{j'} + i\eta} \\
&= \frac{1}{2\pi Q} \sum_{j,j'} \frac{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}}{\omega + E_j - E_{j'} + i\eta} A_{j,j'} B_{j',j}
\end{aligned}$$

3.9 Sum Rule of the Spectral Function

The dissipation-fluctuation theorem tells us that

$$J(\omega) = \frac{-2}{e^{\beta\omega} - \epsilon} \Im G_r(\omega)$$

Employing the Lehmann representation of the GF as derived above we get

$$= \frac{-2}{e^{\beta\omega} - \epsilon} \Im \left(\frac{1}{2\pi Q} \sum_{j,j'} \frac{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}}{\omega + E_j - E_{j'} + i\eta} A_{j,j'} B_{j',j} \right)$$

Using Plemelj theorem to extract the imaginary part we have

$$\begin{aligned}
&= \frac{-2}{e^{\beta\omega} - \epsilon} \frac{1}{2\pi Q} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) A_{j,j'} B_{j',j} (\cancel{\pi} \delta(\omega + E_j - E_{j'})) \\
&= \frac{1}{e^{\beta\omega} - \epsilon} Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) A_{j,j'} B_{j',j} \delta(\omega - (E_{j'} - E_j))
\end{aligned}$$

Now we want to show that the sum rule is true for the case where $A = a$ and $B = a^\dagger$. Integrating it on all the frequency domain we obtain

$$\begin{aligned}
\int_{-\infty}^{+\infty} d\omega J(\omega) &= \int_{-\infty}^{+\infty} d\omega \left(\frac{1}{e^{\beta\omega} - \epsilon} Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) a_{j,j'} a_{j',j}^\dagger \delta(\omega - (E_{j'} - E_j)) \right) \\
&= Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) a_{j,j'} a_{j',j}^\dagger \frac{1}{e^{\beta(E_{j'} - E_j)} - \epsilon} \\
&= Q^{-1} \sum_{j,j'} a_{j,j'} a_{j',j}^\dagger \frac{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}}{e^{\beta E_{j'}} e^{-\beta E_j} - \epsilon} \\
&= Q^{-1} \sum_{j,j'} a_{j,j'} a_{j',j}^\dagger \frac{1}{e^{\beta E_{j'}}} \frac{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}}{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}} \\
&= Q^{-1} \sum_{j,j'} a_{j,j'} a_{j',j}^\dagger e^{-\beta E_{j'}} \\
&= Q^{-1} \sum_{j,j'} \langle j | a | j' \rangle \langle j' | a^\dagger | j \rangle e^{-\beta E_{j'}} \\
&= Q^{-1} \sum_{j,j'} \langle j' | a^\dagger | j \rangle \langle j | a | j' \rangle e^{-\beta E_{j'}} \\
&= Q^{-1} \sum_{j'} \langle j' | a^\dagger \left(\sum_j |j\rangle \langle j| \right) a | j' \rangle e^{-\beta E_{j'}} \\
&= Q^{-1} \sum_{j'} \langle j' | a^\dagger a | j' \rangle e^{-\beta E_{j'}} \\
&= Q^{-1} \sum_{j'} \langle j' | n | j' \rangle e^{-\beta E_{j'}} \\
&= \langle n \rangle
\end{aligned}$$

3.9.1 With a different spectral function

$$A(\omega) = -\frac{1}{\pi} \Im G_r(\omega)$$

Employing the Lehmann representation of the GF as derived above we get

$$= -\frac{1}{\pi} \Im \left(\frac{1}{Q} \sum_{j,j'} \frac{e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}}{\omega + E_j - E_{j'} + i\eta} A_{j,j'} B_{j',j} \right)$$

Using Plemelj theorem to extract the imaginary part we have

$$\begin{aligned}
&= \cancel{\frac{1}{\pi}} \frac{1}{Q} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) A_{j,j'} B_{j',j} (\cancel{\pi} \delta(\omega + E_j - E_{j'})) \\
&= Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) A_{j,j'} B_{j',j} \delta(\omega - (E_{j'} - E_j))
\end{aligned}$$

Now we want to show that the sum rule is true for the case where $A = a$ and $B = a^\dagger$. Integrating it on all the frequency domain we obtain

$$\begin{aligned}
\int_{-\infty}^{+\infty} d\omega A(\omega) &= \int_{-\infty}^{+\infty} d\omega \left(Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) a_{j,j'} a_{j',j}^\dagger \delta(\omega - (E_{j'} - E_j)) \right) \\
&= Q^{-1} \sum_{j,j'} (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) a_{j,j'} a_{j',j}^\dagger \\
&= Q^{-1} \sum_{j,j'} a_{j,j'} a_{j',j}^\dagger (e^{-\beta E_j} - \epsilon e^{-\beta E_{j'}}) \\
&= Q^{-1} \left(\sum_j \langle j| a \left(\sum_{j'} |j'\rangle \langle j'| \right) a^\dagger |j\rangle e^{-\beta E_j} - \epsilon \sum_{j'} \langle j'| a^\dagger \left(\sum_j |j\rangle \langle j| \right) a |j'\rangle e^{-\beta E_{j'}} \right) \\
&= Q^{-1} \sum_j (\langle j| a a^\dagger |j\rangle e^{-\beta E_j} - \epsilon \langle j| a^\dagger a |j\rangle e^{-\beta E_j}) \\
&= Q^{-1} \sum_j \langle j| a a^\dagger - \epsilon a^\dagger a |j\rangle e^{-\beta E_j} \\
&= Q^{-1} \sum_j \langle j| 1 |j\rangle e^{-\beta E_j} \\
&= Q^{-1} Q \\
&= 1
\end{aligned}$$

3.10 Causal GF equation of motion

The causal GF is defined as

$$\begin{aligned}
g_c(A; B|t - t') &= -i \langle \hat{T}_W A(t) B(t') \rangle \\
&= -i \langle \theta(t - t') A(t) B(t') + \theta(t' - t) \epsilon B(t') A(t) \rangle
\end{aligned}$$

To study its equation of motion we, of course, start by differentiating it

$$\begin{aligned}
\frac{d}{dt} g_c(A; B|t - t') &= -i \left\langle \frac{d}{dt} (\theta(t - t') A(t) B(t') + \theta(t' - t) \epsilon B(t') A(t)) \right\rangle \\
&= -i \left\langle \left(\frac{d\theta(t - t')}{dt} A(t) + \theta(t - t') \frac{dA(t)}{dt} \right) B(t') + \epsilon B(t') \left(\frac{d\theta(t' - t)}{dt} A(t) + \theta(t' - t) \frac{dA(t)}{dt} \right) \right\rangle \\
&= -i \left\langle \left(\delta(t - t') A(t) + \theta(t - t') \frac{dA(t)}{dt} \right) B(t') + \epsilon B(t') \left((-\delta(t' - t)) A(t) + \theta(t' - t) \frac{dA(t)}{dt} \right) \right\rangle \\
&= -i \delta(t - t') \langle A(t) B(t) - \epsilon B(t) A(t) \rangle - i \left(\theta(t - t') \frac{dA(t)}{dt} B(t') + \theta(t' - t) \epsilon B(t') \frac{dA(t)}{dt} \right) \\
&= -i \delta(t - t') \langle [A(t), B(t)]_\epsilon \rangle + g_c(dA(t)/dt; B|t - t') \\
&= -i \delta(t - t') \langle [A(t), B(t)]_\epsilon \rangle + i g_c([H, A]; B|t - t')
\end{aligned}$$

For the first term we have that

$$\begin{aligned}
\langle [A(t), B(t)]_\epsilon \rangle &= \langle A(t)B(t) - \epsilon B(t)A(t) \rangle \\
&= \langle e^{+iHt} \cancel{Ae^{-iHt}} e^{+iHt} B e^{-iHt} - \epsilon e^{+iHt} \cancel{B e^{-iHt}} e^{+iHt} A e^{-iHt} \rangle \\
&= \langle e^{+iHt} (AB - \epsilon BA) e^{-iHt} \rangle \\
&= \langle e^{+iHt} [A, B]_\epsilon e^{-iHt} \rangle \\
&= Q^{-1} \text{Tr} (e^{+iHt} [A, B]_\epsilon e^{-iHt} e^{-\beta H}) \\
&= Q^{-1} \text{Tr} (\cancel{e^{-iHt}} e^{+iHt} [A, B]_\epsilon e^{-\beta H}) \\
&= \langle [A, B]_\epsilon \rangle
\end{aligned}$$

Finally

$$\frac{d}{dt} g_c(A; B|t) = -i\delta(t) \langle [A, B]_\epsilon \rangle + i g_c([H, A]; B|t)$$

In frequency space we have

$$\begin{aligned}
&\Leftrightarrow \frac{d}{dt} \int d\omega G(A; B|\omega) e^{-i\omega t} = -i \int d\omega e^{-i\omega t} \langle [A, B]_\epsilon \rangle + i \int d\omega G([H, A]; B|\omega) e^{-i\omega t} \\
&\Leftrightarrow (-i\omega) \int d\omega G(A; B|\omega) e^{-i\omega t} = -i \int d\omega e^{-i\omega t} \langle [A, B]_\epsilon \rangle + i \int d\omega G([H, A]; B|\omega) e^{-i\omega t} \\
&\Rightarrow \omega G(A; B|\omega) = \langle [A, B]_\epsilon \rangle - G([H, A]; B|\omega) e^{-i\omega t}
\end{aligned}$$

3.11 Helmholtz Equation (F3)

The inhomogenous Helmholtz equation is

$$\nabla^2 g(\mathbf{r}, \mathbf{r}') + \kappa^2 g(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}')$$

If we Fourier transform the GF

$$g(\mathbf{r}, \mathbf{r}') = \frac{1}{(2\pi)^3} \int G(\mathbf{k}) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}$$

and the Dirac delta function

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{1}{(2\pi)^3} \int e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k}$$

Substituting we get

$$\begin{aligned}
&(-i\mathbf{k})^2 G(\mathbf{k}) + \kappa^2 G(\mathbf{k}) = 1 \\
&\Leftrightarrow -k^2 G(\mathbf{k}) + \kappa^2 G(\mathbf{k}) = 1 \\
&\Leftrightarrow G(\mathbf{k}) = -\frac{1}{k^2 - \kappa^2} = -\frac{1}{(k - \kappa)(k + \kappa)}
\end{aligned}$$

Which means that

$$\begin{aligned}
g(\mathbf{r} - \mathbf{r}') &= \frac{1}{(2\pi)^3} \int \left(-\frac{1}{k^2 - \kappa^2} \right) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k} \\
&= -\frac{1}{(2\pi)^3} \int_0^{2\pi} d\phi \int_0^{+\infty} k^2 dk \frac{1}{k^2 - \kappa^2} \int_0^\pi \sin(\theta) d\theta e^{-ik|r-r'|\cos(\theta)} \\
&= -\frac{1}{(2\pi)^3} 2\pi \int_0^{+\infty} k^2 dk \frac{1}{k^2 - \kappa^2} \int_{+1}^{-1} d(\cos(\theta)) e^{-ik|r-r'|\cos(\theta)} \\
&= \frac{1}{4\pi^2} \int_0^{+\infty} k^2 dk \frac{1}{k^2 - \kappa^2} \left(\frac{e^{-ik|r-r'|\cos(\theta)}}{-ik|r-r'|} \Big|_{-1}^{+1} \right) \\
&= \frac{+i}{4\pi^2 |r-r'|} \int_0^{+\infty} dk \frac{k}{(k-\kappa)(k+\kappa)} \left(e^{-ik|r-r'|} - e^{ik|r-r'|} \right) \\
&= \frac{+i}{4\pi^2 |r-r'|} \int_{-\infty}^{+\infty} dk \frac{k}{(k-\kappa)(k+\kappa)} (-2i) \sin(k|r-r'|)
\end{aligned}$$

The usual way to solve this type of integral is to extend the variable to the complex plane and chose a semi-circular loop γ around a half plane such that the unwanted contribution of the circular loop γ_{circ} goes to zero for the radius $R \rightarrow +\infty$ leaving only the integral on the real axis from $[-\infty, +\infty]$. Notice, however, that our integral only goes from $[0, +\infty]$. Fortunately, the function we are integrating is even (because both the sine and $k/(k^2 - \kappa^2)$ are odd functions) meaning we can always extend it to $-\infty$ if we add a $1/2$ to remove the excess we just added. We have

$$= \frac{1}{4\pi^2} \frac{1}{|r-r'|} \int_{-\infty}^{+\infty} dk \frac{k}{(k-\kappa)(k+\kappa)} \sin(k|r-r'|)$$

Because we want the solution in terms of an outgoing plane wave it help using the Euler formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$. The integral contribution of the cosine term is zero because it's an odd function evaluate in a simetric domain. We get

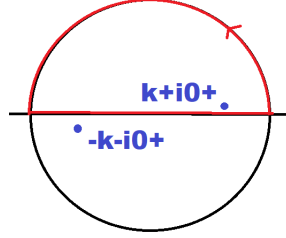
$$= \frac{1}{4\pi^2} \frac{1}{|r-r'|} \int_{-\infty}^{+\infty} dk \frac{k}{(k-\kappa)(k+\kappa)} (-i) e^{ik|r-r'|}$$

Now we can proceed with the complex analysis. The circular loop integral part dies when evaluated at $R \rightarrow +\infty$ because the complex variable z has always a positive imaginary part. We obtain

$$\oint_{\gamma} dz \frac{ze^{iz|r-r'|}}{(z-\kappa)(z+\kappa)} = \int_{-\infty}^{+\infty} dk \frac{ke^{ik|r-r'|}}{(k-\kappa)(k+\kappa)} + \int_{\gamma_{circ}} dz \frac{ze^{iz|r-r'|}}{(z-\kappa)(z+\kappa)} \xrightarrow{R \rightarrow \infty} 0$$

Lets us now analitically continue the variable $\kappa \rightarrow \kappa + i\eta$ meaning that

$$\int_{-\infty}^{+\infty} dk \frac{ke^{ik|r-r'|}}{(k-\kappa)(k+\kappa)} = \oint_{\gamma} dz \frac{ze^{iz|r-r'|}}{(z-(\kappa+i\eta))(z-(-\kappa-i\eta))}$$



Then, by the residue theorem, we have the contribution of only one of the two poles $z = \pm\kappa + i\eta$ as seen in figure

The integral is then

$$= 2\pi i \left(\frac{\kappa e^{i\kappa|r-r'|}}{(\kappa + \kappa)} \right) = \pi i e^{+i\kappa|r-r'|}$$

Substituting next to all the factors we obtain

$$\begin{aligned} g(\mathbf{r} - \mathbf{r}') &= \frac{i}{4\pi^2} \frac{1}{|r - r'|} \pi i e^{+i\kappa|r-r'|} \\ &= -\frac{e^{+i\kappa|r-r'|}}{4\pi|r - r'|} \end{aligned}$$

4 Exact Methods for Green's Functions

4.1 Noninteracting Systems

4.1.1 Retarded and Advanced GF for bosons

The hamiltonian for a noninteracting gas of bosons (or fermions) is given by

$$\mathcal{H} = \sum_{\mathbf{k}} E_{\mathbf{k}} a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}}$$

Hence the equation of motion for the system's GF is

$$\begin{aligned} \omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^{\dagger} | \omega) &= \frac{1}{2\pi} \langle [a_{\mathbf{k}}, a_{\mathbf{k}}^{\dagger}] \rangle - G([\mathcal{H}, a_{\mathbf{k}}]; a_{\mathbf{k}}^{\dagger} | \omega) \\ &= \frac{1}{2\pi} + E_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^{\dagger} | \omega) \end{aligned}$$

The general solution yields

$$\Leftrightarrow G(a_{\mathbf{k}}; a_{\mathbf{k}}^{\dagger} | \omega) = \left(\frac{1}{2\pi} \right) \frac{1}{\omega - E_{\mathbf{k}}} + f\delta(\omega - E_{\mathbf{k}})$$

The next step is to analytically continue the real-time GF. Because it has a discontinuity on the real axis and it is analytic elsewhere in the complex plane $\tilde{\omega}$ we can write it as

$$G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \tilde{\omega}) = \left(\frac{1}{2\pi} \right) \frac{1}{\tilde{\omega} - E_{\mathbf{k}}}$$

For the retarded GF we use that $G_r(\omega) = G(\omega + i\eta)$. Therefore, we have

$$G_r(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \left(\frac{1}{2\pi} \right) \frac{1}{\omega - E_{\mathbf{k}} + i\eta}$$

Using the Sokhotski–Plemelj identity

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x \pm i\eta} = \mathcal{P} \left(\frac{1}{x} \right) \mp i\pi\delta(x)$$

we obtain

$$\begin{aligned} G_r(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) &= \left(\frac{1}{2\pi} \right) \mathcal{P} \frac{1}{\omega - E_{\mathbf{k}} + i\eta} - \left(\frac{1}{2\pi} \right) i\pi\delta(\omega - E_{\mathbf{k}}) \\ &= \left(\frac{1}{2\pi} \right) \mathcal{P} \frac{1}{\omega - E_{\mathbf{k}} + i\eta} - \left(\frac{i}{2} \right) \delta(\omega - E_{\mathbf{k}}) \end{aligned}$$

Comparing with the above expression we see that $f = -i/2$. For the advanced GF the treatment is analogous, we instead use that $G_a(\omega) = G(\omega - i\eta)$ and obtain by comparison that $f = +i/2$.

4.1.2 Causal GF for Bosons

To get f for the causal GF we need to go back to the definition.

$$g_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | t - t') = -i \left\langle \hat{T}_W a_{\mathbf{k}}(t) a_{\mathbf{k}}^\dagger(t') \right\rangle$$

The GF equation of motion in the time space

$$[i\partial_t - E_{\mathbf{k}}] g_c(t - t') = \delta(t - t')$$

gives the solution for $t \neq t'$

$$g_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | t - t') = \begin{cases} A e^{-iE_{\mathbf{k}}(t-t')} & , t > t' \\ B e^{+iE_{\mathbf{k}}(t-t')} & , t < t' \end{cases}$$

To find the values of the constants A and B and use that the thermal average for bosons is (as shown in 4.1.3),

$$\left\langle a_{\mathbf{k}}^\dagger(t') a_{\mathbf{k}}(t') \right\rangle = n_B(\epsilon_{\mathbf{k}})$$

The causal GF for $t \rightarrow t' + 0^+$ and $t \rightarrow t' - 0^+$ is

$$\begin{aligned} g_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | 0^+) &= -i \left\langle \hat{T}_W a_{\mathbf{k}}(t' + 0^+) a_{\mathbf{k}}^\dagger(t') \right\rangle = -i \left\langle a_{\mathbf{k}}(t') a_{\mathbf{k}}^\dagger(t') \right\rangle = -i(1 + n_B(\epsilon_{\mathbf{k}})) \\ &\Rightarrow A = -i(1 + n_B(\epsilon_{\mathbf{k}})) \end{aligned}$$

$$g_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | -0^+) = -i \left\langle \hat{T}_W a_{\mathbf{k}}(t' - 0^+) a_{\mathbf{k}}^\dagger(t') \right\rangle = -i \left\langle a_{\mathbf{k}}^\dagger(t') a_{\mathbf{k}}(t') \right\rangle = -in_B(\epsilon_{\mathbf{k}})$$

$$\Rightarrow B = -in_B(\epsilon_{\mathbf{k}})$$

Now we can represent our causal GF in the time space as

$$g_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | t - t') = -i\theta(t - t') (1 + n_B(\epsilon_{\mathbf{k}})) e^{-iE_{\mathbf{k}}(t - t')} - i\theta(t' - t) n_B(\epsilon_{\mathbf{k}}) e^{+iE_{\mathbf{k}}(t - t')}$$

and, by doing the Fourier transform, in the frequency space as

$$G_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} \frac{1}{\omega + i0^+ + E_{\mathbf{k}}} (1 + n_B(\epsilon_{\mathbf{k}})) - \frac{1}{2\pi} \frac{1}{\omega - i0^+ + E_{\mathbf{k}}} n_B(\epsilon_{\mathbf{k}})$$

Again, by the Sokhotski–Plemelj identity we can obtain the term with the delta and compare it to the original real-time GF wto find the f unknown

$$G_c(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} \mathcal{P} \left(\frac{1}{\omega + E_{\mathbf{k}}} \right) + \frac{1}{2\pi} (- (1 + n_B(\epsilon_{\mathbf{k}})) - n_B(\epsilon_{\mathbf{k}})) i\pi \delta(\omega - E_{\mathbf{k}})$$

$$= \frac{1}{2\pi} \mathcal{P} \left(\frac{1}{\omega + E_{\mathbf{k}}} \right) + \frac{-i}{2} (1 + 2n_B(\epsilon_{\mathbf{k}})) \delta(\omega - E_{\mathbf{k}})$$

Where

$$1 + 2n_B(\epsilon_{\mathbf{k}}) = 1 + 2 \frac{1}{e^{+\beta E_{\mathbf{k}}} - 1} = \frac{e^{+\beta E_{\mathbf{k}}} + 1}{e^{+\beta E_{\mathbf{k}}} - 1} = \frac{e^{+\beta E_{\mathbf{k}}/2} (e^{+\beta E_{\mathbf{k}}/2} + e^{-\beta E_{\mathbf{k}}/2})}{e^{+\beta E_{\mathbf{k}}/2} (e^{+\beta E_{\mathbf{k}}/2} - e^{-\beta E_{\mathbf{k}}/2})}$$

$$= \frac{\cosh(\beta E_{\mathbf{k}}/2)}{\sinh(\beta E_{\mathbf{k}}/2)} = \coth(\beta E_{\mathbf{k}}/2)$$

And so

$$f_c = -\frac{i}{2} \coth(\frac{1}{2}\beta E_{\mathbf{k}})$$

4.1.3 Thermal average for boson ladder operators

By the definition of thermal average

$$\begin{aligned} \left\langle a_{\mathbf{k}}^\dagger(t) a_{\mathbf{k}}(t) \right\rangle &= \frac{1}{\text{Tr}(e^{-\beta \mathcal{H}})} \text{Tr} \left(a_{\mathbf{k}}^\dagger(t) a_{\mathbf{k}}(t) e^{-\beta H} \right) \\ &= \frac{1}{Z_0} \text{Tr} \left(a_{\mathbf{k}}^\dagger a_{\mathbf{k}} e^{-\beta \sum_{\mathbf{k}'} E_{\mathbf{k}'} a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}} \right) \\ &= \frac{1}{Z_0} \frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \text{Tr} \left(e^{-\beta \sum_{\mathbf{k}'} E_{\mathbf{k}'} a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}} \right) \\ &= \frac{1}{Z_0} \frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \text{Tr} \left(\prod_{\mathbf{k}'} e^{-\beta E_{\mathbf{k}'} a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}} \right) \\ &= \frac{1}{Z_0} \frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \prod_{\mathbf{k}'} \text{Tr} (e^{-\beta E_{\mathbf{k}'} n_{\mathbf{k}'}}) \end{aligned}$$

Because, for boson the occupation number can be whatever (while for fermions can only be 0 or 1) we have

$$= \frac{1}{Z_0} \frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \prod_{\mathbf{k}'} (1 + e^{-\beta E_{\mathbf{k}'}} + e^{-2\beta E_{\mathbf{k}'}} + \dots)$$

This sum is nothing more than the geometric series $1/(1-z) = 1 + z + z^2 + \dots$ and so

$$\begin{aligned} &= \frac{1}{Z_0} \frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \prod_{\mathbf{k}'} \frac{1}{1 - e^{-\beta E_{\mathbf{k}'}}} \\ &= \frac{1}{Z_0} \left(\prod_{\mathbf{k}' \neq \mathbf{k}} \frac{1}{1 - e^{-\beta E_{\mathbf{k}'}}} \right) \left(\frac{\partial}{\partial(-\beta E_{\mathbf{k}})} \frac{1}{1 - e^{-\beta E_{\mathbf{k}}}} \right) \\ &= \frac{1}{Z_0} \left(\prod_{\mathbf{k}' \neq \mathbf{k}} \frac{1}{1 - e^{-\beta E_{\mathbf{k}'}}} \right) (-1) \frac{e^{-\beta E_{\mathbf{k}}}}{(1 - e^{-\beta E_{\mathbf{k}}})^2} \\ &= \frac{-1}{Z_0} \left(\prod_{\mathbf{k}'} \frac{1}{1 - e^{-\beta E_{\mathbf{k}'}}} \right) \frac{e^{-\beta E_{\mathbf{k}}}}{1 - e^{-\beta E_{\mathbf{k}}}} \\ &= \frac{-1}{Z_0} \left(\prod_{\mathbf{k}'} \frac{1}{1 - e^{-\beta E_{\mathbf{k}'}}} \right) \frac{1}{e^{+\beta E_{\mathbf{k}}} - 1} \end{aligned}$$

Because the total partition function of a system is the product of all the individual systems (in this case denoted by $\mathbf{k}$) we have that

$$Z_0 = \prod_{\mathbf{k}} \sum_i e^{-\beta E_{\mathbf{k}} n_i} = \prod_{\mathbf{k}} \frac{1}{1 - e^{-\beta E_{\mathbf{k}}}}$$

such that the Z_0 in denominators cancels out

$$\langle a_{\mathbf{k}}^\dagger(t) a_{\mathbf{k}}(t) \rangle = \frac{1}{e^{+\beta E_{\mathbf{k}}} - 1} = n_B(E_{\mathbf{k}})$$

4.2 Matsubara GF for Noninteracting Systems

The Matsubara GF can be expressed more simply as

$$g_M(\tau) = -e^{-E_k \tau} \{ \theta(\tau) [1 + n_k] + \theta(-\tau) n_k \}$$

To express in the frequency space we do a inverse Fourier transform

$$\begin{aligned}
G(i\omega_m) &= \int_0^\beta d\tau e^{i\omega_m \tau} g_M(\tau) \\
&= \int_0^\beta d\tau e^{i\omega_m \tau} (-e^{-E_k \tau} \{\theta(\tau)[1 + n_k] + \theta(-\tau)n_k\}) \\
&= -[1 + n_k] \int_0^\beta d\tau e^{(i\omega_m - E_k)\tau} \\
&= -[1 + n_k] \frac{e^{(i\omega_m - E_k)\tau}}{(i\omega_m - E_k)} \Big|_0^\beta \\
&= -[1 + n_k] \frac{e^{(i\omega_m - E_k)\beta} - 1}{(i\omega_m - E_k)}
\end{aligned}$$

The bosonic frequencies is given by $\omega_n = \frac{1}{\beta} 2\pi n$ and so $e^{i\omega_m \beta} = 1$. This way

$$\begin{aligned}
G(i\omega_m) &= - \left[1 + \frac{1}{e^{\beta E_k} - 1} \right] \frac{e^{-\beta E_k} - 1}{(i\omega_m - E_k)} \\
&= - \left[\frac{e^{\beta E_k} - 1}{e^{\beta E_k} - 1} \right] \frac{e^{-\beta E_k} - 1}{(i\omega_m - E_k)} \\
&= - \left[\frac{1}{1 - e^{-\beta E_k}} \right] \frac{e^{-\beta E_k} - 1}{(i\omega_m - E_k)} \\
&= + \left[\frac{1}{e^{-\beta E_k} - 1} \right] \frac{e^{-\beta E_k} - 1}{(i\omega_m - E_k)} \\
&= \frac{1}{(i\omega_m - E_k)}
\end{aligned}$$

4.3 GF equation of motion for Noninteracting System

4.3.1 First GF's equation of motion

Consider N weakly interacting boson particles with most of the particles condensed into the $\mathbf{k} = 0$ state

$$\mathcal{H}_R = \sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) (a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}})$$

where the prime on the summation means that we are excluding the terms with $\mathbf{k} = 0$ are excluded. The equation of motion is given by

$$\omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} \langle [a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] \rangle - G([\mathcal{H}_R, a_{\mathbf{k}}]; a_{\mathbf{k}}^\dagger | \omega)$$

To solve this we first need to calculate the commutators. The first one is trivial giving $[a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] = 1$, for the second one we have

$$\begin{aligned} [\mathcal{H}_{\mathcal{R}}, a_{\mathbf{k}'}] &= \left[\sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) (a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}), a_{\mathbf{k}'} \right] \\ &= \sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] [a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{\mathbf{k}'}] + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) ([a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}] + [\cancel{a_{\mathbf{k}} a_{-\mathbf{k}}}, a_{\mathbf{k}'}]) \end{aligned}$$

The last one gives directly zero. For the other ones we use the relation $[AB, C] = A[B, C] + [A, C]B$ as follows

$$\begin{aligned} [a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{\mathbf{k}'}] &= a_{\mathbf{k}}^\dagger [\cancel{a_{\mathbf{k}}, a_{\mathbf{k}'}}] + [a_{\mathbf{k}}^\dagger, a_{\mathbf{k}'}] a_{\mathbf{k}} \\ &= -\delta_{\mathbf{k}, \mathbf{k}'} a_{\mathbf{k}} \end{aligned}$$

and analogously

$$\begin{aligned} [a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}] &= a_{\mathbf{k}}^\dagger [a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}'}] + [a_{\mathbf{k}}^\dagger, a_{\mathbf{k}'}] a_{-\mathbf{k}}^\dagger \\ &= a_{\mathbf{k}}^\dagger (-\delta_{-\mathbf{k}, \mathbf{k}'} - \delta_{\mathbf{k}, \mathbf{k}'} a_{-\mathbf{k}}^\dagger) \end{aligned}$$

Substituting in the original commutator we obtain

$$\begin{aligned} [\mathcal{H}_{\mathcal{R}}, a_{\mathbf{k}'}] &= \sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] (-\delta_{\mathbf{k}, \mathbf{k}'} a_{\mathbf{k}}) + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) (a_{\mathbf{k}}^\dagger (-\delta_{-\mathbf{k}, \mathbf{k}'} - \delta_{\mathbf{k}, \mathbf{k}'} a_{-\mathbf{k}}^\dagger)) \\ &= -[\epsilon_{\mathbf{k}'} + Nv(\mathbf{k}')] a_{\mathbf{k}'} - \frac{1}{2} N (v(-\mathbf{k}') a_{-\mathbf{k}'}^\dagger + v(\mathbf{k}') a_{-\mathbf{k}'}^\dagger) \\ &= -[\epsilon_{\mathbf{k}'} + Nv(\mathbf{k}')] a_{\mathbf{k}'} - Nv(\mathbf{k}') a_{-\mathbf{k}'}^\dagger \end{aligned}$$

where we use the simmetric property of the potential $v(-\mathbf{k}') = v(\mathbf{k}')$. Going back to the equation of motion we get

$$\omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + Nv(\mathbf{k}) G(a_{-\mathbf{k}}^\dagger; a_{\mathbf{k}}^\dagger | \omega)$$

4.3.2 Second GF's equation of motion

It appears a new GF which now we have to calculate to obtain a closed set of equation. Its equation of motions is

$$\begin{aligned} \omega G(a_{-\mathbf{k}}^\dagger; a_{\mathbf{k}}^\dagger | \omega) &= \frac{1}{2\pi} \langle \cancel{[a_{-\mathbf{k}}^\dagger, a_{\mathbf{k}}^\dagger]_\epsilon} \rangle - G([\mathcal{H}_{\mathcal{R}}, a_{-\mathbf{k}}^\dagger]; a_{\mathbf{k}}^\dagger | \omega) \\ &= -G([\mathcal{H}_{\mathcal{R}}, a_{-\mathbf{k}}^\dagger]; a_{\mathbf{k}}^\dagger | \omega) \end{aligned}$$

Now we need again to calculate the new commutator

$$\begin{aligned} [\mathcal{H}_{\mathcal{R}}, a_{-\mathbf{k}}^\dagger] &= \left[\sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) (a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger + a_{\mathbf{k}} a_{-\mathbf{k}}), a_{-\mathbf{k}}^\dagger \right] \\ &= \sum_{\mathbf{k}}' [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] [a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{-\mathbf{k}}^\dagger] + \frac{1}{2} \sum_{\mathbf{k}}' Nv(\mathbf{k}) ([\cancel{a_{\mathbf{k}}^\dagger a_{-\mathbf{k}}^\dagger}, a_{-\mathbf{k}}^\dagger] + [a_{\mathbf{k}} a_{-\mathbf{k}}, a_{-\mathbf{k}}^\dagger]) \end{aligned}$$

where

$$\begin{aligned}
\left[a_{\mathbf{k}}^\dagger a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] &= a_{\mathbf{k}}^\dagger \left[a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] + \cancel{\left[a_{\mathbf{k}}^\dagger, a_{-\mathbf{k}'}^\dagger \right]} a_{\mathbf{k}} \\
&= a_{\mathbf{k}}^\dagger \left[a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] \\
&= a_{-\mathbf{k}'}^\dagger \delta_{\mathbf{k}, -\mathbf{k}'}
\end{aligned}$$

and

$$\begin{aligned}
\left[a_{\mathbf{k}} a_{-\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] &= a_{\mathbf{k}} \left[a_{-\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] + \left[a_{\mathbf{k}}, a_{-\mathbf{k}'}^\dagger \right] a_{-\mathbf{k}} \\
&= a_{\mathbf{k}'} \delta_{\mathbf{k}, \mathbf{k}'} + \delta_{\mathbf{k}, -\mathbf{k}'} a_{\mathbf{k}'}
\end{aligned}$$

Substituting we obtain

$$\begin{aligned}
[\mathcal{H}_{\mathcal{R}}, a_{-\mathbf{k}'}^\dagger] &= [\epsilon_{-\mathbf{k}'} + Nv(-\mathbf{k}')] a_{-\mathbf{k}'}^\dagger + \frac{1}{2} N (v(\mathbf{k}') a_{\mathbf{k}'} + v(-\mathbf{k}') a_{\mathbf{k}'}) \\
&= [\epsilon_{\mathbf{k}'} + Nv(\mathbf{k}')] a_{-\mathbf{k}'}^\dagger + Nv(\mathbf{k}') a_{\mathbf{k}'}
\end{aligned}$$

And so, the new equation of motion gives

$$\begin{aligned}
\omega G(a_{-\mathbf{k}}^\dagger; a_{\mathbf{k}}^\dagger | \omega) &= -[\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] G(a_{-\mathbf{k}}^\dagger; a_{\mathbf{k}}^\dagger | \omega) - Nv(\mathbf{k}) G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) \\
\Leftrightarrow G(a_{-\mathbf{k}}^\dagger; a_{\mathbf{k}}^\dagger | \omega) &= \frac{-Nv(\mathbf{k}) G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)}{\omega + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})]}
\end{aligned}$$

This GF depends only on the first one we already calculated and so the system of equations closes. We can now backtrack and substitute it in its solution

$$\begin{aligned}
\omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) &= \frac{1}{2\pi} + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + Nv(\mathbf{k}) \frac{-Nv(\mathbf{k}) G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)}{\omega + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})]} \\
&= \frac{1}{2\pi} + \left([\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] - \frac{(Nv(\mathbf{k}))^2}{\omega + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})]} \right) G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) \\
\Leftrightarrow G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) &= \frac{1/2\pi}{\omega - \left([\epsilon_{\mathbf{k}} + Nv(\mathbf{k})] - \frac{(Nv(\mathbf{k}))^2}{\omega + [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})]} \right)}
\end{aligned}$$

Let us define $\tilde{\epsilon}_{\mathbf{k}} = [\epsilon_{\mathbf{k}} + Nv(\mathbf{k})]$ and $\tilde{\omega}_{\mathbf{k}} = Nv(\mathbf{k})$ in order to simplify the expression for our real-time GF as

$$G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi}{\omega - \tilde{\epsilon}_{\mathbf{k}} + \frac{|\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}}}$$

4.3.3 Finding the Spectrum

Analitically continnuing the GF to the upper complex plain i.e $\omega = \omega + i0^+$ we get the corresponding retarded GF

$$G_r(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi}{\left(\omega - \tilde{\epsilon}_{\mathbf{k}} + \frac{|\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}} \right) + i0^+}$$

where the spectrum is given by the poles

$$\begin{aligned}
\omega - \tilde{\epsilon}_{\mathbf{k}} + \frac{|\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}} &= \frac{(\omega - \tilde{\epsilon}_{\mathbf{k}})(\omega + \tilde{\epsilon}_{\mathbf{k}}) + |\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}} \\
&= \frac{\omega^2 - (\epsilon_{\mathbf{k}} + \tilde{\omega}_{\mathbf{k}})^2 + \tilde{\omega}_{\mathbf{k}}}{\omega + \tilde{\epsilon}_{\mathbf{k}}} = \frac{\omega^2 - \epsilon_{\mathbf{k}}^2 - 2\epsilon_{\mathbf{k}}\tilde{\omega}_{\mathbf{k}} - \tilde{\omega}_{\mathbf{k}}^2 + |\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}} \\
&= \frac{\omega^2 - \epsilon_{\mathbf{k}}^2 - 2\epsilon_{\mathbf{k}}\tilde{\omega}_{\mathbf{k}} - \tilde{\omega}_{\mathbf{k}}^2 + |\tilde{\omega}_{\mathbf{k}}|^2}{\omega + \tilde{\epsilon}_{\mathbf{k}}} = \frac{\omega^2 - \epsilon_{\mathbf{k}}(\epsilon_{\mathbf{k}} - 2\tilde{\omega}_{\mathbf{k}})}{\omega + \tilde{\epsilon}_{\mathbf{k}}}
\end{aligned}$$

We have two poles corresponding to

$$\begin{aligned}
\omega^2 - \epsilon_{\mathbf{k}}(\epsilon_{\mathbf{k}} - 2\tilde{\omega}_{\mathbf{k}}) &= 0 \\
\Rightarrow \pm E_{\mathbf{k}} &= \pm \sqrt{\epsilon_{\mathbf{k}}(\epsilon_{\mathbf{k}} - 2\tilde{\omega}_{\mathbf{k}})}
\end{aligned}$$

Substituting in the retarded GF we obtain

$$G_r(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi (\omega + \tilde{\epsilon}_{\mathbf{k}})}{(\omega - E_{\mathbf{k}})(\omega + E_{\mathbf{k}}) + i0^+}$$

4.3.4 Finding the Spectral Function

To eventually use the Sokhotski–Plemelj identity we need to separate the poles from each other. For this we try to find two variables A and B such that

$$\begin{aligned}
\frac{(\omega + \tilde{\epsilon}_{\mathbf{k}})}{(\omega - E_{\mathbf{k}})(\omega + E_{\mathbf{k}})} &\stackrel{!}{=} \frac{A}{(\omega - E_{\mathbf{k}})} + \frac{B}{(\omega + E_{\mathbf{k}})} \\
&= \frac{A(\omega + E_{\mathbf{k}}) + B(\omega - E_{\mathbf{k}})}{(\omega - E_{\mathbf{k}})(\omega + E_{\mathbf{k}})} = \frac{\omega(A + B) + E_{\mathbf{k}}(A - B)}{(\omega - E_{\mathbf{k}})(\omega + E_{\mathbf{k}})} \\
&\Rightarrow \begin{cases} A + B = 1 & , \text{for the } \omega \\ A - B = \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} & , \text{for the } \tilde{\epsilon}_{\mathbf{k}} \end{cases} \\
&\Leftrightarrow \begin{cases} A = 1 - B \Leftrightarrow A = -\frac{1}{2} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}}\right) & , \text{for the } \omega \\ 1 - 2B = \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \Leftrightarrow B = \frac{1}{2} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}}\right) & , \text{for the } \tilde{\epsilon}_{\mathbf{k}} \end{cases}
\end{aligned}$$

Our retarded GF can then rewritten as

$$G_r(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi \times -\frac{1}{2} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}{(\omega - E_{\mathbf{k}}) + i0^+} + \frac{1/2\pi \times \frac{1}{2} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}}\right)}{(\omega + E_{\mathbf{k}}) + i0^+}$$

Only now we can use fluctuation-dissipation theorem

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \Im G_r(\omega)$$

and the Sokhotski–Plemelj identity

$$\frac{1}{x \pm i0^+} = \mp i\pi\delta(x) + \mathcal{P}\left(\frac{1}{x}\right)$$

to obtain the spectral function $J(\omega)$

$$\begin{aligned} J(\omega) &= \frac{\cancel{2\pi}}{(e^{\beta\omega} - 1) 2\pi} \left(-\frac{1}{2} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega - E_{\mathbf{k}}) + \frac{1}{2} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega + E_{\mathbf{k}}) \right) \\ &= \left(\frac{1}{e^{-\beta E_{\mathbf{k}}} - 1} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega - E_{\mathbf{k}}) - \frac{1}{e^{\beta E_{\mathbf{k}}} - 1} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega + E_{\mathbf{k}}) \right) \end{aligned}$$

4.3.5 Calculating the thermal average

The initial definition of the spectral function was it being the Fourier spectrum of the correlation function i.e

$$\langle B(t')A(t) \rangle \equiv F_{BA}(t - t') = \int_{-\infty}^{+\infty} J(\omega) e^{-i\omega(t-t')} d\omega$$

So, the thermal average asked at times $t = t' = 0$ is given by

$$\begin{aligned} \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \rangle &= \int_{-\infty}^{+\infty} J(\omega) d\omega \\ &= -\frac{1}{2} \int_{-\infty}^{+\infty} \left(\frac{1}{e^{-\beta E_{\mathbf{k}}} - 1} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega - E_{\mathbf{k}}) - \frac{1}{e^{\beta E_{\mathbf{k}}} - 1} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \delta(\omega + E_{\mathbf{k}}) \right) d\omega \\ &= -\frac{1}{2} \frac{1}{e^{-\beta E_{\mathbf{k}}} - 1} \left(1 + \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) - \frac{1}{e^{\beta E_{\mathbf{k}}} - 1} \left(1 - \frac{\tilde{\epsilon}_{\mathbf{k}}}{E_{\mathbf{k}}} \right) \end{aligned}$$

4.4 GF for Noninteracting Bosons in a monoatomic lattice

The quantized longitudinal vibrations of a monoatomic lattice of atoms can be described in terms of non-interacting phonons with the Hamiltonian

$$H = \sum_k \omega_k (a_k^\dagger a_k + 1/2)$$

We want to derive the retarded GF $G_r(a_k; a_k^\dagger | \omega)$. For this we start with the equation of motion

$$\omega G(a_k; a_k^\dagger | \omega) = \frac{1}{2\pi} \langle [a_k, a_k^\dagger] \rangle - G([\mathcal{H}, a_k]; a_k^\dagger | \omega)$$

where

$$\begin{aligned}
[\mathcal{H}, a_k] &= \left[\sum_{k'} \omega_{k'} (a_{k'}^\dagger a_{k'} + 1/2), a_k \right] \\
&= \sum_{k'} \omega_{k'} [a_{k'}^\dagger a_{k'}, a_k] \\
&= \sum_{k'} \omega_{k'} \left(0 + [a_{k'}^\dagger, a_k] a_{k'} \right) \\
&= \sum_{k'} \omega_{k'} (-\delta_{k,k'}) a_k = -\omega_k a_k
\end{aligned}$$

Substituting with obtain

$$\omega G(a_k; a_k^\dagger | \omega) = \frac{1}{2\pi} + \omega_k G(a_k; a_k^\dagger | \omega)$$

To obtain the retarded GF we just need to do analitical continuation of $\omega \rightarrow \omega + i0^+$ obtaining

$$\Rightarrow G_r(a_k; a_k^\dagger | \omega) = \frac{1/2\pi}{\omega - \omega_k + i0^+}$$

From this we can deduce another retarded GF

$$G_r(\{a_k - a_{-k}^\dagger\}; \{a_k^\dagger - a_{-k}\} | \omega) = G_r(a_k; a_k^\dagger | \omega) - \cancel{G_r(a_k; a_{-k} | \omega)} - \cancel{G_r(a_{-k}^\dagger; a_k^\dagger | \omega)} + G_r(a_{-k}^\dagger; a_{-k} | \omega)$$

If we use the proprety $G_r^*(A; B | \omega) = \epsilon G_r(A^\dagger; B^\dagger | -\omega)$ (in this case $\epsilon = 1$ since phonons are bosons) derived in section 3 we obtain

$$\begin{aligned}
&= G_r(a_k; a_k^\dagger | \omega) + G_r^*(a_{-k}; a_{-k}^\dagger | -\omega) \\
&= \frac{1/2\pi}{\omega - \omega_k + i0^+} + \frac{1/2\pi}{-\omega - \omega_{-k} - i0^+} \\
&= \frac{1/2\pi}{\omega - \omega_k + i0^+} - \frac{1/2\pi}{\omega + \omega_{-k} + i0^+} \\
&= \frac{1}{2\pi} \frac{\cancel{\omega} + \omega_{-k} + i0^+ - \cancel{\omega} + \omega_k - i0^+}{(\omega - \omega_k + i0^+)(\omega + \omega_{-k} + i0^+)} \\
&= \frac{1}{2\pi} \frac{2\omega_k}{(\omega + i0^+)^2 - \omega_k^2}
\end{aligned}$$

4.5 Jaynes-Cummings Model Hamiltonean

4.6 Jaynes-Cummings Model Spectral Function

The Jaynes-Cummings Hamiltonean is given by

$$H = \omega_f a^\dagger a + \frac{1}{2} \omega_a \sigma_z + g(\sigma^+ a + \sigma^- a^\dagger)$$

We pretend to solve the equation of motion for the following GF

$$\omega G(\sigma^-, \sigma^+ | \omega) = \frac{1}{2\pi} \langle [\sigma^-, \sigma^+] \rangle - G([H, \sigma^-]; \sigma^+ | \omega)$$

knowing the commutator relations of the spin operators

$$\begin{cases} [\sigma^+, \sigma^-] = \sigma^z \\ [\sigma^z, \sigma^\pm] = \pm 2\sigma^\pm \end{cases}$$

We start by calculating the commutator with the Hamiltonian

$$\begin{aligned} [H, \sigma^-] &= \left[\omega_f a^\dagger a + \frac{1}{2} \omega_a \sigma_z + g(\sigma^+ a + \sigma^- a^\dagger), \sigma^- \right] \\ &= \frac{1}{2} \omega_a [\sigma_z, \sigma^-] + g a [\sigma^+, \sigma^-] + a^\dagger [\cancel{\sigma^-, \sigma^-}] \\ &= -\omega_a \sigma^- + g a \sigma^z \end{aligned}$$

Substituting we obtain

$$\omega G(\sigma^-, \sigma^+ | \omega) = -\frac{1}{2\pi} \langle \sigma^z \rangle - \omega_a G(\sigma^-; \sigma^+ | \omega) - g G(a \sigma^z; \sigma^+ | \omega)$$

For the last GF we do the decoupling (RPA)

$$G(a \sigma^z; \sigma^+ | \omega) \approx \langle \sigma^z \rangle G(a; \sigma^+ | \omega)$$

Futhermore, at zero temperatura the expectation value of $\langle \sigma^z \rangle$ is -1 so we assume it throughout the rest of the calculation. We are left with

$$\omega G(\sigma^-, \sigma^+ | \omega) = \frac{1}{2\pi} - \omega_a G(\sigma^-; \sigma^+ | \omega) + g G(a; \sigma^+ | \omega) \Rightarrow G(\sigma^-, \sigma^+ | \omega) = \frac{1/2\pi + g G(a; \sigma^+ | \omega)}{\omega - \omega_a}$$

We now need to calculate the new GF hoping that they form a closed set of equation. Its equation of motion is

$$\omega G(a, \sigma^+ | \omega) = \frac{1}{2\pi} \langle [\cancel{a}, \sigma^+] \rangle - G([H, a]; \sigma^+ | \omega)$$

where

$$\begin{aligned} [H, a] &= \left[\omega_f a^\dagger a + \frac{1}{2} \omega_a \sigma_z + g(\sigma^+ a + \sigma^- a^\dagger), a \right] \\ &= \omega_f [a^\dagger a, a] + g \sigma^+ [\cancel{a}, a] + g \sigma^- [a^\dagger, a] \\ &= -\omega_f a - g \sigma^- \end{aligned}$$

And so

$$\omega G(a, \sigma^+ | \omega) = \omega_f G(a; \sigma^+ | \omega) + g G(\sigma^-; \sigma^+ | \omega) \Rightarrow G(a, \sigma^+ | \omega) = \frac{g G(\sigma^-; \sigma^+ | \omega)}{\omega - \omega_f}$$

Substituting in the first equation of motion we obtain

$$G(\sigma^-, \sigma^+|\omega) = \frac{1/2\pi + g \frac{gG(\sigma^-; \sigma^+|\omega)}{\omega - \omega_f}}{\omega - \omega_a}$$

$$\Leftrightarrow G(\sigma^-, \sigma^+|\omega) = \frac{1/2\pi}{\omega - \omega_a - \frac{g^2}{\omega - \omega_f}} = \frac{1/2\pi(\omega - \omega_f)}{(\omega - \omega_a)(\omega - \omega_f) - g^2}$$

In order to use the Plemelj theorem to get the imaginary part of the GF for the fluctuation-dissipation theorem we need to isolate the poles. First we find the energy spectrums by evaluating the denominator

$$0 = (\omega - \omega_a)(\omega - \omega_f) - g^2 = \omega^2 - \omega(\omega_a + \omega_f) - (g^2 - \omega_a\omega_f)$$

$$\Rightarrow \omega = \frac{1}{2}(\omega_a + \omega_f) \pm \frac{1}{2}\sqrt{(\omega_a + \omega_f)^2 + 4(g^2 - \omega_a\omega_f)}$$

$$= \frac{1}{2}\omega_a + \frac{1}{2}\omega_f \left\{ -\frac{1}{2}\omega_a + \frac{1}{2}\omega_a \right\} \pm \frac{1}{2}\sqrt{\omega_a^2 + \omega_f^2 + 2\omega_a\omega_f - 4\omega_a\omega_f + 4g^2}$$

$$= \omega_a - \frac{1}{2}(\omega_a - \omega_f) \pm \frac{1}{2}\sqrt{(\omega_a - \omega_f)^2 + 4g^2}$$

It's usual to define a detuning parameter $\delta \equiv (\omega_a - \omega_f)$ and defining $\omega_R \equiv \sqrt{\delta^2 + 4g^2}$ as the called Rabi splitting. The solutions are then

$$\omega^\pm = \left(\omega_a - \frac{\delta}{2} \right) \pm \frac{1}{2}\omega_R$$

We can rewrite our GF as

$$G(\sigma^-, \sigma^+|\omega) = \frac{1/2\pi(\omega - \omega_f)}{(\omega - \omega^+)(\omega - \omega^-)}$$

By isolating the Poles —

To isolate the poles we try to find the value of constants A and B such that

$$\frac{(\omega - \omega_f)}{(\omega - \omega^+)(\omega - \omega^-)} \stackrel{!}{=} \frac{A}{(\omega - \omega^+)} + \frac{B}{(\omega - \omega^-)}$$

$$= \frac{A(\omega - \omega^-) + B(\omega - \omega^+)}{(\omega - \omega^+)(\omega - \omega^-)}$$

$$= \frac{\omega(A + B) - (A\omega^- + B\omega^+)}{(\omega - \omega^+)(\omega - \omega^-)}$$

$$\Rightarrow \begin{cases} A + B = 1 & , \text{for } \omega \\ A\omega^- + B\omega^+ = \omega_f & , \text{for } \omega_f \end{cases}$$

$$\Rightarrow \begin{cases} A = 1 - B = 1 - \frac{1}{2\omega_R}(\omega_f - \omega^-) \\ \omega_f = \omega^- - B(\omega^- - \omega^+) = \omega^- + 2B\frac{1}{2}\omega_R = \omega^- + 2B\omega_R \Rightarrow B = \frac{1}{2\omega_R}(\omega_f - \omega^-) \end{cases}$$

We can now analitically continue our GF and rewrite it as

$$G_r(\sigma^-, \sigma^+|\omega) = \frac{1}{2\pi} \left(\frac{A}{(\omega - \omega^+) + i0^+} + \frac{B}{(\omega - \omega^-) + i0^+} \right)$$

Only now we can use flutuaction-dissipation theorem

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \Im G_r(\omega)$$

and the Sokhotski–Plemelj identity

$$\frac{1}{x \pm i0^+} = \mp i\pi\delta(x) + \mathcal{P} \left(\frac{1}{x} \right)$$

to obtain the spectral function $J(\omega)$

$$\begin{aligned} J(\omega) &= \frac{-2\pi}{(e^{\beta\omega} - 1)} \frac{1}{2\pi} \left(\frac{A}{(\omega - \omega^+)} \delta(\omega - \omega^-) + \frac{B}{(\omega - \omega^-)} \delta(\omega - \omega^-) \right) \\ &= \frac{-1}{(e^{\beta\omega^+} - 1)} \frac{1 - \frac{1}{2\omega_R}(\omega_f - \omega^-)}{(\omega - \omega^+)} \delta(\omega - \omega^-) + \frac{-1}{(e^{\beta\omega^-} - 1)} \frac{\frac{1}{2\omega_R}(\omega_f - \omega^-)}{(\omega - \omega^-)} \delta(\omega - \omega^-) \end{aligned}$$

Or by the Sokhotski–Plemelj identity —

We can rewrite our GF as

$$G_r(\sigma^-, \sigma^+|\omega) = \frac{1}{2\pi} (\omega - \omega_f) \frac{1}{(\omega - \omega^+) + i0^+} \frac{1}{(\omega - \omega^-) + i0^+}$$

By the Sokhotski–Plemelj identity

$$\frac{1}{x \pm i0^+} = \mathcal{P} \left(\frac{1}{x} \right) \mp i\pi\delta(x)$$

we have that

$$\begin{aligned} \frac{1}{(\omega - \omega^+) + i0^+} \frac{1}{(\omega - \omega^-) + i0^+} &= \left(\mathcal{P} \left(\frac{1}{\omega - \omega^+} \right) - i\pi\delta(\omega - \omega^+) \right) \left(\mathcal{P} \left(\frac{1}{\omega - \omega^-} \right) - i\pi\delta(\omega - \omega^-) \right) \\ &= \mathcal{P} \left(\frac{1}{\omega - \omega^+} \right) \mathcal{P} \left(\frac{1}{\omega - \omega^-} \right) - i\pi \left[\mathcal{P} \left(\frac{1}{\omega - \omega^+} \right) \delta(\omega - \omega^-) + \mathcal{P} \left(\frac{1}{\omega - \omega^-} \right) \delta(\omega - \omega^+) \right] + 0 \end{aligned}$$

We can use flutuaction-dissipation theorem

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \Im G_r(\omega)$$

to obtain the spectral function $J(\omega)$

$$\begin{aligned} J(\omega) &= \frac{-2\pi}{(e^{\beta\omega} - 1)} \frac{1}{2\pi} (\omega - \omega_f) \left(\mathcal{P} \left(\frac{1}{\omega - \omega^+} \right) \delta(\omega - \omega^-) + \mathcal{P} \left(\frac{1}{\omega - \omega^-} \right) \delta(\omega - \omega^+) \right) \\ &= \frac{-1}{(e^{\beta\omega^-} - 1)} \frac{(\omega^- - \omega_f)}{(\omega^- - \omega^+)} \delta(\omega - \omega^-) + \frac{-1}{(e^{\beta\omega^+} - 1)} \frac{(\omega^+ - \omega_f)}{(\omega^+ - \omega^-)} \delta(\omega - \omega^+) \end{aligned}$$

4.7 Dicke Model

The Dicke Model Hamiltonian in RWA can be written as

$$H_D = \omega_f a^\dagger a + \omega_a S^z + g(S^+ a + S^- a^\dagger)$$

Here ω_a is the frequency of any one of the atoms, while $S^{\pm, z}$ are collective atomic operators defined by

$$S^\pm = \sum_{i=1}^N \sigma_{(i)}^\pm \quad \text{and} \quad S^z = \frac{1}{2} \sum_{i=1}^N \sigma_{(i)}^z$$

The Green's Function equation of motions is

$$\omega G(S^-; S^+ | \omega) = \frac{1}{2\pi} \langle [S^-, S^+]_\epsilon \rangle - G([H, S^-]; S^+ | \omega)$$

First, we evaluate the commutator

$$[S^-, S^+]_\epsilon = \sum_{i=1}^N \sum_{j=1}^N [\sigma_{(i)}^-, \sigma_{(j)}^+] = -\sigma_{(i)}^z \delta_{ij}$$

4.8 Exact Solution GF Paramagnetic System

We assume that the spin Hamiltonian has the form

$$H = -g\mu_B B_0 \sum_i S_i^z - D_0 \sum_i (S_i^z)^2$$

We want to calculate the equation of motion for the GF

$$G(\{S_n^z S_n^+ + S_n^+ S_n^z\}, S_m^- | \omega) = G(S_n^z S_n^+, S_m^- | \omega) + G(S_n^+ S_n^z, S_m^- | \omega)$$

knowing the commutators

$$\begin{cases} [S_i^+, S_j^-] = 2S_i^z \delta_{i,j} \\ [S_j^z, S_i^\pm] = \pm S_i^\pm \delta_{i,j} \end{cases}$$

For the first GF we have the equation of motion —

$$\omega G(S_n^z S_n^+, S_m^- | \omega) = \frac{1}{2\pi} \langle [S_n^z S_n^+, S_m^-] \rangle - G([\mathcal{H}, S_n^z S_n^+]; S_m^- | \omega)$$

The first commutator gives

$$\begin{aligned} [S_n^z S_n^+, S_m^-] &= S_n^z [S_n^+, S_m^-] + [S_n^z, S_m^-] S_n^+ = 2S_n^z S_n^z \delta_{n,m} - S_n^- S_n^+ \delta_{n,m} \\ &= (2(S_n^z)^2 - S_n^- S_n^+) \delta_{n,m} \end{aligned}$$

And the second one gives

$$\begin{aligned} [\mathcal{H}, S_n^z S_n^+] &= \left[-g\mu_B B_0 \sum_i S_i^z - D_0 \sum_i (S_i^z)^2, S_n^z S_n^+ \right] \\ &= -g\mu_B B_0 \sum_i [S_i^z, S_n^z S_n^+] - D_0 \sum_i [(S_i^z)^2, S_n^z S_n^+] \end{aligned}$$

The first commutator gives

$$- [S_n^z S_n^+, S_i^z] = - (S_n^z [S_n^+, S_i^z] + [S_n^z, S_i^z] S_n^+) = S_n^z S_n^+ \delta_{n,i}$$

For the second one we use the relation

$$[AB, CD] = A[B, C]D + [A, C]BD + CA[B, D] + C[A, D]B$$

giving

$$\begin{aligned} [S_i^z S_i^z, S_n^z S_n^+] &= S_i^z [S_i^z, S_n^z] S_n^+ + [S_i^z, S_n^z] S_i^z S_n^+ + S_n^z S_i^z [S_i^z, S_n^+] + S_n^z [S_i^z, S_n^+] S_i^z \\ &= S_i^z S_i^z S_n^+ \delta_{i,n} + S_n^z S_n^+ S_i^z \delta_{i,n} \\ &= ((S_n^z)^2 S_n^+ + S_n^z S_n^+ S_n^z) \delta_{i,n} \end{aligned}$$

Putting everything back we obtain

$$\begin{aligned} \omega G(S_n^z S_n^+, S_m^- | \omega) &= \frac{1}{2\pi} (\langle (S_n^z)^2 \rangle - \langle S_n^- S_n^+ \rangle) \delta_{n,m} + g\mu_B B_0 \sum_i \delta_{n,i} G(S_n^z S_n^+; S_m^- | \omega) \\ &\quad + D_0 \sum_i \delta_{i,n} G((S_n^z)^2 S_n^+ + S_n^z S_n^+ S_n^z; S_m^- | \omega) \end{aligned}$$

We need to go way back we calculate the equation of motion for the other GF —

$$\omega G(S_n^+ S_n^z, S_m^- | \omega) = \frac{1}{2\pi} \langle [S_n^+ S_n^z, S_m^-] \rangle - G([\mathcal{H}, S_n^+ S_n^z]; S_m^- | \omega)$$

The first commutator gives

$$\begin{aligned} [S_n^+ S_n^z, S_m^-] &= S_n^+ [S_n^z, S_m^-] + [S_n^+, S_m^-] S_n^z = -S_n^+ S_m^- \delta_{n,m} + 2S_n^z S_n^z \delta_{n,m} \\ &= (2(S_n^z)^2 - S_n^+ S_m^-) \delta_{n,m} \end{aligned}$$

And the second one gives

$$\begin{aligned} [\mathcal{H}, S_n^+ S_n^z] &= \left[-g\mu_B B_0 \sum_i S_i^z - D_0 \sum_i (S_i^z)^2, S_n^+ S_n^z \right] \\ &= -g\mu_B B_0 \sum_i [S_i^z, S_n^+ S_n^z] - D_0 \sum_i [(S_i^z)^2, S_n^+ S_n^z] \end{aligned}$$

The first commutator gives

$$- [S_n^+ S_n^z, S_i^z] = - (S_n^+ [S_n^z, S_i^z] + [S_n^+, S_i^z] S_n^z) = S_n^+ S_n^z \delta_{n,i}$$

For the second one we use the relation

$$[AB, CD] = A[B, C]D + [A, C]BD + CA[B, D] + C[A, D]B$$

giving

$$\begin{aligned} [S_i^z S_i^z, S_n^+ S_n^z] &= S_i^z [S_i^z, S_n^+] S_n^z + [S_i^z, S_n^+] S_i^z S_n^z + S_n^+ S_i^z [S_i^z, S_n^z] + S_n^+ [S_i^z, S_n^z] S_i^z \\ &= S_i^z S_i^+ S_n^z \delta_{i,n} + S_n^+ S_i^z S_n^z \delta_{i,n} \\ &= (S_n^+ (S_n^z)^2 + S_n^z S_n^+ S_n^z) \delta_{i,n} \end{aligned}$$

Putting everything back we obtain

$$\begin{aligned} \omega G(S_n^+ S_n^z, S_m^- | \omega) &= \frac{1}{2\pi} (2 \langle (S_n^z)^2 \rangle - \langle S_n^+ S_n^- \rangle) \delta_{n,m} + g\mu_B B_0 \sum_i \delta_{n,i} G(S_n^+ S_n^z; S_m^- | \omega) \\ &\quad + D_0 \sum \delta_{i,n} G((S_n^+ (S_n^z)^2 + S_n^z S_n^+ S_n^z); S_m^- | \omega) \end{aligned}$$

Summing the two together we obtain —

$$\begin{aligned} \omega G(\{S_n^z S_n^+ + S_n^+ S_n^z\}, S_m^- | \omega) &= \omega G(S_n^z S_n^+, S_m^- | \omega) + \omega G(S_n^+ S_n^z, S_m^- | \omega) \\ &= \frac{1}{2\pi} (\langle 2(S_n^z)^2 \rangle - \langle S_n^- S_n^+ \rangle) \delta_{n,m} + g\mu_B B_0 \sum_i \delta_{n,i} G(S_n^z S_n^+; S_m^- | \omega) \\ &\quad + D_0 \sum \delta_{i,n} G((S_n^z)^2 S_n^+ + S_n^z S_n^+ S_n^z; S_m^- | \omega) \\ &\quad + \frac{1}{2\pi} (2 \langle (S_n^z)^2 \rangle - \langle S_n^+ S_n^- \rangle) \delta_{n,m} + g\mu_B B_0 \sum_i \delta_{n,i} G(S_n^+ S_n^z; S_m^- | \omega) \\ &\quad + D_0 \sum \delta_{i,n} G((S_n^+ (S_n^z)^2 + S_n^z S_n^+ S_n^z); S_m^- | \omega) \\ &= \frac{1}{2\pi} \{4 \langle (S_n^z)^2 \rangle - \langle S_n^- S_n^+ \rangle - \langle S_n^+ S_n^- \rangle\} \delta_{n,m} \\ &\quad + g\mu_B B_0 G(\{S_n^z S_n^+ + S_n^+ S_n^z\}; S_m^- | \omega) \\ &\quad + D_0 G((S_n^z)^2 S_n^+ + 2S_n^z S_n^+ S_n^z + S_n^+ (S_n^z)^2; S_m^- | \omega) \end{aligned}$$

Remembering the definition of the total spin

$$\begin{aligned} S^2 &= S_x^2 + S_y^2 + S_z^2 \\ &= S^+ S^- - S_z + S_z^2 \end{aligned}$$

and that its mean values is given by

$$\langle S^2 \rangle = S(S+1)$$

we have that

$$\begin{aligned} &= 4 \langle (S_n^z)^2 \rangle - \langle S_n^- S_n^+ \rangle - \langle S_n^+ S_n^- \rangle \\ &= 4 \langle (S_n^z)^2 \rangle - \langle S_n^+ S_n^- - [S_n^+, S_n^-] \rangle - \langle S_n^+ S_n^- \rangle \\ &= 4 \langle (S_n^z)^2 \rangle + 2 \langle S_n^z \rangle - 2 \langle S_n^+ S_n^- \rangle \\ &= 4 \langle (S_n^z)^2 \rangle + \{ 2 \langle (S_n^z)^2 \rangle - 2 \langle (S_n^z)^2 \rangle \} + 2 \langle S_n^z \rangle - 2 \langle S_n^+ S_n^- \rangle \\ &= 6 \langle (S_n^z)^2 \rangle - 2 (\langle S_n^+ S_n^- \rangle - \langle S_n^z \rangle + \langle (S_n^z)^2 \rangle) \end{aligned}$$

Our final result is then

$$\begin{aligned} &\omega G(\{S_n^z S_n^+ + S_n^+ S_n^z\}, S_m^- | \omega) = \\ &= \frac{1}{2\pi} \{ 4 \langle (S_n^z)^2 \rangle - \langle S_n^- S_n^+ \rangle - \langle S_n^+ S_n^- \rangle \} \delta_{n,m} + g\mu_B B_0 G(\{S_n^z S_n^+ + S_n^+ S_n^z\}; S_m^- | \omega) + D_0 G((S_n^z)^2 S_n^+ + 2S_n^z S_n^+ S_n^z) \end{aligned}$$

4.9 Paramagnet with spin S=1

4.9.1 The spectral function

The analytically continued GF for a paramagnet with spin $S = 1$ is given by

$$G(S_n^+; S_m^- | \tilde{\omega}) = \frac{\langle S_n^z \rangle (\tilde{\omega} - g\mu_B B_0) + \{ 3 \langle (S_n^z)^2 \rangle - 2 \} D_0}{\pi (\tilde{\omega} - g\mu_B B_0 - D_0) (\tilde{\omega} - g\mu_B B_0 + D_0)} \delta_{n,m}.$$

To make more obvious the following treatment we can define $\omega^\pm = -g\mu_B B_0 \pm D_0$ and rewrite our GF as

$$\begin{aligned} G(S_n^+; S_m^- | \tilde{\omega}) &= \frac{1}{\pi} \langle S_n^z \rangle (\tilde{\omega} - g\mu_B B_0) \delta_{n,m} \frac{1}{(\tilde{\omega} - \omega^-)} \frac{1}{(\tilde{\omega} - \omega^+)} \\ &\quad + \frac{1}{\pi} \{ 3 \langle (S_n^z)^2 \rangle - 2 \} D_0 \delta_{n,m} \frac{1}{(\tilde{\omega} - \omega^-)} \frac{1}{(\tilde{\omega} - \omega^+)} \end{aligned}$$

By the Sokhotski–Plemelj identity

$$\frac{1}{x \pm i0^+} = \mathcal{P} \left(\frac{1}{x} \right) \mp i\pi \delta(x)$$

we have that

$$\frac{1}{(\omega - \omega^+) + i0^+} \frac{1}{(\omega - \omega^-) + i0^+} = \left(\mathcal{P} \left(\frac{1}{\omega - \omega^+} \right) - i\pi \delta(\omega - \omega^+) \right) \left(\mathcal{P} \left(\frac{1}{\omega - \omega^-} \right) - i\pi \delta(\omega - \omega^-) \right)$$

$$\begin{aligned}
&= \mathcal{P}\left(\frac{1}{\omega - \omega^+}\right) \mathcal{P}\left(\frac{1}{\omega - \omega^-}\right) - i\pi \left[\mathcal{P}\left(\frac{1}{\omega - \omega^+}\right) \delta(\omega - \omega^-) + \mathcal{P}\left(\frac{1}{\omega - \omega^-}\right) \delta(\omega - \omega^+) \right] + 0 \\
&== \mathcal{P}\left(\frac{1}{\omega - \omega^+}\right) \mathcal{P}\left(\frac{1}{\omega - \omega^-}\right) - i\pi \left[\left(\frac{1}{\omega^- - \omega^+}\right) \delta(\omega - \omega^-) + \left(\frac{1}{\omega^+ - \omega^-}\right) \delta(\omega - \omega^+) \right]
\end{aligned}$$

We can use fluctuation-dissipation theorem

$$J(\omega) = \frac{-2}{(e^{\beta\omega} - \epsilon)} \Im G_r(\omega)$$

to obtain the spectral function $J(\omega)$

$$\begin{aligned}
J(\omega) &= \frac{-2}{(e^{\beta\omega} - 1)} \frac{1}{\pi} \langle S_n^z \rangle (\omega - g\mu_B B_0) \delta_{n,m}(-\pi) \left[\left(\frac{1}{\omega^- - \omega^+}\right) \delta(\omega - \omega^-) + \left(\frac{1}{\omega^+ - \omega^-}\right) \delta(\omega - \omega^+) \right] \\
&\quad + \frac{-2}{(e^{\beta\omega} - 1)} \frac{1}{\pi} \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \delta_{n,m}(-\pi) \left[\left(\frac{1}{\omega^- - \omega^+}\right) \delta(\omega - \omega^-) + \left(\frac{1}{\omega^+ - \omega^-}\right) \delta(\omega - \omega^+) \right] \\
&= \frac{2}{(e^{\beta\omega^-} - 1)} \langle S_n^z \rangle \delta_{n,m} \left(\frac{\omega^- - g\mu_B B_0}{\omega^- - \omega^+} \right) \delta(\omega - \omega^-) \\
&\quad + \frac{2}{(e^{\beta\omega^+} - 1)} \langle S_n^z \rangle (\omega - g\mu_B B_0) \delta_{n,m} \left(\frac{\omega^+ - g\mu_B B_0}{\omega^+ - \omega^-} \right) \delta(\omega - \omega^+) \\
&\quad + \frac{2}{(e^{\beta\omega^-} - 1)} \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \delta_{n,m} \left(\frac{\omega^- - g\mu_B B_0}{\omega^- - \omega^+} \right) \delta(\omega - \omega^-) \\
&\quad + \frac{2}{(e^{\beta\omega^+} - 1)} \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \delta_{n,m} \left(\frac{\omega^+ - g\mu_B B_0}{\omega^+ - \omega^-} \right) \delta(\omega - \omega^+) \\
&= \frac{2}{(e^{\beta\omega^-} - 1)} \left[\langle S_n^z \rangle + \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \right] \delta_{n,m} \left(\frac{\omega^- - g\mu_B B_0}{\omega^- - \omega^+} \right) \delta(\omega - \omega^-) \\
&\quad + \frac{2}{(e^{\beta\omega^+} - 1)} \left[\langle S_n^z \rangle + \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \right] \delta_{n,m} \left(\frac{\omega^+ - g\mu_B B_0}{\omega^+ - \omega^-} \right) \delta(\omega - \omega^+)
\end{aligned}$$

Our spectral function takes the form

$$\begin{aligned}
J(\omega) &= 2 \left[\langle S_n^z \rangle + \left\{ 3 \langle (S_n^z)^2 \rangle - 2 \right\} D_0 \right] \delta_{n,m} \\
&\quad \times \left[\frac{1}{e^{\beta\omega^-} - 1} \left(\frac{\omega^- - g\mu_B B_0}{\omega^- - \omega^+} \right) \delta(\omega - \omega^-) + \frac{1}{e^{\beta\omega^+} - 1} \left(\frac{\omega^+ - g\mu_B B_0}{\omega^+ - \omega^-} \right) \delta(\omega - \omega^+) \right]
\end{aligned}$$

Substituting for the energies $\omega^\pm = -g\mu_B B_0 \pm D_0$ the terms in curve brackets simplify too

$$\frac{\omega^- - g\mu_B B_0}{\omega^- - \omega^+} = \frac{-g\mu_B B_0 - D_0 - g\mu_B B_0}{-2D_0} = \frac{1}{2} + \frac{g\mu_B B_0}{D_0}$$

and

$$\frac{\omega^+ - g\mu_B B_0}{\omega^+ - \omega^-} = \frac{-g\mu_B B_0 + D_0 - g\mu_B B_0}{+2D_0} = \frac{1}{2} - \frac{g\mu_B B_0}{D_0}$$

4.9.2 Limit when $D_0 \rightarrow 0$

When we take $D_0 \rightarrow 0$ the term $\left\{3 \left\langle (S_n^z)^2 \right\rangle - 2\right\} D_0$ vanishes but more importantly we have that $\omega^+ = \omega^-$. Because of this the the bose-einsteins distribution give the same contributions $e^{\beta\omega^-} = e^{\beta\omega^+}$ and the spectral function reduces to

$$J(\omega) \approx 2 \langle S_n^z \rangle \delta_{n,m} \frac{1}{e^{-\beta g \mu_B B_0} - 1} \left(\frac{g \mu_B B_0}{D_0} \right) [\delta(\omega - \omega^-) - \delta(\omega - \omega^+)]$$

But, because $\omega^+ = \omega^-$ the δ 's are the same and they cancel out. Is it to expect that for a null single-ion uniaxial anisotropy D_0 the Hamiltonean is just the Zeeman ???????

4.9.3 Limit when $\beta \rightarrow \infty$

When we take $\beta \rightarrow \infty$ the exponential terms decays very rapidly and so

$$\frac{1}{e^{\beta\omega^\pm} - 1} \approx e^{+\beta\omega^\pm}$$

the spectral function reduces to

$$J(\omega) = 2 \left[\langle S_n^z \rangle + \left\{3 \left\langle (S_n^z)^2 \right\rangle - 2\right\} D_0 \right] \delta_{n,m} \\ \times e^{\beta\omega} \left[e^{+\beta D_0} \left(\frac{\omega^- - g \mu_B B_0}{\omega^- - \omega^+} \right) \delta(\omega - \omega^-) + \left(\frac{\omega^+ - g \mu_B B_0}{\omega^+ - \omega^-} \right) \delta(\omega - \omega^+) \right]$$

4.10 Coupling between magnon and phonons

The coupling between spin-waves and phonons may be described bt the Hamiltonian

$$H = \sum_{\mathbf{k}} \left[\omega_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \Omega_{\mathbf{k}} b_{\mathbf{k}}^\dagger b_{\mathbf{k}} + c_{\mathbf{k}} \left(a_{\mathbf{k}}^\dagger b_{\mathbf{k}} + b_{\mathbf{k}}^\dagger a_{\mathbf{k}} \right) \right]$$

We want to solve the GF's equation of motion

$$\omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} \left\langle [a_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] \right\rangle - G([\mathcal{H}, a_{\mathbf{k}}]; a_{\mathbf{k}}^\dagger | \omega)$$

For this we need to know the commutator with the Hamiltonean which gives

$$\begin{aligned} [H, a_{\mathbf{k}}] &= \left[\sum_{\mathbf{k}'} \left[\omega_{\mathbf{k}'} a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} + \Omega_{\mathbf{k}'} b_{\mathbf{k}'}^\dagger b_{\mathbf{k}'} + c_{\mathbf{k}'} \left(a_{\mathbf{k}'}^\dagger b_{\mathbf{k}'} + b_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} \right) \right], a_{\mathbf{k}} \right] \\ &= \sum_{\mathbf{k}'} \omega_{\mathbf{k}'} [a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}}] + c_{\mathbf{k}'} \left([a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}}] b_{\mathbf{k}'} + b_{\mathbf{k}'}^\dagger [a_{\mathbf{k}}, a_{\mathbf{k}}] \right) \\ &= \sum_{\mathbf{k}'} \omega_{\mathbf{k}'} [a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}}] + c_{\mathbf{k}'} [a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}}] b_{\mathbf{k}'} \\ &= \sum_{\mathbf{k}'} (-\omega_{\mathbf{k}'} a_{\mathbf{k}'} + c_{\mathbf{k}'} b_{\mathbf{k}'}) \delta_{\mathbf{k}\mathbf{k}'} \\ &= -(\omega_{\mathbf{k}} a_{\mathbf{k}} + c_{\mathbf{k}} b_{\mathbf{k}}) \end{aligned}$$

Substituting we obtain

$$\omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} + \omega_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + c_{\mathbf{k}} G(b_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)$$

We need now to solve a equation of motion

$$\omega G(b_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1}{2\pi} \langle [b_{\mathbf{k}}, a_{\mathbf{k}}^\dagger] \rangle - G([H, b_{\mathbf{k}}]; a_{\mathbf{k}}^\dagger | \omega)$$

where

$$\begin{aligned} [H, b_{\mathbf{k}}] &= \left[\sum_{\mathbf{k}'} \left[\omega_{\mathbf{k}'} a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'} + \Omega_{\mathbf{k}'} b_{\mathbf{k}'}^\dagger b_{\mathbf{k}'} + c_{\mathbf{k}'} (a_{\mathbf{k}'}^\dagger b_{\mathbf{k}'} + b_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}) \right], b_{\mathbf{k}} \right] \\ &= \sum_{\mathbf{k}'} \omega_{\mathbf{k}'} [\cancel{a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}}, b_{\mathbf{k}}] + \Omega_{\mathbf{k}'} [b_{\mathbf{k}'}^\dagger b_{\mathbf{k}'}, b_{\mathbf{k}}] + c_{\mathbf{k}'} (a_{\mathbf{k}'}^\dagger [\cancel{b_{\mathbf{k}'}, b_{\mathbf{k}}}] + a_{\mathbf{k}'} [b_{\mathbf{k}'}, b_{\mathbf{k}}]) \\ &= \sum_{\mathbf{k}'} \Omega_{\mathbf{k}'} [b_{\mathbf{k}'}^\dagger, b_{\mathbf{k}}] b_{\mathbf{k}'} + c_{\mathbf{k}'} a_{\mathbf{k}'} [b_{\mathbf{k}'}, b_{\mathbf{k}}] \\ &= \sum_{\mathbf{k}'} \Omega_{\mathbf{k}'} [b_{\mathbf{k}'}^\dagger, b_{\mathbf{k}}] b_{\mathbf{k}'} + c_{\mathbf{k}'} a_{\mathbf{k}'} [b_{\mathbf{k}'}, b_{\mathbf{k}}] \\ &= -(\Omega_{\mathbf{k}} b_{\mathbf{k}} + c_{\mathbf{k}} a_{\mathbf{k}}) \delta_{\mathbf{k}, \mathbf{k}'} \end{aligned}$$

Giving

$$\omega G(b_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \Omega_{\mathbf{k}} G(b_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + c_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) \Rightarrow G(b_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{c_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)}{(\omega - \Omega_{\mathbf{k}})}$$

Substituting in the original equation of motion we have

$$\begin{aligned} \omega G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) &= \frac{1}{2\pi} + \omega_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + c_{\mathbf{k}} \frac{c_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)}{(\omega - \Omega_{\mathbf{k}})} \\ &= \frac{1}{2\pi} + \omega_{\mathbf{k}} G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) + c_{\mathbf{k}}^2 \frac{G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega)}{(\omega - \Omega_{\mathbf{k}})} \\ &\Rightarrow G(a_{\mathbf{k}}; a_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi}{\omega - \omega_{\mathbf{k}} - \frac{c_{\mathbf{k}}^2}{\omega - \Omega_{\mathbf{k}}}} \end{aligned}$$

Analogously, if we do the same treatment for the phonons we obtain

$$G(b_{\mathbf{k}}; b_{\mathbf{k}}^\dagger | \omega) = \frac{1/2\pi}{\omega - \Omega_{\mathbf{k}} - \frac{c_{\mathbf{k}}^2}{\omega - \omega_{\mathbf{k}}}} = \frac{1/2\pi(\omega - \omega_{\mathbf{k}})}{(\omega - \Omega_{\mathbf{k}})(\omega - \omega_{\mathbf{k}}) - c_{\mathbf{k}}^2}$$

The spectrum is given by the poles of the either of the GF

$$0 = (\omega - \Omega_{\mathbf{k}})(\omega - \omega_{\mathbf{k}}) - c_0^2 = \omega^2 - \omega(\Omega_{\mathbf{k}} + \omega_{\mathbf{k}}) + (\Omega_{\mathbf{k}}\omega_{\mathbf{k}} - c_{\mathbf{k}}^2)$$

$$\begin{aligned}
\Rightarrow \omega_{\pm} &= \frac{1}{2}(\Omega_{\mathbf{k}} + \omega_{\mathbf{k}}) \pm \frac{1}{2}\sqrt{(\Omega_{\mathbf{k}} + \omega_{\mathbf{k}})^2 - 4(\Omega_{\mathbf{k}}\omega_{\mathbf{k}} - c_{\mathbf{k}}^2)} \\
&= \frac{1}{2}(\Omega_{\mathbf{k}} + \omega_{\mathbf{k}}) \pm \frac{1}{2}\sqrt{(\Omega_{\mathbf{k}} - \omega_{\mathbf{k}})^2 + 4c_{\mathbf{k}}^2}
\end{aligned}$$

We can define the detuning frequency $\delta = \omega_{\mathbf{k}} - \Omega_{\mathbf{k}}$ and obtain

$$\begin{aligned}
\omega_{\pm} &= \frac{1}{2}(-\delta + 2\omega_{\mathbf{k}}) \pm \frac{1}{2}\sqrt{\delta^2 + 4c_{\mathbf{k}}^2} \\
&= \omega_{\mathbf{k}} - \frac{1}{2}\delta \pm \frac{1}{2}\sqrt{\delta^2 + 4c_{\mathbf{k}}^2}
\end{aligned}$$

For the case of an acoustic phonon (with $\Omega_{\mathbf{k}} = v|\mathbf{k}|$ where v is the sound velocity) and a magnon (approximated by $\omega_{\mathbf{k}} = \omega_0$). From this we have that $\delta = \omega_{\mathbf{k}} - \Omega_{\mathbf{k}} = \omega_0 - v|\mathbf{k}|$. The interaction term you may assume that $c_{\mathbf{k}} = c_0$ and also $c_0/\omega_0 \ll 1$.

4.11 Absence of an interaction term

In the absence of an interaction term in the Hamiltonian we would have that $c_0 = 0$ and so

$$\Rightarrow \begin{cases} \omega_{\text{phonons}}(\mathbf{k}) = v|\mathbf{k}| \\ \omega_{\text{magnons}}(\mathbf{k}) = \omega_0 \end{cases},$$

As for the stability condition for the lower branch when $c_0 \neq 0$ we have

$$\omega_- = \frac{\omega_0 + vk}{2} - \frac{1}{2}\sqrt{(\omega_0 - vk)^2 + 4c_0^2} > 0.$$

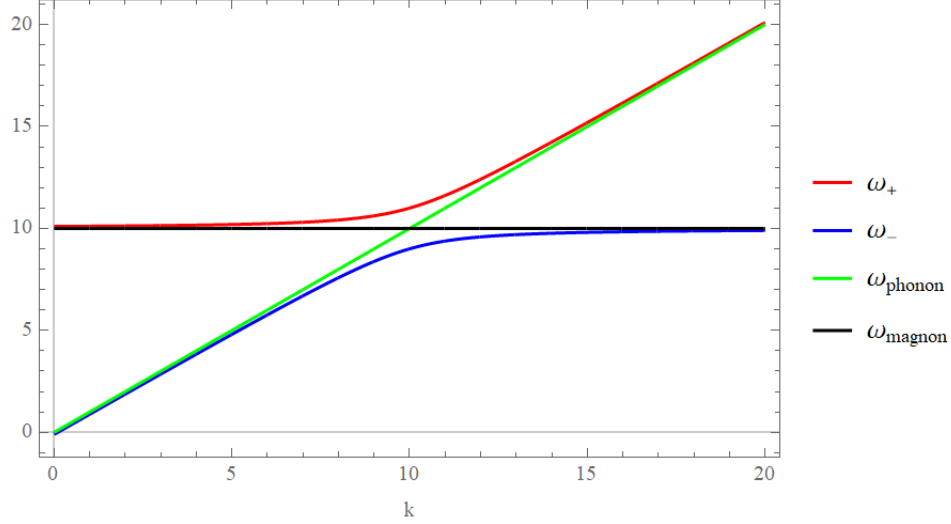
The condition for the point where this quantity becomes negative is

$$\begin{aligned}
&\frac{\omega_0 + vk}{2} - \frac{1}{2}\sqrt{(\omega_0 - vk)^2 + 4c_0^2} = 0 \\
&\Leftrightarrow \frac{(\omega_0 + vk)^2}{4} = \frac{1}{4}(\omega_0 - vk)^2 + c_0^2 \\
&\Leftrightarrow (\omega_0 + vk)^2 - (\omega_0 - vk)^2 = 4c_0^2 \\
&\Leftrightarrow 4\omega_0 vk = 4c_0^2 \\
&\Leftrightarrow k = \frac{c_0^2}{\omega_0 v}.
\end{aligned}$$

This concludes the problem, as we must have

$$k > \frac{c_0^2}{\omega_0 v}.$$

We can plot this expression with $v = c_0^2 = 1$ and $\omega_0 = 10$ as shown in the figure below. We can see that the dispersion including the interaction term has some sort of repulsion of levels along the



4.12 Paramagnet's GF

4.13 Noninteracting gas' GF

In the absence of any interaction terms, the Hamiltonian is given by

$$H = \sum_{\mathbf{k}} \frac{k^2}{2m} a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}}$$

The density operator $\rho_{\mathbf{q}}^{\dagger}(\mathbf{k})$ is defined by

$$\rho_{\mathbf{q}}^{\dagger}(\mathbf{k}) = a_{\mathbf{k}+\mathbf{q}}^{\dagger} a_{\mathbf{k}}$$

which means that, because $(AB)^{\dagger} = B^{\dagger} A^{\dagger}$, also

$$\rho_{\mathbf{q}}(\mathbf{k}) = a_{\mathbf{k}}^{\dagger} a_{\mathbf{k}+\mathbf{q}}$$

We want to solve the GF's equation of motion

$$\omega G(\rho_{\mathbf{q}}^{\dagger}(\mathbf{k}); \rho_{\mathbf{q}}(\mathbf{k}) | \omega) = \frac{1}{2\pi} \langle [\rho_{\mathbf{q}}^{\dagger}(\mathbf{k}), \rho_{\mathbf{q}}(\mathbf{k})] \rangle - G([\mathcal{H}, \rho_{\mathbf{q}}^{\dagger}(\mathbf{k})]; \rho_{\mathbf{q}}(\mathbf{k}) | \omega)$$

For this we will need to use the 4 operator commutator property in terms of the anti-commutators because they are fermionic operators

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

The first commutator then gives

$$\begin{aligned}
[\rho_{\mathbf{q}}^\dagger(\mathbf{k}), \rho_{\mathbf{q}}(\mathbf{k}')] &= [a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'+\mathbf{q}}] \\
&= -a_{\mathbf{k}+\mathbf{q}}^\dagger \overbrace{a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}'+\mathbf{q}}, a_{\mathbf{k}}\}} + a_{\mathbf{k}+\mathbf{q}}^\dagger \{a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}}\} a_{\mathbf{k}'+\mathbf{q}} - a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}'+\mathbf{q}}, a_{\mathbf{k}+\mathbf{q}}^\dagger\} a_{\mathbf{k}} + \overbrace{\{a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}+\mathbf{q}}^\dagger\} a_{\mathbf{k}'+\mathbf{q}} a_{\mathbf{k}}} \\
&= a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} - a_{\mathbf{k}'}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}} \\
&= a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} - a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}}
\end{aligned}$$

Where we used that, because it is a anti-commutator, $\{A, B\} = \{B, A\}$. Notice also that q ????. This means that

$$\left\langle [\rho_{\mathbf{k}}^\dagger(\mathbf{k}), \rho_{\mathbf{k}}(\mathbf{k})] \right\rangle = \left\langle a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} \right\rangle - \left\langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}} \right\rangle = n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0$$

The second one gives

$$\begin{aligned}
[\mathcal{H}(\mathbf{k}'), \rho_{\mathbf{q}}^\dagger(\mathbf{k})] &= \sum_{\mathbf{k}'} \frac{k'^2}{2m} [a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}^\dagger, a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}] \\
&= \sum_{\mathbf{k}'} \frac{k'^2}{2m} \left(-a_{\mathbf{k}'}^\dagger a_{\mathbf{k}+\mathbf{q}}^\dagger \overbrace{\{a_{\mathbf{k}}, a_{\mathbf{k}'}\}} + a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}+\mathbf{q}}^\dagger, a_{\mathbf{k}}\} a_{\mathbf{k}} - a_{\mathbf{k}+\mathbf{q}}^\dagger \{a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger\} a_{\mathbf{k}'} + \overbrace{\{a_{\mathbf{k}+\mathbf{q}}^\dagger, a_{\mathbf{k}'}^\dagger\} a_{\mathbf{k}} a_{\mathbf{k}'} \right) \\
&= \sum_{\mathbf{k}} \frac{k'^2}{2m} \left(a_{\mathbf{k}'}^\dagger (\delta_{\mathbf{k}+\mathbf{q}, \mathbf{k}'} a_{\mathbf{k}} - a_{\mathbf{k}+\mathbf{q}}^\dagger (\delta_{\mathbf{k}, \mathbf{k}'} a_{\mathbf{k}'} \right) \\
&= \frac{(k+q)^2}{2m} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}} - \frac{k^2}{2m} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}} \\
&= (E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}}) \rho_{\mathbf{q}}^\dagger(\mathbf{k})
\end{aligned}$$

The equation of motion is then

$$\begin{aligned}
\omega G(\rho_{\mathbf{k}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) &= \frac{1}{2\pi} (n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0) - (E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}}) G(\rho_{\mathbf{q}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) \\
&\Leftrightarrow G(\rho_{\mathbf{k}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) = \frac{1}{2\pi} \frac{n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0}{\omega + (E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}})} \\
&\Rightarrow G_r(\rho_{\mathbf{k}}^\dagger; \rho_{\mathbf{k}}|\omega) = \frac{1}{2\pi} \sum_{\mathbf{k}} \frac{n_{\mathbf{k}+\mathbf{q}}^0 + n_{\mathbf{k}}^0}{E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}} + \omega + i0^+}
\end{aligned}$$

5 Green's Functions Using Decoupling Methods

5.1 Commutation relation from the exchange-dominated ferromagnet hamiltonean

The Heisenberg Hamiltonian in terms of bosons operators is given by

$$H = -\frac{1}{2} \sum_{i,j} J_{i,j} \left\{ \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) + S_i^z S_j^z \right\} - g\mu_B B_0 \sum_i S_i^z$$

We need to calculate

$$\begin{aligned} [H, S_n^+] &= \left[-\frac{1}{2} \sum_{i,j} J_{i,j} \left\{ \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) + S_i^z S_j^z \right\} - g\mu_B B_0 \sum_i S_i^z, S_n^+ \right] \\ &= -\frac{1}{4} \sum_{i,j} J_{i,j} [S_i^+ S_j^-, S_n^+] - \frac{1}{4} \sum_{i,j} J_{i,j} [S_i^- S_j^+, S_n^+] - \frac{1}{2} \sum_{i,j} J_{i,j} [S_i^z S_j^z, S_n^+] - g\mu_B B_0 \sum_i [S_i^z, S_n^+] \end{aligned}$$

Using the commutator relation $[AB, C] = A[B, C] + [A, C]B$ and the spin commutation relations

$$\begin{cases} [S_i^+, S_j^-] = 2S_i^z \delta_{i,j} \\ [S_i^z, S_j^\pm] = \pm S_j^\pm \delta_{i,j} \end{cases}$$

we obtain

$$\begin{aligned} &= -\frac{1}{4} \sum_{i,j} J_{i,j} S_i^+ [S_j^-, S_n^+] - \frac{1}{4} \sum_{i,j} J_{i,j} [S_i^-, S_n^+] S_j^+ - \frac{1}{2} \sum_{i,j} J_{i,j} (S_i^z [S_j^z, S_n^+] + [S_i^z, S_n^+] S_j^z) - g\mu_B B_0 \sum_i (+S_i^z) \\ &= -\frac{1}{4} \sum_{i,j} J_{i,j} S_i^+ (-2S_n^z \delta_{j,n}) - \frac{1}{4} \sum_{i,j} J_{i,j} (-2S_n^z \delta_{i,n}) S_j^+ - \frac{1}{2} \sum_{i,j} J_{i,j} (S_i^z (+S_n^+ \delta_{n,j}) + (+S_n^+ \delta_{n,i}) S_j^z) - g\mu_B B_0 S_n^+ \\ &= \frac{1}{2} \sum_i J_{i,n} S_i^+ S_n^z + \frac{1}{2} \sum_i J_{n,i} S_n^z S_i^+ - \frac{1}{2} \sum_i J_{i,n} S_i^z S_n^+ - \frac{1}{2} \sum_i J_{n,i} S_n^+ S_i^z - g\mu_B B_0 S_n^+ \end{aligned}$$

Because the interaction term is symmetric $J_{i,n} = J_{n,i}$ we have

$$\begin{aligned} &= \frac{1}{2} \sum_i J_{i,n} (S_i^+ S_n^z + S_n^z S_i^+) - \frac{1}{2} \sum_i J_{i,n} (S_i^z S_n^+ + S_n^+ S_i^z) - g\mu_B B_0 S_n^+ \\ &= \frac{1}{2} \sum_i J_{i,n} (\{-S_i^+ \delta_{i,n} + S_n^z S_i^+\} + S_n^z S_i^+) - \frac{1}{2} \sum_i J_{i,n} (S_i^z S_n^+ + \{-S_n^+ \delta_{i,n} + S_n^z S_i^+\}) - g\mu_B B_0 S_n^+ \\ &= \sum_i J_{i,n} S_n^z S_i^+ - \frac{1}{2} \cancel{S_n^+} - \sum_i J_{i,n} S_n^z S_i^+ + \frac{1}{2} \cancel{S_n^+} - g\mu_B B_0 S_n^+ \\ &= \sum_i J_{i,n} (S_n^z S_i^+ - S_n^z S_i^+) - g\mu_B B_0 S_n^+ \end{aligned}$$

5.2 Curie Temperature

Starting from the equation of state

$$\frac{1}{2} - \langle S^z \rangle = 2 \langle S^z \rangle \frac{1}{N} \sum_{\mathbf{k}} \frac{1}{e^{\beta E_{\mathbf{k}}} - 1}$$

Introducing a scaled magnetic susceptibility χ defined by

$$\chi = \lim_{b \rightarrow 0} \frac{\langle S^z \rangle}{b}$$

where for convenience we denote $b = g\mu_B B_0$. Because this is a zero-field susceptibility, the preceding definition of χ is equivalent to taking a derivative with respect to b . Also the energy $E_{\mathbf{k}}$ is given by

$$E_{\mathbf{k}} = g\mu_B B_0 + \langle S^z \rangle \{J(0) - J(\mathbf{k})\}$$

The equation above then becomes

$$\frac{1}{2} - \chi b = \frac{1}{N} \sum_{\mathbf{k}} \frac{2\chi b}{e^{\beta b + \beta \chi b \{J(0) - J(\mathbf{k})\}} - 1}$$

For small enough b the denominator can employ a power-series expansion of the exponential series expansion to yield

$$\begin{aligned} \frac{1}{2} &= \frac{1}{N} \sum_{\mathbf{k}} \frac{2\chi b}{\beta b + \beta \chi b \{J(0) - J(\mathbf{k})\}} \\ &= \frac{1}{N} \sum_{\mathbf{k}} \frac{2b}{\frac{1}{\chi} \beta b + \beta b \{J(0) - J(\mathbf{k})\}} \end{aligned}$$

Applying the condition for χ to diverge as the temperature T approaches T_C we get

$$\frac{1}{2} = \frac{1}{N} \sum_{\mathbf{k}} \frac{2\cancel{b}}{\beta_C \cancel{b} \{J(0) - J(\mathbf{k})\}} = \frac{1}{N} \frac{2}{k_B T_C} \sum_{\mathbf{k}} \frac{1}{\{J(0) - J(\mathbf{k})\}}$$

Hence the final result is

$$T_C = \frac{1}{4k_B} \left[\frac{1}{N} \sum_{\mathbf{k}} \frac{1}{J(0) - J(\mathbf{k})} \right]^{-1}$$

5.3 Hyperfine Heisenberg Ferromagnet

5.4 RPA Heisenberg Antiferromagnet with $B_0 = 0$

The four coupled equations of motion for a Heisenberg antiferromagnet in zero applied field ($B_0 = 0$) after RPA decoupling are

$$\begin{cases} \{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AA}(\omega) - \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{BA}(\omega) = \frac{\langle S^z \rangle}{\pi} \\ \{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AB}(\omega) - \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{BB}(\omega) = 0 \\ \{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{BB}(\omega) + \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{AB}(\omega) = -\frac{\langle S^z \rangle}{\pi} \\ \{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{BA}(\omega) + \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{AA}(\omega) = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} \{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AA}(\omega) - \langle S^z \rangle J(\mathbf{k}) \left(\frac{-\langle S^z \rangle J(\mathbf{k})}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} \right) G_{\mathbf{k}}^{AA}(\omega) = \frac{\langle S^z \rangle}{\pi}, \\ \{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AB}(\omega) - \langle S^z \rangle J(\mathbf{k}) \left(\frac{-\langle S^z \rangle 1/\pi - \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{AB}(\omega)}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} \right) = 0 \\ G_{\mathbf{k}}^{BB}(\omega) = \frac{-\langle S^z \rangle 1/\pi - \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{AB}(\omega)}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} \\ G_{\mathbf{k}}^{BA}(\omega) = \frac{-\langle S^z \rangle J(\mathbf{k})}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} G_{\mathbf{k}}^{AA}(\omega) \end{cases}$$

Firstly we solve for $G_{\mathbf{k}}^{AA}(\omega)$, we obtain

$$\begin{aligned} G_{\mathbf{k}}^{AA}(\omega) &= \frac{\langle S^z \rangle / \pi}{\{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} + \left(\frac{\{\langle S^z \rangle J(\mathbf{k})\}^2}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} \right)} \\ &= \frac{\langle S^z \rangle / \pi \{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}}{\{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} \{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\} + \{\langle S^z \rangle J(\mathbf{k})\}^2} \\ &= \frac{\langle S^z \rangle \{g\mu_B B_{an} + \langle S^z \rangle J(0) + \omega\}}{\omega^2 - \{g\mu_B B_{an} - \langle S^z \rangle J(0)\}^2 + \{\langle S^z \rangle J(\mathbf{k})\}^2} \\ &= \frac{\langle S^z \rangle \{g\mu_B B_{an} + \langle S^z \rangle J(0) + \omega\}}{\omega^2 - \left(\{g\mu_B B_{an} - \langle S^z \rangle J(0)\}^2 - \{\langle S^z \rangle J(\mathbf{k})\}^2 \right)} \end{aligned}$$

If we define the energy

$$E_{\mathbf{k}}^2 = +\{g\mu_B B_{an} - \langle S^z \rangle J(0)\}^2 - \{\langle S^z \rangle J(\mathbf{k})\}^2$$

and analitically continue ω to a complex frequency $\tilde{\omega}$ we obtain

$$G_{\mathbf{k}}^{AA}(\tilde{\omega}) = \frac{\langle S^z \rangle \{g\mu_B B_{an} + \langle S^z \rangle J(0) + \tilde{\omega}\}}{\pi [\tilde{\omega}^2 - E_{\mathbf{k}}^2]}$$

Now we solve for $G_{\mathbf{k}}^{AB}(\omega)$, we get

$$\{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AB}(\omega) - \langle S^z \rangle J(\mathbf{k}) \left(\frac{-\langle S^z \rangle 1/\pi - \langle S^z \rangle J(\mathbf{k}) G_{\mathbf{k}}^{AB}(\omega)}{\{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\}} \right) = 0$$

$$\Leftrightarrow \{\omega - g\mu_B B_{an} - \langle S^z \rangle J(0)\} \{\omega + g\mu_B B_{an} + \langle S^z \rangle J(0)\} G_{\mathbf{k}}^{AB}(\omega) + \langle S^z \rangle^2 1/\pi J(\mathbf{k}) + \{\langle S^z \rangle J(\mathbf{k})\}^2 G_{\mathbf{k}}^{AB}(\omega) = 0$$

$$\Leftrightarrow \left(\omega^2 - \{g\mu_B B_{an} - \langle S^z \rangle J(0)\}^2 + \{\langle S^z \rangle J(\mathbf{k})\}^2 \right) G_{\mathbf{k}}^{AB}(\omega) + \langle S^z \rangle^2 1/\pi J(\mathbf{k}) = 0$$

$$\Leftrightarrow \left(\omega^2 - \left(\{g\mu_B B_{an} - \langle S^z \rangle J(0)\}^2 - \{\langle S^z \rangle J(\mathbf{k})\}^2 \right) \right) G_{\mathbf{k}}^{AB}(\omega) = \frac{-\langle S^z \rangle^2 J(\mathbf{k})}{\pi}$$

$$\Rightarrow G_{\mathbf{k}}^{AB}(\tilde{\omega}) = \frac{-\langle S^z \rangle^2 J(\mathbf{k})}{\pi [\tilde{\omega}^2 - E_{\mathbf{k}}^2]}$$

5.5 Dispersion relation for the electronic bands in a 3D s.c solid

As shown in the figure a simple cubic (s.c) solid is characterized with the basic lattice vectors

$$\mathbf{a}_1 = a\hat{\mathbf{x}}, \quad \mathbf{a}_2 = a\hat{\mathbf{y}}, \quad \text{and} \quad \mathbf{a}_3 = a\hat{\mathbf{z}}$$

We can construct vectors the reciprocal lattice via the translation symmetry of, for example, the electronic density n i.e

$$n(\mathbf{r}) = n(\mathbf{r} + \mathbf{R})$$

where $\mathbf{R} = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + u_3\mathbf{a}_3$ is the translation vector and u_1, u_2 and u_3 are integers. We can use a Fourier series to express this symmetry as

$$n(\mathbf{r}) = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{-i\mathbf{G} \cdot \mathbf{r}}$$

where the summation in $\mathbf{G}$ is over k_x, k_y and k_z . We must find the reciprocal lattice vectors $\mathbf{G} = v_1\mathbf{b}_1 + v_2\mathbf{b}_2 + v_3\mathbf{b}_3$. From the symmetry relation above we have

$$\begin{aligned} n(\mathbf{r}) &= n(\mathbf{r} + \mathbf{R}) \\ \Leftrightarrow \sum_{\mathbf{G}} n_{\mathbf{G}} e^{-i\mathbf{G} \cdot \mathbf{r}} &= \sum_{\mathbf{G}} n_{\mathbf{G}} e^{-i\mathbf{G} \cdot (\mathbf{r} + \mathbf{R})} = \sum_{\mathbf{G}} n_{\mathbf{G}} e^{-i\mathbf{G} \cdot \mathbf{r}} e^{-i\mathbf{G} \cdot \mathbf{R}} \\ \Rightarrow e^{-i\mathbf{G} \cdot \mathbf{R}} &= 1 \Rightarrow \mathbf{G} \cdot \mathbf{R} = 2\pi \Rightarrow \mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{i,j} \end{aligned}$$

This can be achieved if we define the reciprocal lattice vectors as

$$\mathbf{b}_i = 2\pi \frac{\varepsilon_{ijk} a_j a_k \hat{\mathbf{i}}}{a_1 \times (a_2 \cdot a_3)}$$

For this particular case we then have

$$\mathbf{b}_i = 2\pi \frac{a^2}{a^3} \hat{\mathbf{i}} = \frac{2\pi}{a} \hat{\mathbf{i}}$$

We obtain our dispersion relation we first need to write our Hamiltonian in second quantization. If we only allow hopping between next-nearest-neighbours (N-N-N) we will have three hopping possibilities which we can describe as

$$H = \frac{1}{2} \sum_{\langle i,j \rangle} t'_{i,j} (c_i^\dagger c_j + c_j^\dagger c_i) = -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i)$$

Here the $t'_{i,j}$ is the hopping parameter where, given the simple cubic symmetry, we can assume is the same in every direction (the extra minus sign is to guarantee stability, a negative hopping term in the Hamiltonian favors it from happening) and $\langle i,j \rangle$ refer to the N-N-N sites. The annihilation operator c and creation

operator $c^\dagger$ translates in second quantization the idea of an electron hopping from one site i (annihilation c_i) to another site j (creation $c_j^\dagger$). It's better to express the Hamiltonian in reciprocal space, for this we make the Fourier transform

$$c_i = c(\mathbf{r}_i) = \sum_{\mathbf{k}} c_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i}$$

where the summation in $\mathbf{k}$ is in k_x, k_y and k_z . Substituting we obtain

$$\begin{aligned} H &= -t \sum_{\langle i,j \rangle} (c_i^\dagger c_j + c_j^\dagger c_i) \\ &= -t \sum_{\langle i,j \rangle} \sum_{\mathbf{k}, \mathbf{k}'} \left(c_{\mathbf{k}}^\dagger e^{+i\mathbf{k} \cdot \mathbf{r}_i} c_{\mathbf{k}'} e^{-i\mathbf{k}' \cdot \mathbf{r}_j} + c_{\mathbf{k}}^\dagger e^{+i\mathbf{k} \cdot \mathbf{r}_j} c_{\mathbf{k}'} e^{-i\mathbf{k}' \cdot \mathbf{r}_i} \right) \end{aligned}$$

We can always define the position vector $\mathbf{r}_j$ in terms of $\mathbf{r}_i$ by doing a translation inside the unit cell by $\mathbf{a}_j$ i.e

$$\mathbf{r}_j = \mathbf{r}_i + \mathbf{a}_j$$

Substituting we obtain

$$\begin{aligned} &= -t \sum_{\langle i,j \rangle} \sum_{\mathbf{k}, \mathbf{k}'} \left(c_{\mathbf{k}}^\dagger e^{+i\mathbf{k} \cdot \mathbf{r}_i} c_{\mathbf{k}'} e^{-i\mathbf{k}' \cdot (\mathbf{r}_i + \mathbf{a}_j)} + c_{\mathbf{k}}^\dagger e^{+i\mathbf{k} \cdot (\mathbf{r}_i + \mathbf{a}_j)} c_{\mathbf{k}'} e^{-i\mathbf{k}' \cdot \mathbf{r}_i} \right) \\ &= -t \sum_{\langle i,j \rangle} \sum_{\mathbf{k}, \mathbf{k}'} e^{+i\mathbf{r}_i \cdot (\mathbf{k} - \mathbf{k}')} \left(e^{-i\mathbf{k}' \cdot \mathbf{a}_j} + e^{+i\mathbf{k} \cdot \mathbf{a}_j} \right) c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \end{aligned}$$

Using the Dirac delta Fourier series

$$\delta_{\mathbf{k}, \mathbf{k}'} = \left(\frac{a}{2\pi} \right)^3 \sum_i e^{+i\mathbf{r}_i \cdot (\mathbf{k} - \mathbf{k}')}$$

we let off the summation in $\mathbf{k}'$, we get

$$\begin{aligned} H &= -t \left(\frac{2\pi}{a} \right)^3 \sum_{\mathbf{k}'} \delta_{\mathbf{k}, \mathbf{k}'} \sum_{\mathbf{k}} \left(e^{-i\mathbf{k}' \cdot \mathbf{a}_j} + e^{+i\mathbf{k} \cdot \mathbf{a}_j} \right) c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \\ &= -t \left(\frac{2\pi}{a} \right)^3 \sum_{\mathbf{k}} \left(e^{-i\mathbf{k} \cdot \mathbf{a}_j} + e^{+i\mathbf{k} \cdot \mathbf{a}_j} \right) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \\ &= -2t \left(\frac{2\pi}{a} \right)^3 \sum_{\mathbf{k}} \cos(\mathbf{k} \cdot \mathbf{a}_j) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \\ &= -2t \left(\frac{2\pi}{a} \right)^3 \sum_{k_x, k_y, k_z} (\cos(k_x a) + \cos(k_y a) + \cos(k_z a)) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \end{aligned}$$

5.6 Ladder spin operator

The fermion ladder operators commutators give

$$\begin{cases} [S_i^+, S_j^-] = 2S_i^z \delta_{i,j} \\ [S_j^z, S_i^\pm] = \pm S_i^\pm \delta_{i,j} \end{cases}$$

Lets see if this holds for the choice of

$$\begin{cases} S_i^+ = a_{i,\uparrow}^\dagger a_{i,\downarrow} \\ S_i^- = a_{i,\downarrow}^\dagger a_{i,\uparrow} \\ S_i^z = \frac{1}{2} (a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow}) \end{cases}$$

5.6.1 Proof for $[S_i^+, S_j^-] = 2S_i^z \delta_{i,j}$

Substituting directly we obtain

$$[S_i^+, S_j^-] = [a_{i,\uparrow}^\dagger a_{i,\downarrow}, a_{j,\downarrow}^\dagger a_{j,\uparrow}]$$

Using the already proven formula

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

and substituting we obtain

$$[S_i^+, S_j^-] = -a_{i,\uparrow}^\dagger a_{j,\downarrow}^\dagger \{a_{j,\uparrow}, a_{i,\downarrow}\} + a_{i,\uparrow}^\dagger \{a_{j,\downarrow}^\dagger, a_{i,\downarrow}\} a_{j,\uparrow} - a_{j,\downarrow}^\dagger \{a_{j,\uparrow}, a_{i,\uparrow}^\dagger\} a_{i,\downarrow} + \{a_{j,\downarrow}^\dagger, a_{i,\uparrow}^\dagger\} a_{j,\uparrow} a_{i,\downarrow}$$

Because this is electronic (fermionic) operators we have that

$$\{a_{i,\sigma}, a_{j,\sigma'}^\dagger\} = \{a_{j,\sigma'}^\dagger, a_{i,\sigma}\} = \delta_{\sigma,\sigma'} \delta_{i,j}$$

where

$$\delta_{\sigma,\sigma'} = \begin{cases} 1 & \text{if } \sigma = \sigma' \\ 0 & \text{if } \sigma \neq \sigma' \end{cases}$$

This means that

$$\begin{aligned} [S_i^+, S_j^-] &= a_{i,\uparrow}^\dagger (\delta_{\downarrow\downarrow} \delta_{i,j}) a_{j,\uparrow} - a_{j,\downarrow}^\dagger (\delta_{\uparrow\uparrow} \delta_{i,j}) a_{i,\downarrow} \\ &= (a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow}) \delta_{i,j} \\ &= 2S_i^z \delta_{i,j} \end{aligned}$$

5.6.2 Proof for $[S_i^z, S_j^+] = +S_j^+ \delta_{i,j}$

Substituting directly we obtain

$$[S_i^z, S_j^+] = \left[\frac{1}{2} (a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow}), a_{j,\uparrow}^\dagger a_{j,\downarrow} \right]$$

Again, using the already proven formula

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

and substituting we obtain

$$\begin{aligned}
[S_i^z, S_j^\pm] &= \frac{1}{2} \left(-a_{i,\uparrow}^\dagger a_{j,\uparrow}^\dagger \cancel{\{a_{j,\downarrow}, a_{i,\uparrow}\}} + a_{i,\uparrow}^\dagger \{a_{j,\uparrow}^\dagger, a_{i,\uparrow}\} a_{j,\downarrow} - a_{j,\uparrow}^\dagger \cancel{\{a_{j,\downarrow}, a_{i,\uparrow}^\dagger\}} a_{i,\uparrow} + \cancel{\{a_{j,\uparrow}^\dagger, a_{i,\uparrow}^\dagger\}} a_{j,\downarrow} a_{i,\uparrow} \right) \\
&\quad - \frac{1}{2} \left(-a_{i,\downarrow}^\dagger a_{j,\uparrow}^\dagger \cancel{\{a_{j,\downarrow}, a_{i,\downarrow}\}} + a_{i,\downarrow}^\dagger \cancel{\{a_{j,\uparrow}^\dagger, a_{i,\downarrow}\}} a_{j,\downarrow} - a_{j,\uparrow}^\dagger \{a_{j,\downarrow}, a_{i,\downarrow}^\dagger\} a_{i,\downarrow} + \cancel{\{a_{j,\uparrow}^\dagger, a_{i,\downarrow}^\dagger\}} a_{j,\downarrow} a_{i,\downarrow} \right) \\
&= \frac{1}{2} \left(a_{i,\uparrow}^\dagger \{a_{j,\uparrow}^\dagger, a_{i,\uparrow}\} a_{j,\downarrow} + a_{j,\uparrow}^\dagger \{a_{j,\downarrow}, a_{i,\downarrow}^\dagger\} a_{i,\downarrow} \right) \\
&= \frac{1}{2} \left(a_{i,\uparrow}^\dagger (\delta_{i,j} \delta_{\uparrow\uparrow}) a_{j,\downarrow} + a_{j,\uparrow}^\dagger (\delta_{i,j} \delta_{\downarrow\downarrow}) a_{i,\downarrow} \right) \\
&= \frac{1}{2} \left(a_{i,\uparrow}^\dagger a_{i,\downarrow} + a_{i,\uparrow}^\dagger a_{i,\downarrow} \right) \delta_{i,j} \\
&= S_i^+ \delta_{i,j}
\end{aligned}$$

5.6.3 Proof for $[S_i^z, S_j^-] = -S_j^- \delta_{i,j}$

Its analogous to the above.

5.6.4 Proof that the Hamiltoneans are the same

5.6.5 Expansion of the second Hamiltonean

Let's calculate the term

$$\mathbf{S}_i \cdot \mathbf{S}_j = S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z$$

where

$$\begin{cases} S^+ = S^x + iS^y \\ S^- = S^x - iS^y \end{cases} \Leftrightarrow \begin{cases} S^x = S^+ - iS^y = S^+ + S^- - S^x \\ S^y = i(S^- - S^x) = i(S^- - \frac{1}{2}(S^+ + S^-)) \end{cases}$$

or better the other way around

$$\Leftrightarrow \begin{cases} S^x = \frac{1}{2}(S^+ + S^-) \\ S^y = \frac{1}{2i}(S^+ - S^-) \end{cases}$$

Substituting we obtain

$$\begin{aligned}
\mathbf{S}_i \cdot \mathbf{S}_j &= S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z \\
&= \frac{1}{2} (S_i^+ + S_i^-) \frac{1}{2} (S_j^+ + S_j^-) + \frac{1}{2i} (S_i^+ - S_i^-) \frac{1}{2i} (S_j^+ - S_j^-) + S_i^z S_j^z \\
&= \frac{1}{4} \left(\cancel{S_i^+ S_j^+} + S_i^+ S_j^- + S_i^- S_j^+ - \cancel{S_i^- S_j^-} \right) - \frac{1}{4} \left(\cancel{S_i^+ S_j^+} - S_i^+ S_j^- - S_i^- S_j^+ - \cancel{S_i^- S_j^-} \right) + S_i^z S_j^z \\
&= \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) + S_i^z S_j^z
\end{aligned}$$

The wanted Hamiltonian then gives

$$\begin{aligned} H^{(2)} &= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left\{ \mathbf{S}_i \cdot \mathbf{S}_j - \frac{1}{4} \right\} = \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left\{ \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) + S_i^z S_j^z - \frac{1}{4} \right\} \\ &= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left\{ - (S_i^+ S_j^- + S_i^- S_j^+) - 2S_i^z S_j^z + \frac{1}{2} \right\} \end{aligned}$$

5.6.6 Expansion of the second Hamiltonian

For future query we rewrite again the second quantization expression of the spins

$$\begin{cases} S_i^+ = a_{i,\uparrow}^\dagger a_{i,\downarrow} \\ S_i^- = a_{i,\downarrow}^\dagger a_{i,\uparrow} \\ S_i^z = \frac{1}{2} (a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow}) \end{cases}$$

The Hubbard Hamiltonian is given by

$$\begin{aligned} H^{(2)} &= -\frac{2t^2}{U} \sum_{\langle i,j \rangle} \sum_{\sigma'} \sum_{\sigma} a_{i,\sigma'}^\dagger a_{j,\sigma'} a_{j,\sigma}^\dagger a_{i,\sigma} \\ &= -\frac{2t^2}{U} \sum_{\langle i,j \rangle} \sum_{\sigma'} \left(a_{i,\sigma'}^\dagger a_{j,\sigma'} a_{j,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\sigma'}^\dagger a_{j,\sigma'} a_{j,\downarrow}^\dagger a_{i,\downarrow} \right) \\ &= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left(a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} \right) \end{aligned}$$

Our mission is to commute the operators in the same site together but also in the order $a^\dagger a$ so we can reconstruct in terms of the spin operators

$$\begin{aligned} &a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} \\ &= a_{i,\uparrow}^\dagger a_{j,\uparrow} (\delta_{i,j} - a_{i,\uparrow} a_{j,\uparrow}^\dagger) + a_{i,\downarrow}^\dagger a_{j,\downarrow} (\delta_{i,j} - a_{i,\downarrow} a_{j,\downarrow}^\dagger) + a_{i,\uparrow}^\dagger a_{j,\uparrow} (\delta_{i,j} - a_{i,\downarrow} a_{j,\downarrow}^\dagger) + a_{i,\downarrow}^\dagger a_{j,\downarrow} (\delta_{i,j} - a_{i,\uparrow} a_{j,\uparrow}^\dagger) \end{aligned}$$

We regroup the terms with $\delta_{i,j}$ and substitute j for i leaving us with

$$= 2\delta_{i,j} (a_{i,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow}) - (a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{i,\uparrow} a_{j,\uparrow}^\dagger + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{i,\downarrow} a_{j,\downarrow}^\dagger + a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{i,\downarrow} a_{j,\downarrow}^\dagger + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{i,\uparrow} a_{j,\uparrow}^\dagger)$$

Because in the Hubbard model we work with a lattice of spin, in each site we can only have one electron with a given spin, this means that $(a_{i,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow}) = 1$. Also, if we look closely, the two operators on the middle in each term always anticommute because they are all a 's. We can switch them as long we put a global minus sign in that term

$$= 2\delta_{i,j} + (a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow} a_{j,\uparrow}^\dagger + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow} a_{j,\downarrow}^\dagger + a_{i,\uparrow}^\dagger a_{i,\downarrow} a_{j,\downarrow} a_{j,\uparrow}^\dagger + a_{i,\downarrow}^\dagger a_{i,\uparrow} a_{j,\uparrow} a_{j,\downarrow}^\dagger)$$

Now we just have to commute the a_j with the $a_j^\dagger$ to get the desire form. Notice that some of them have opposite spins and so they anticommute because $\delta_{\uparrow\downarrow} = \delta_{\downarrow\uparrow} = 0$. We get

$$\begin{aligned}
&= 2\delta_{i,j} + \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} \left(1 - a_{j,\uparrow}^\dagger a_{j,\uparrow} \right) + a_{i,\downarrow}^\dagger a_{i,\uparrow} \left(-a_{j,\uparrow}^\dagger a_{j,\downarrow} \right) + a_{i,\uparrow}^\dagger a_{i,\downarrow} \left(-a_{j,\downarrow}^\dagger a_{j,\uparrow} \right) + a_{i,\downarrow}^\dagger a_{i,\downarrow} \left(1 - a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \right) \\
&= 2\delta_{i,j} + \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} \right) + \left(-a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\downarrow} - a_{i,\uparrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= 2\delta_{i,j} + 1 - \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\downarrow} + a_{i,\uparrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= 2\delta_{i,j} + 1 - \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + S_i^- S_j^+ + S_i^+ S_j^- + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= 2\delta_{i,j} + 1 - \left(S_i^- S_j^+ + S_i^+ S_j^- \right) - \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \right)
\end{aligned}$$

Let's, for now, focus only to the last term. On other hand the product $S_i^z S_j^z$ gives

$$\begin{aligned}
S_i^z S_j^z &= \frac{1}{4} \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow} \right) \left(a_{j,\uparrow}^\dagger a_{j,\uparrow} - a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= \frac{1}{4} \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} - a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} \right)
\end{aligned}$$

Meaning that

$$a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} = 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\uparrow}^\dagger a_{j,\uparrow}$$

We can try subtract and add terms like

$$\begin{aligned}
& a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} + \left(+a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} - a_{i,\uparrow}^\dagger a_{i,\uparrow} a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) + a_{i,\downarrow}^\dagger a_{i,\downarrow} a_{j,\uparrow}^\dagger a_{j,\uparrow} \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} \left(a_{j,\downarrow}^\dagger a_{j,\downarrow} + a_{j,\uparrow}^\dagger a_{j,\uparrow} \right) + \left(-a_{i,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{i,\downarrow} \right) a_{j,\uparrow}^\dagger a_{j,\uparrow} \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} - 2S_i^z a_{j,\uparrow}^\dagger a_{j,\uparrow} + \left(S_i^z a_{j,\downarrow}^\dagger a_{j,\downarrow} - S_i^z a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} - S_i^z a_{j,\uparrow}^\dagger a_{j,\uparrow} - S_i^z \left(a_{j,\uparrow}^\dagger a_{j,\uparrow} - a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) - S_i^z a_{j,\downarrow}^\dagger a_{j,\downarrow} \\
&= 4S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} - S_i^z a_{j,\uparrow}^\dagger a_{j,\uparrow} - 2S_i^z S_j^z - S_i^z a_{j,\downarrow}^\dagger a_{j,\downarrow} \\
&= 2S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} - S_i^z a_{j,\uparrow}^\dagger a_{j,\uparrow} - S_i^z a_{j,\downarrow}^\dagger a_{j,\downarrow} \\
&= 2S_i^z S_j^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} - S_i^z \left(a_{j,\uparrow}^\dagger a_{j,\uparrow} + a_{j,\downarrow}^\dagger a_{j,\downarrow} \right) \\
&= 2S_i^z S_j^z - S_i^z + a_{i,\uparrow}^\dagger a_{i,\uparrow} \\
&= 2S_i^z S_j^z - \frac{1}{2} \left(a_{i,\uparrow}^\dagger a_{i,\uparrow} - a_{i,\downarrow}^\dagger a_{i,\downarrow} \right) + a_{i,\uparrow}^\dagger a_{i,\uparrow} \\
&= 2S_i^z S_j^z + \frac{1}{2} a_{i,\downarrow}^\dagger a_{i,\downarrow} + \frac{1}{2} a_{i,\uparrow}^\dagger a_{i,\uparrow} \\
&= 2S_i^z S_j^z + \frac{1}{2} \left(a_{i,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\uparrow}^\dagger a_{i,\uparrow} \right) \\
&= 2S_i^z S_j^z + \frac{1}{2}
\end{aligned}$$

Putting all the terms together we obtain

$$\begin{aligned}
H^{(2)} &= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left(a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\uparrow}^\dagger a_{j,\uparrow} a_{j,\downarrow}^\dagger a_{i,\downarrow} + a_{i,\downarrow}^\dagger a_{j,\downarrow} a_{j,\uparrow}^\dagger a_{i,\uparrow} \right) \\
&= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left(2\delta_{i,j} + 1 - (S_i^- S_j^+ + S_i^+ S_j^-) - \left(2S_i^z S_j^z + \frac{1}{2} \right) \right) \\
&= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left(- (S_i^- S_j^+ + S_i^+ S_j^-) - 2S_i^z S_j^z + \frac{1}{2} \right) - \frac{4t^2}{U} \sum_{\langle i,j \rangle} \delta_{i,j} \\
&= \frac{4t^2}{U} \sum_{\langle i,j \rangle} \left(-\frac{1}{2} \right) \left(- (S_i^- S_j^+ + S_i^+ S_j^-) - 2S_i^z S_j^z + \frac{1}{2} \right)
\end{aligned}$$

Where, because we are only doing first neighbours in the summation we have necessarily that $i \neq j$ meaning that the last summation is zero.

5.7 Spectral function of the Anderson model with $V_{\mathbf{k}} = 0$ but $U \neq 0$

The total Anderson Hamiltonian is given by

$$H = H_0 + H_d + H_{sd} + H_U$$

$$= \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + \sum_{\sigma} \epsilon_d d_{\sigma}^{\dagger} d_{\sigma} + \sum_{\mathbf{k}, \sigma} \left(V_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} d_{\sigma} + V_{\mathbf{k}}^* a_{\mathbf{k}, \sigma} d_{\sigma}^{\dagger} \right) + U n_{d, \uparrow} n_{d, \downarrow}$$

If we include the onsite Coulomb repulsion term ($U \neq 0$) but not the interaction between the conducting electrons and the localized states ($V_{\mathbf{k}} = 0 = V_{\mathbf{k}}^*$) our Hamiltonian reduced to

$$H = \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + \sum_{\sigma} \epsilon_d d_{\sigma}^{\dagger} d_{\sigma} + U n_{d, \uparrow} n_{d, \downarrow}$$

$$= \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + \sum_{\sigma} \epsilon_d n_{d, \sigma} + U (\langle n_{d, \uparrow} \rangle n_{d, \downarrow} + \langle n_{d, \downarrow} \rangle n_{d, \uparrow})$$

$$= \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + \epsilon_d (n_{d, \uparrow} + n_{d, \downarrow}) + U (\langle n_{d, \uparrow} \rangle n_{d, \downarrow} + \langle n_{d, \downarrow} \rangle n_{d, \uparrow})$$

$$= \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + n_{\uparrow, \sigma} (\epsilon_d + \langle n_{d, \downarrow} \rangle U) + n_{\downarrow, \sigma} (\epsilon_d + \langle n_{d, \uparrow} \rangle U)$$

We want to calculate the spectral function of this system so, for this, we will use the GF's equation of motion method. We start with

$$\omega G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) = \frac{1}{2\pi} \langle \{d_{\sigma}, d_{\sigma}^{\dagger}\} \rangle - G([H, d_{\sigma}], d_{\sigma}^{\dagger} | \omega)$$

$$= \frac{1}{2\pi} - G([H, d_{\sigma}], d_{\sigma}^{\dagger} | \omega)$$

For the last term we need to calculate the commutator first. It's given by

$$[H, d_{\sigma'}] = \left[\sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma} + \sum_{\sigma} \epsilon_d d_{\sigma}^{\dagger} d_{\sigma} + U n_{d, \uparrow} n_{d, \downarrow}, d_{\sigma'} \right]$$

$$= \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} \left[a_{\mathbf{k}, \sigma}^{\dagger} a_{\mathbf{k}, \sigma}, d_{\sigma'} \right] + \sum_{\sigma} \epsilon_d [d_{\sigma}^{\dagger} d_{\sigma}, d_{\sigma'}] + U [(d_{\uparrow}^{\dagger} d_{\uparrow}) (d_{\downarrow}^{\dagger} d_{\downarrow}), d_{\sigma'}]$$

The operators d always anticommute with the operators a so the first term is directly zero. For the other we use the property

$$[AB, C] = A\{B, C\} - \{A, C\}B$$

$$= A(BC + CB) - (AC + CA)B$$

$$= ABC + ACB - ACB - CAB$$

$$= (AB)C - C(AB)$$

$$= [AB, C]$$

we obtain

$$\begin{aligned}
[H, d_{\sigma'}] &= \sum_{\sigma} \epsilon_d (d_{\sigma}^{\dagger} \{d_{\sigma}, d_{\sigma'}\} - \{d_{\sigma}^{\dagger}, d_{\sigma'}\} d_{\sigma}) + U (d_{\uparrow}^{\dagger} d_{\uparrow} \{d_{\downarrow}^{\dagger} d_{\downarrow}, d_{\sigma'}\} + \{d_{\uparrow}^{\dagger} d_{\uparrow}, d_{\sigma'}\} d_{\downarrow}^{\dagger} d_{\downarrow}) \\
&= - \sum_{\sigma} \epsilon_d d_{\sigma} \delta_{\sigma, \sigma'} + U (d_{\uparrow}^{\dagger} d_{\uparrow} (d_{\downarrow}^{\dagger} \{d_{\downarrow}, d_{\sigma'}\} - \{d_{\downarrow}^{\dagger}, d_{\sigma'}\} d_{\downarrow}) + (d_{\uparrow}^{\dagger} \{d_{\uparrow}, d_{\sigma'}\} - \{d_{\uparrow}^{\dagger}, d_{\sigma'}\} d_{\uparrow}) d_{\downarrow}^{\dagger} d_{\downarrow}) \\
&= -\epsilon_d d_{\sigma} - U (d_{\uparrow}^{\dagger} d_{\uparrow} d_{\downarrow} \delta_{\downarrow, \sigma'} + d_{\uparrow} d_{\downarrow}^{\dagger} \delta_{\uparrow, \sigma'})
\end{aligned}$$

In the last term we can switch $d_{\uparrow}$ to last (but we must include two minus sign that cancel each other from the anticommutation) so we get a more symmetric form

$$[H, d_{\sigma'}] = -\epsilon_d d_{\sigma}^{\dagger} - U (n_{d, \uparrow} d_{\downarrow} \delta_{\downarrow, \sigma'} + n_{d, \downarrow} d_{\uparrow} \delta_{\uparrow, \sigma'})$$

Reconstructing the GF's equation of motion we obtain

$$\omega G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) = \frac{1}{2\pi} + \epsilon_d G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) + U (G(n_{d, \uparrow} d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) \delta_{\downarrow, \sigma} + G(n_{d, \downarrow} d_{\uparrow}, d_{\sigma}^{\dagger} | \omega) \delta_{\uparrow, \sigma})$$

Seems to appear we have two new GF to calculate. To solve for this one we use the decoupling approximation i.e

$$G(AB, C) = \langle A \rangle G(B, C) + \langle B \rangle G(A, C)$$

We obtain

$$G(n_{\uparrow} d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) = \langle n_{d, \uparrow} \rangle G(d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) + \langle n_{d, \downarrow} \rangle G(n_{d, \uparrow}, d_{\sigma}^{\dagger} | \omega)$$

Because of conservation of particles we have that $\langle d_{\downarrow} \rangle = 0$. We are left with

$$\begin{cases} G(n_{\uparrow} d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) = \langle n_{d, \uparrow} \rangle G(d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) \\ G(n_{\downarrow} d_{\uparrow}, d_{\sigma}^{\dagger} | \omega) = \langle n_{d, \downarrow} \rangle G(d_{\uparrow}, d_{\sigma}^{\dagger} | \omega) \end{cases}$$

and so, our GF's equation of motion reduces to

$$\begin{aligned}
\omega G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) &= \frac{1}{2\pi} + \epsilon_d G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) + U (\langle n_{d, \uparrow} \rangle G(d_{\downarrow}, d_{\sigma}^{\dagger} | \omega) \delta_{\downarrow, \sigma} + \langle n_{d, \downarrow} \rangle G(d_{\uparrow}, d_{\sigma}^{\dagger} | \omega) \delta_{\uparrow, \sigma}) \\
&= \frac{1}{2\pi} + \epsilon_d G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) + U \sum_{\sigma'} \langle n_{d, \bar{\sigma}'} \rangle G(d_{\sigma'}, d_{\sigma}^{\dagger} | \omega) \delta_{\sigma', \sigma} \\
&= \frac{1}{2\pi} + \epsilon_d G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega) + U \langle n_{d, \bar{\sigma}} \rangle G(d_{\sigma}, d_{\sigma}^{\dagger} | \omega)
\end{aligned}$$

where the $\bar{\sigma}$ notation means the spin in the opposite direction of σ .

$$\Leftrightarrow G(d_{\sigma}^{\dagger}, d_{\sigma} | \omega) = \frac{1/2\pi}{\omega - \epsilon_d - U \langle n_{d, \bar{\sigma}} \rangle}$$

Now we use the dissipation-fluctuation theorem to obtain the spectral function.

$$J_{\sigma}(\omega) = \frac{-2}{e^{\beta\omega} + 1} \Im G_r(d_{\sigma}^{\dagger}, d_{\sigma} | \omega)$$

First, however, to get the retarded GF we do analitical continuation as $\omega \rightarrow \tilde{\omega} = \omega + i0^+$. We can use Plemelj theorem

$$\frac{1}{x \pm i0^+} = \mathcal{P} \left(\frac{1}{x} \right) \mp i\pi\delta(x)$$

to extract the GF's imaginary part. We get

$$\begin{aligned} J_\sigma(\omega) &= \frac{-2}{e^{\beta\omega} + 1} \left[-\pi \frac{1}{2\pi} \delta(\omega - (\epsilon_d + U \langle n_{d,\bar{\sigma}} \rangle)) \right] \\ &= \frac{1}{e^{\beta\omega} + 1} \delta(\omega - (\epsilon_d + U \langle n_{d,\bar{\sigma}} \rangle)) \end{aligned}$$

For an electron of, for example, spin $\sigma = \uparrow$, one of the kronecker deltas gives zero and so we get

$$J_\uparrow(\omega) = \frac{1}{e^{\beta\omega} + 1} \delta(\omega - (\epsilon_d + U \langle n_{d,\downarrow} \rangle))$$

and for $\sigma = \downarrow$

$$J_\downarrow(\omega) = \frac{1}{e^{\beta\omega} + 1} \delta(\omega - (\epsilon_d + U \langle n_{d,\uparrow} \rangle))$$

And so

$$J_{tot}(\omega) = J_\uparrow(\omega) + J_\downarrow(\omega)$$

5.8 Regions of magnetic behavior of the Anderson model

The coupled equations for the spin's occupation are given by

$$\begin{cases} \cot(\pi n_{d,\uparrow}) = \left(n_{d,\downarrow} + \frac{\epsilon'_d}{U} \right) \frac{U}{\Gamma} \\ \cot(\pi n_{d,\downarrow}) = \left(n_{d,\uparrow} + \frac{\epsilon'_d}{U} \right) \frac{U}{\Gamma} \end{cases} \Leftrightarrow \begin{cases} n_{d,\uparrow} = \frac{1}{\pi} \operatorname{arccot} \left(\left(n_{d,\downarrow} - x \right) \frac{\pi}{y} \right) \\ n_{d,\downarrow} = \frac{1}{\pi} \operatorname{arccot} \left(\left(n_{d,\uparrow} - x \right) \frac{\pi}{y} \right) \end{cases} \equiv \begin{cases} n_{d,\uparrow} = \frac{1}{\pi} \left(\frac{\pi}{2} - \arctan \left(\frac{\pi}{y} (n_{d,\downarrow} - x) \right) \right) \\ n_{d,\downarrow} = \frac{1}{\pi} \left(\frac{\pi}{2} - \arctan \left(\frac{\pi}{y} (n_{d,\uparrow} - x) \right) \right) \end{cases}$$

We present the code and the results of the solution of the given coupled equations

5.9 Relationship between $|BCS\rangle$ and $|0\rangle$

5.10 BCS Hamiltonian GF

5.11 Interacting fermion gas GF

Opposing to what we did in Exercice 4.13 we now consider the presence of the interaction term in the Hamiltonian. This means that

$$\begin{aligned} H &= \sum_{\mathbf{k}} \frac{k^2}{2m} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} v(\mathbf{q}) a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger a_{\mathbf{k}_2 + \mathbf{q}} a_{\mathbf{k}_1 - \mathbf{q}} \\ &\equiv H_0 + H_1 \end{aligned}$$

```

> import numpy as np
> import matplotlib.pyplot as plt

> def f(x,y,no):
    return np.abs( ( 1/np.pi ) * ( (np.pi/2) - np.arctan( (np.pi/y) * (no-x) ) ) )

> def sol(x,y,nu0,nd0):
    nu = f(x,y,nd0)
    nd = f(x,y,nu0)
    nuold, ndold = 0, 0
    while np.any( np.abs( nu-nuold ) > 1e-6 ):
        nuold = nu
        ndold = nd
        nu = f(x,y,nd)
        nd = f(x,y,nu)
    return nu, nd

> y = np.linspace(0.01, 2, 1000)
    nu1, nd1 = sol(0.25,y,0.7,0.3)
    nu2, nd2 = sol(0.5,y,0.7,0.3)

> plt.plot(y,nu1)
    plt.plot(y,nd1)
    plt.plot(y,nu2)
    plt.plot(y,nd2)
    plt.show()

> x = np.linspace(0.01,1,100)
    y = np.linspace(0.01,1,100)

    x1, y1 = np.meshgrid(x,y)

    nu, nd = sol(x1,y1,0.7,0.3)

    m = np.abs( nu - nd )

> plt.imshow( m,extent=[0,1,0,1],origin="lower" )
    plt.show()

```



The density operator $\rho_{\mathbf{q}}^\dagger(\mathbf{k})$ is defined by

$$\rho_{\mathbf{q}}^\dagger(\mathbf{k}) = a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}$$

which means that, because $(AB)^\dagger = B^\dagger A^\dagger$, also

$$\rho_{\mathbf{q}}(\mathbf{k}) = a_{\mathbf{k}}^\dagger a_{\mathbf{k}+\mathbf{q}}$$

We want to solve the GF's equation of motion

$$\omega G(\rho_{\mathbf{q}}^\dagger(\mathbf{k}); \rho_{\mathbf{q}}(\mathbf{k})|\omega) = \frac{1}{2\pi} \langle [\rho_{\mathbf{q}}^\dagger(\mathbf{k}), \rho_{\mathbf{q}}(\mathbf{k})] \rangle - G([\mathcal{H}, \rho_{\mathbf{q}}^\dagger(\mathbf{k})]; \rho_{\mathbf{q}}(\mathbf{k})|\omega)$$

For this we will need to use the 4 operator commutator property in terms of the anti-commutators because they are fermionic operators

$$[AB, CD] = -AC\{D, B\} + A\{C, B\}D - C\{D, A\}B + \{C, A\}DB$$

The first commutator then gives

$$\begin{aligned} [\rho_{\mathbf{q}}^\dagger(\mathbf{k}), \rho_{\mathbf{q}}(\mathbf{k}')] &= [a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'+\mathbf{q}}] \\ &= -a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}'+\mathbf{q}}, a_{\mathbf{k}}\} + a_{\mathbf{k}+\mathbf{q}}^\dagger \{a_{\mathbf{k}'}, a_{\mathbf{k}}\} a_{\mathbf{k}'+\mathbf{q}} - a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}'+\mathbf{q}}, a_{\mathbf{k}+\mathbf{q}}^\dagger\} a_{\mathbf{k}} + \{a_{\mathbf{k}'}, a_{\mathbf{k}+\mathbf{q}}^\dagger\} a_{\mathbf{k}'+\mathbf{q}} a_{\mathbf{k}} \\ &= a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}'+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} - a_{\mathbf{k}'}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}} \\ &= a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} - a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}} \end{aligned}$$

Where we used that, because it is a anti-commutator, $\{A, B\} = \{B, A\}$. This means that

$$\langle [\rho_{\mathbf{k}}^\dagger(\mathbf{k}), \rho_{\mathbf{k}}(\mathbf{k})] \rangle = \langle a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}+\mathbf{q}} \delta_{\mathbf{k}', \mathbf{k}} \rangle - \langle a_{\mathbf{k}}^\dagger a_{\mathbf{k}} \delta_{\mathbf{k}'+\mathbf{q}, \mathbf{k}+\mathbf{q}} \rangle = n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0$$

For the second commutator we can separate the Hamiltonian in the non-interacting part H_0 and the interacting part H_1 . For H_0 we obtain

$$\begin{aligned} [\mathcal{H}_0(\mathbf{k}'), \rho_{\mathbf{q}}^\dagger(\mathbf{k})] &= \sum_{\mathbf{k}'} \frac{k'^2}{2m} [a_{\mathbf{k}'}^\dagger a_{\mathbf{k}'}, a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}}] \\ &= \sum_{\mathbf{k}'} \frac{k'^2}{2m} \left(-a_{\mathbf{k}'}^\dagger a_{\mathbf{k}+\mathbf{q}}^\dagger \{a_{\mathbf{k}}, a_{\mathbf{k}'}\} + a_{\mathbf{k}'}^\dagger \{a_{\mathbf{k}+\mathbf{q}}^\dagger, a_{\mathbf{k}'}\} a_{\mathbf{k}} - a_{\mathbf{k}+\mathbf{q}}^\dagger \{a_{\mathbf{k}}, a_{\mathbf{k}'}\} a_{\mathbf{k}'} + \{a_{\mathbf{k}+\mathbf{q}}^\dagger, a_{\mathbf{k}'}\} a_{\mathbf{k}'} a_{\mathbf{k}} \right) \\ &= \sum_{\mathbf{k}} \frac{k'^2}{2m} \left(a_{\mathbf{k}'}^\dagger (\delta_{\mathbf{k}+\mathbf{q}, \mathbf{k}'} a_{\mathbf{k}} - a_{\mathbf{k}+\mathbf{q}}^\dagger (\delta_{\mathbf{k}, \mathbf{k}'} a_{\mathbf{k}'} \right) \\ &= \frac{(k+q)^2}{2m} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}} - \frac{k^2}{2m} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{k}} \\ &= (E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}}) \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \\ &\equiv \omega_0 \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \end{aligned}$$

5.11.1 Interaction Hamiltonian commutator, decoupling approximation and conservation of wave-vector and number of particles in thermal equilibrium

And now for the interacting part of the Hamiltonian

$$[\mathcal{H}_1, \rho_{\mathbf{q}}^\dagger(\mathbf{k})] = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} v(\mathbf{q}) [a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}, \rho_{\mathbf{q}}^\dagger(\mathbf{k})]$$

For this we use the four operator commutator property as

$$[AB, C] = A\{B, C\} - \{A, C\}B$$

we get

$$\begin{aligned} & \left[\left(a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger \right) (a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}), \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \right] \\ &= \left(a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger \right) \{ (a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}), \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} - \left\{ \left(a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger \right), \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \right\} (a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}) \end{aligned}$$

Giving that

$$\{AB, C\} = A\{B, C\} + \{A, C\}B$$

we obtain

$$\begin{aligned} &= \left(a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger \right) \{ a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} - \left\{ a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \right\} (a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}) \\ &= \left(a_{\mathbf{k}_1}^\dagger a_{\mathbf{k}_2}^\dagger \right) (a_{\mathbf{k}_2+\mathbf{q}} \{ a_{\mathbf{k}_1-\mathbf{q}}, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} + \{ a_{\mathbf{k}_2+\mathbf{q}}, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} a_{\mathbf{k}_1-\mathbf{q}}) \\ &\quad - \left(a_{\mathbf{k}_1}^\dagger \{ a_{\mathbf{k}_2}^\dagger, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} + \{ a_{\mathbf{k}_1}^\dagger, \rho_{\mathbf{q}}^\dagger(\mathbf{k}) \} a_{\mathbf{k}_2}^\dagger \right) (a_{\mathbf{k}_2+\mathbf{q}} a_{\mathbf{k}_1-\mathbf{q}}) \end{aligned}$$

5.11.2 Equation of motion assemble

Substituting back in the equation of motion we obtain

$$\begin{aligned} \omega G(\rho_{\mathbf{k}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) &= \frac{1}{2\pi} (n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0) - \omega_0 G(\rho_{\mathbf{q}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) \\ &\Leftrightarrow G(\rho_{\mathbf{k}}^\dagger(\mathbf{k}); \rho_{\mathbf{k}}(\mathbf{k})|\omega) = \frac{1}{2\pi} \frac{n_{\mathbf{k}+\mathbf{q}}^0 - n_{\mathbf{k}}^0}{\omega + (E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}})} \\ &\Rightarrow G_r(\rho_{\mathbf{k}}^\dagger; \rho_{\mathbf{k}}|\omega) = \frac{1}{2\pi} \sum_{\mathbf{k}} \frac{n_{\mathbf{k}+\mathbf{q}}^0 + n_{\mathbf{k}}^0}{E_{\mathbf{k}+\mathbf{q}} - E_{\mathbf{k}} + \omega + i0^+} \end{aligned}$$

6 Linear Response Theory

6.1 Thermal Average in the frequency space

The thermal average in the frequency space

$$\begin{aligned}
\langle A \rangle_\omega &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \langle A \rangle e^{i\omega t} dt \\
&= -\frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g_r(A; B | t - t') f(t') \right) e^{i\omega t} dt' dt \\
&= -\frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} G(A; B | \omega') e^{-i\omega'(t-t')} d\omega' \right\} \left\{ \int_{-\infty}^{\infty} F(\omega'') e^{-i\omega''t'} d\omega'' \right\} \right) e^{i\omega t} dt' dt \\
&= -\frac{1}{2\pi} \int \int \int \int G(A; B | \omega') F(\omega'') e^{-i\omega'(t-t')} e^{-i\omega''t'} e^{i\omega t} dt dt' d\omega' d\omega'' \\
&= -\frac{1}{2\pi} \int \int \int \int G(A; B | \omega') F(\omega'') e^{-i(\omega' - \omega)t} e^{-i(\omega'' - \omega')t'} dt dt' d\omega' d\omega'' \\
&= -\frac{1}{2\pi} \int F(\omega'') d\omega'' \int G(A; B | \omega') d\omega' \int e^{-i(\omega' - \omega)t} dt \int e^{-i(\omega'' - \omega')t'} dt'
\end{aligned}$$

Using the delta function definition

$$\delta(\omega - \omega') = \frac{1}{2\pi} \int e^{-i(\omega - \omega')t} dt$$

we obtain

$$\begin{aligned}
&= -2\pi \int F(\omega'') d\omega'' \int G(A; B | \omega') d\omega' \delta(\omega' - \omega) dt \delta(\omega'' - \omega') \\
&= -2\pi G(A; B | \omega) F(\omega)
\end{aligned}$$

6.2 Average energy dissipation

We will consider a case in which the function $f(t)$ is real and takes the simple form

$$f(t) = \frac{1}{2}(f_0 e^{-i\omega t} + f_0^* e^{+i\omega t})$$

The average of an operator A , using the definition of $\chi(\omega)$ is given by

$$\langle A(t) \rangle = \frac{1}{2}(\chi(\omega) f_0 e^{-i\omega t} + \chi^*(\omega) f_0^* e^{+i\omega t})$$

The average energy dissipation per unit time dW/dt is given by

$$\begin{aligned}
\frac{dW}{dt} &= \frac{dH_1}{dt} = -\frac{df(t)}{dt} \langle A(t) \rangle \\
&= -\frac{d\frac{1}{2}(f_0 e^{-i\omega t} + f_0^* e^{+i\omega t})}{dt} \frac{1}{2}(\chi(\omega) f_0 e^{-i\omega t} + \chi^*(\omega) f_0^* e^{+i\omega t}) \\
&= -\frac{1}{2}((-i\omega) f_0 e^{-i\omega t} + (+i\omega) f_0^* e^{+i\omega t}) \left(\frac{1}{2}(\chi(\omega) f_0 e^{-i\omega t} + \chi^*(\omega) f_0^* e^{+i\omega t}) \right)
\end{aligned}$$

Averaging over the period $2\pi/\omega$ of the force we find

$$\begin{aligned}
&= -\frac{1}{4} ((-i\omega)f_0 + i\omega f_0^*) (\chi(\omega)f_0 + \chi^*(\omega)f_0^*) \\
&= -\frac{1}{4} ((-i\omega)\chi(\omega)f_0^2 + (-i\omega)\chi^*(\omega)|f_0|^2 + i\omega\chi(\omega)|f_0|^2 + i\omega\chi^*(\omega)(f_0^*)^2) \\
&= \frac{i\omega}{4} (\chi^*(\omega) - \chi(\omega)) |f_0|^2 + \frac{i\omega}{4} (\chi(\omega)f_0^2 - \chi^*(\omega)(f_0^*)^2)
\end{aligned}$$

The last term is zero because $\chi(\omega) = \chi^*(-\omega)$ and so

$$\begin{aligned}
\chi(\omega)f_0^2 - \chi^*(\omega)(f_0^*)^2 &= \chi(\omega)|f_0|^2 e^{2i\omega t} - \chi^*(\omega)|f_0|^2 e^{-2i\omega t} \\
&= (\chi(\omega) - \chi^*(\omega)) |f_0|^2 \\
&=
\end{aligned}$$

The final result is then

$$\frac{dW}{dt} = \frac{\omega}{2} \Im(\chi(\omega)) |f_0|^2$$

6.3 Retarded GF property

We start from the definition of the retarded Gf in real-time

$$g_r(A; B|t - t') = -i\theta(t - t') \langle [A(t), B(t')] \rangle$$

where the definition of the thermal average is

$$\langle A \rangle = \frac{1}{Q} \text{Tr} \{ A e^{-\beta H} \}$$

assuming that $H = H_0$ we have

$$\begin{aligned}
\langle [A(t)B(t')] \rangle &= Q^{-1} \text{Tr} ([A(t)B(t')] e^{-\beta H_0}) \\
&= Q^{-1} \text{Tr} ((A(t)B(t') - B(t')A(t)) e^{-\beta H_0}) \\
&= Q^{-1} \text{Tr} (A(t)B(t') e^{-\beta H_0}) - \text{Tr} (-B(t')A(t) e^{-\beta H_0})
\end{aligned}$$

Using the cyclic property of the trace $\text{Tr}(ABC) = \text{Tr}(CAB) = \text{Tr}(BCA)$ This means that

$$\begin{aligned}
&= Q^{-1} \text{Tr} (B(t') e^{-\beta H_0} A(t)) - \text{Tr} (e^{-\beta H_0} B(t') A(t)) \\
&= Q^{-1} \text{Tr} ([B(t') e^{-\beta H_0}] A(t))
\end{aligned}$$

Using the property that

$$[B, e^{-\beta H_0}] = -ie^{-\beta H_0} \int_0^\beta e^{\lambda H_0} \frac{dB}{d\lambda} e^{-\lambda H_0} d\lambda$$

we have

$$\begin{aligned}
&= \left\langle e^{+\beta H_0} \left(-i e^{-\beta H_0} \int_0^\beta e^{\lambda H_0} \left(\frac{dB(t')}{dt'} \right) e^{-\lambda H_0} d\lambda \right) A(t) \right\rangle \\
&= -i \int_0^\beta d\lambda \left\langle e^{\lambda H_0} \left(\frac{dB(t')}{dt'} \right) e^{-\lambda H_0} A(t) \right\rangle
\end{aligned}$$

Our final result is then

$$\begin{aligned}
g_r(A; B|t - t') &= -i\theta(t - t') \langle [A(t), B(t')] \rangle \\
&= -\theta(t - t') \int_0^\beta d\lambda \left\langle e^{\lambda H_0} \left(\frac{dB(t')}{dt'} \right) e^{-\lambda H_0} A(t) \right\rangle
\end{aligned}$$

6.4 Particle in a viscous fluid

Consider a particle of mass m moving along a fixed direction with velocity v in a viscous fluid. The equation of motion is

$$\frac{dv(t)}{dt} + \frac{\Gamma}{m}v(t) = \frac{1}{m}f(t)$$

If we consider the driving force is a delta function $f(t) = \delta(t)$ then what is the impulse response function i.e the Green's function of the system? The GF definition is, for a given linear operator $\mathcal{L}$

$$\mathcal{L}(t)G(t, t') = \delta(t - t')$$

such that, when we integrate it with the given driving force we obtain

$$\int dt' (\mathcal{L}(t)G(t, t')) f(t') = \int dt' \delta(t - t') f(t') = f(t)$$

On the other hand, because $\mathcal{L}$ is a linear operator we can take it out of the integral obtaining

$$\int \mathcal{L}(t)G(t, t') f(t') dt' = \mathcal{L}(t) \int dt' G(t, t') f(t') \stackrel{!}{=} f(t)$$

and so, our solution is

$$v(t) = + \int_{-\infty}^{+\infty} dt' G(t, t') f(t')$$

The first step is then find the GF of the system

$$\frac{dG(t, t')}{dt} + \frac{\Gamma}{m}G(t, t') = \frac{1}{m}\delta(t - t')$$

We assume a solution of the type

$$G(t, t') = \begin{cases} Ae^{r(t-t')} & t > t' \\ Be^{r(t-t')} & t < t' \end{cases}$$

For $t > t'$ (or for $t < t'$ if we exchange B for A), if we substitute directly in the ODE we obtain

$$\cancel{A}re^{pt} + \frac{\Gamma}{m}\cancel{A}e^{pt} = 0, \quad \text{for } t \neq t'$$

$$\Leftrightarrow r = -\frac{\Gamma}{m} \equiv \frac{1}{\tau}$$

To find A and B we need to impose some conditions. Firstly, if we agree that the initial time is given by $t' = 0$ then, for $t < t'$, the exponential term $e^{-t/\tau}$ explodes giving an infinite contribution that is not allow. We can bypass this problem if we assume that, in fact, $B = 0$ to conserve causality. Our GF is then

$$G(t) = \begin{cases} Ae^{-t/\tau} & t > 0 \\ 0 & t < 0 \end{cases}$$

If we integrate the ODE near the discontinuity at $t = 0$ we obtain the boundary condition

$$\lim_{\epsilon \rightarrow 0} \left\{ \int_{-\epsilon}^{+\epsilon} dt \frac{dG(t)}{dt} + \int_{-\epsilon}^{+\epsilon} dt \frac{\Gamma}{m} G(t) = \int_{-\epsilon}^{+\epsilon} dt \frac{1}{m} \delta(t) \right\}$$

$$\Leftrightarrow \lim_{\epsilon \rightarrow 0} \left\{ G(t)|_{-\epsilon}^{+\epsilon} + \frac{\Gamma}{m} \left(-\frac{m}{\Gamma} A \right) e^{-\frac{\Gamma}{m}t}|_{-\epsilon}^{+\epsilon} = \frac{1}{m} \right\}$$

$$\Leftrightarrow \left(G(0^+) - \cancel{G(0^-)} \right)^0 - \cancel{A(e^0 - e^0)} = \frac{1}{m}$$

$$\Leftrightarrow Ae^0 = \frac{1}{m} \Rightarrow A = \frac{1}{m}$$

Our final solution is then

$$G(t - t') = \begin{cases} \frac{1}{m} e^{-\frac{\Gamma}{m}(t-t')} & t > t' \\ 0 & t < t' \end{cases}$$

Suppose now we have a harmonic driving force with $f(t) = f_0(\omega)e^{-i\omega t}$. The corresponding response is

$$\langle v(t) \rangle = \int_{-\infty}^{+\infty} dt' G(t - t') f(t')$$

We now know, however, that causality requires that $g(t - t') = 0$ for $t < t'$

$$= \int_{-\infty}^t dt' G(t - t') f(t')$$

If we change the variable to $\xi = t - t'$ and thus $d\xi = -dt'$ we get

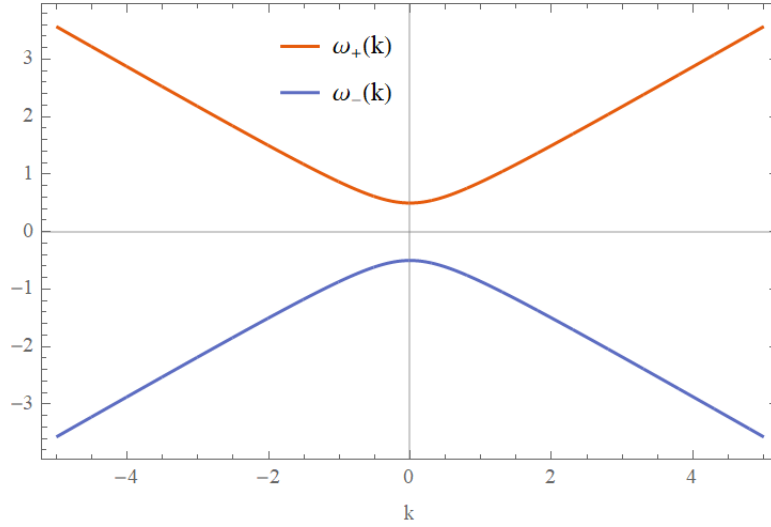
$$\begin{aligned}
&= - \int_{\infty}^0 d\xi G(\xi) f(t - \xi) \\
&= + \int_0^{+\infty} d\xi \left(\frac{1}{m} e^{-\frac{\Gamma}{m}\xi} \right) \left(f_0(\omega) e^{-i\omega(t-\xi)} \right) \\
&= \frac{1}{m} f_0(\omega) e^{-i\omega t} \int_0^{+\infty} d\xi e^{-(\frac{\Gamma}{m} + i\omega)\xi} \\
&= \frac{1}{m} f_0(\omega) e^{-i\omega t} \frac{1}{-(\frac{\Gamma}{m} + i\omega)} (0 - 1) \\
&= \left(\frac{1}{\Gamma + i\omega m} f_0(\omega) \right) e^{-i\omega t} \\
&= v_0(\omega) e^{-i\omega t}
\end{aligned}$$

Hence the response produced in the velocity is also harmonic. The generalized susceptibility by definition is just

$$\begin{aligned}
\chi(\omega) &= \left(-\frac{1}{2\pi} \right) G(\omega) \\
&= \left(-\frac{1}{2\pi} \right) \left(\frac{1}{2\pi} \int_{-\infty}^{+\infty} dt G(t - t') e^{i\omega t} \right) \\
&= - \int_0^{+\infty} dt G(t - t') e^{i\omega t} \\
&= - \int_{-t'}^{+\infty} d\xi G(\xi) e^{i\omega(\xi+t')} \\
&= -e^{+i\omega t'} \int_{-t'}^{+\infty} d\xi \left(\frac{1}{m} e^{-\frac{\Gamma}{m}\xi} \right) e^{i\omega\xi} \\
&= -\frac{1}{m} e^{+i\omega t'} \int_{-t'}^{+\infty} d\xi e^{-(\frac{\Gamma}{m} - i\omega)\xi} \\
&= -\frac{1}{m} e^{+i\omega t'} \left(\frac{1}{-(\frac{\Gamma}{m} - i\omega)} e^{(\frac{\Gamma}{m} - i\omega)t'} \right) \\
&= \frac{1}{m} \frac{1}{\frac{\Gamma}{m} - i\omega} e^{\frac{\Gamma}{m}t'}
\end{aligned}$$

Choosing again $t' = 0$ and using the definition of the lifetime $\tau = m/\Gamma$ we obtain

$$\begin{aligned}
\chi(\omega) &= \frac{1}{m} \frac{1}{1/\tau - i\omega} \times \frac{1/\tau + i\omega}{1/\tau\tau + i\omega} \\
&= \frac{1}{m} \frac{1/\tau + i\omega}{1/\tau^2 + \omega^2} \\
&= \frac{\tau}{m} \left(\frac{1 + i\omega\tau}{1 + \omega^2\tau^2} \right)
\end{aligned}$$



The real and imaginary part of this quantity is

$$\begin{cases} \Re(\chi(\omega)) = \frac{\tau}{m} \frac{1}{1+\omega^2\tau^2} \\ \Im(\chi(\omega)) = \frac{\tau^2}{m} \frac{\omega}{1+\omega^2\tau^2} \end{cases}$$

Choosing $\tau = 1.5$ and $m = 0.2$ for the plot

6.5 Driven Harmonic Oscillator $xf(t)$

6.5.1 Equation of motions

Consider a driven harmonic oscillator with a external force given by $xf(t)$. The Hamiltonian of the system is given by

$$H = \frac{p^2}{2m} + \frac{1}{2}\kappa x^2 - xf(t)$$

To obtain the average displacement we need to first get its equation of motion by the Heisenberg equation

$$\begin{aligned} \frac{dx(t)}{dt} &= i[H, x] \\ &= i \left[\frac{p^2}{2m} + \frac{1}{2}\kappa x^2 - xf(t), x \right] \\ &= \frac{i}{2m} [pp, x] + 0 + 0 \\ &= \frac{i}{2m} (p(-i) + (-i)p) \\ &= \frac{p}{m} \end{aligned}$$

We have in turn to also calculate the equation of motion for the momentum

$$\begin{aligned}
\frac{dp(t)}{dt} &= i[H, p] \\
&= i \left[\frac{p^2}{2m} + \frac{1}{2}\kappa x^2 - xf(t), p \right] \\
&= 0 + \frac{i}{2}\kappa [xx, p] - if(t)[x, p] \\
&= \frac{i}{2}\kappa (2ix) - if(t)i \\
&= -\kappa x + f(t)
\end{aligned}$$

If we derive x 's equation of motion in order of time and substitute we get

$$\frac{d^2 x(t)}{dt^2} = \frac{1}{m} \frac{dp}{dt} = \frac{1}{m} (-\kappa x + f(t))$$

Which result in a ODE for x in terms of t

$$\Rightarrow \left(\frac{d^2}{dt^2} - \omega_0^2 \right) x(t) = \frac{1}{m} f(t)$$

where $\omega_0^2 = \kappa/m$.

6.5.2 Susceptibility

As we saw, the GF is defined as

$$\left(\frac{d^2}{dt^2} - \omega_0^2 \right) g(t - t') = \frac{1}{m} \delta(t - t')$$

It's easier to solve in frequency space so we Fourier transform the GF as

$$g(t - t') = \int G(\omega) e^{-i\omega(t-t')} d\omega$$

and the Dirac delta as

$$\delta(t - t') = \frac{1}{2\pi} \int e^{-i\omega(t-t')} d\omega$$

For the left-hand side we obtain

$$\begin{aligned}
\int \left(\frac{d^2}{dt^2} - \omega_0^2 \right) G(\omega) e^{-i\omega(t-t')} d\omega &= \int ((-i\omega^2) - \omega_0^2) G(\omega) e^{-i\omega(t-t')} d\omega \\
&= \int (\omega^2 - \omega_0^2) G(\omega) e^{-i\omega(t-t')} d\omega
\end{aligned}$$

Equating to the Dirac delta we obtain

$$\int (\omega^2 - \omega_0^2) G(\omega) e^{-i\omega(t-t')} d\omega = \frac{1}{2\pi} \int e^{-i\omega(t-t')} d\omega$$

$$\Rightarrow G(\omega) = \frac{1}{m} \frac{1/2\pi}{\omega_0^2 - \omega^2}$$

We can obtain the retarded GF by analitically continuing the real-frequency GF as $\omega \rightarrow \omega + i0^+$

$$G_r(\omega) = \frac{-1}{2\pi m} \frac{1}{(\omega - \omega_0 + i0^+)(\omega + \omega_0 + i0^+)}$$

To obtain the retarded GF is time space we just need to do a Fourier Transform

$$g_r(t) = \int_{-\infty}^{+\infty} G_r(\omega) e^{-i\omega t} d\omega$$

$$= \frac{-1}{2\pi m} \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{(\omega + i0^+ - \omega_0)(\omega + i0^+ + \omega_0)} d\omega$$

We now need to integrate in the complex plane in a semi-circular loop who goes around all the lower complex plane. It can be shown that the circular part goes to zero as its radius $R \rightarrow \infty$ and so we are left with

$$\oint_{\gamma} f(z) dz = \int_{-\infty}^{+\infty} f(z) dz + \int_{\gamma_{circ}} f(z) dz$$

$$\Rightarrow \int_{-\infty}^{+\infty} \frac{e^{-i\omega t}}{(\omega + i0^+ - \omega_0)(\omega + i0^+ + \omega_0)} d\omega = \oint_{\gamma} \frac{e^{-izt}}{(z - (\omega_0 - i0^+))(z - (-\omega_0 - i0^+))} dz$$

In the lower complex plane we have two different poles at $z = \pm\omega_0 - i0^+$. By the residue theorem the integral is then evaluated as

$$\frac{1}{2\pi i} \oint_{\gamma} \frac{e^{-izt}}{(z - (\omega_0 - i0^+))(z - (-\omega_0 - i0^+))} dz = 2\pi i \sum_p \text{Res}(z, p)$$

$$= 2\pi i \left(\lim_{z \rightarrow \omega_0 - i0^+} \cancel{(z - (\omega_0 - i0^+))} \times \frac{e^{-izt}}{\cancel{(z - (\omega_0 - i0^+))} (z - (-\omega_0 - i0^+))} + \lim_{z \rightarrow -\omega_0 - i0^+} \cancel{(z - (-\omega_0 - i0^+))} \times \frac{e^{-izt}}{\cancel{(z - (-\omega_0 - i0^+))} (z - (\omega_0 - i0^+))} \right)$$

$$= 2\pi i \left(\frac{e^{-i\omega_0 t} e^{\pm i0^+ t}}{((\omega_0 - i0^+) - (-\omega_0 - i0^+))} + \frac{e^{+i\omega_0 t} e^{\pm i0^+ t}}{((- \omega_0 - i0^+) - (\omega_0 - i0^+))} \right)$$

$$= 2\pi i \frac{1}{2\omega_0} \frac{(e^{-i\omega_0 t} - e^{+i\omega_0 t})}{2i} 2i = \frac{-2\pi}{\omega_0} \sin(\omega_0 t)$$

Our final result of the GF is time space is then

$$g(t) = \frac{1}{\cancel{(-2\pi)}m} \frac{\cancel{(-2\pi)}}{\omega_0} \sin(\omega_0 t) = \frac{1}{m\omega_0} \sin(\omega_0 t)$$

We can calculate the average displacement using

$$\langle x(t) \rangle = \int g(t - t') f(t') dt'$$

6.5.3 Another way to get the GF

As we saw, the GF is defined as the function that satisfies

$$\left(\frac{d^2}{dt^2} - \omega_0^2\right)g(t-t') = \frac{1}{m}\delta(t-t')$$

At $t \neq t'$ the solution will be $\propto e^{rt}$ which means that $r = \pm\omega_0$. We now however that causality requires that $g_r(t-t') = 0$ for $t < t'$ and so, our solution will be

$$g_r(t) = \begin{cases} Ae^{+i\omega_0 t} + Be^{-i\omega_0 t} & t > 0 \\ 0 & t < 0 \end{cases}$$

We now need to know the value of the constants A and B . Firstly, by the initial condition that $g_r(0) = 0$ we have that $A = -B$. This way we have that

$$g_r(t) = \begin{cases} C \sin(\omega_0 t) & t > 0 \\ 0 & t < 0 \end{cases}$$

where $C = 2iA$. If we integrate the ODE near the discontinuity at $t = 0$ we obtain the boundary condition

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \left\{ \int_{-\epsilon}^{+\epsilon} dt \frac{d^2 g(t)}{dt^2} - \int_{-\epsilon}^{+\epsilon} dt \omega_0^2 g(t) \right\} &= \int_{-\epsilon}^{+\epsilon} dt \delta(t) \\ \Leftrightarrow \lim_{\epsilon \rightarrow 0} \left\{ \frac{dG(t)}{dt} \Big|_{-\epsilon}^{+\epsilon} - \omega_0^2 g(t) \Big|_{-\epsilon}^{+\epsilon} \right\} &= \frac{1}{m} \\ \Leftrightarrow (C\omega_0(\cos(0) - 0)) - \omega_0^2 \overbrace{C(\sin(0) - 0)}^{\cancel{0}} &= \frac{1}{m} \\ \Leftrightarrow C\omega_0 = \frac{1}{m} \Rightarrow C &= \frac{1}{m\omega_0} \end{aligned}$$

Our final solution is then

$$g_r(t-t') = \frac{1}{m\omega_0} \sin(\omega_0 t)$$

The same result as we obtain doing first in the frequency space and then reverse transform! Let's check the other way around. In frequency space we have

$$\begin{aligned}
G_r(\omega) &= \frac{1}{2\pi} \int g_r(t) e^{+i\omega t} dt \\
&= \frac{1}{2\pi} \frac{1}{m\omega_0} \int_0^\infty \sin(\omega_0 t) e^{+i\omega t} dt \\
&= \frac{1}{2\pi m} \frac{1}{\omega_0} \frac{1}{2i} \left(\int_0^\infty dt e^{+i(\omega+\omega_0+i\eta)t} - \int_0^\infty dt e^{+i(\omega-\omega_0+i\eta)t} \right) \\
&= \frac{1}{2\pi m} \frac{1}{\omega_0} \frac{1}{2i} \left(\frac{(0-1)}{i(\omega+\omega_0)} - \frac{(0-1)}{i(\omega-\omega_0)} \right) \\
&= \frac{1}{2\pi m} \frac{1}{\omega_0} \frac{1}{2} \left(\frac{\cancel{\omega} - \omega_0 - (\cancel{\omega} + \omega_0)}{(\omega+\omega_0)(\omega-\omega_0)} \right) \\
&= \frac{-1}{2\pi m} \frac{1}{(\omega+\omega_0)(\omega-\omega_0)} \\
&= \frac{1}{m} \frac{1/2\pi}{\omega_0^2 - \omega^2}
\end{aligned}$$

6.6 Conductivity

6.6.1 Uniform electric field

The current density is given by

$$J = \frac{dQ}{dt} = \frac{dQ}{dx} \frac{dx}{dt} = env$$

Where the velocity is given by

$$\begin{aligned}
\frac{dv}{dt} &= \frac{eE}{m} \Leftrightarrow \int_0^{\delta t} \frac{dv}{dt} dt = \int_0^{\delta t} \frac{eE}{m} dt \\
\Leftrightarrow v(\delta t) - v(0) &= \frac{eE}{m} (\delta t - 0) \Rightarrow v = \frac{eE}{m} \delta t
\end{aligned}$$

and so

$$J = en \frac{eE}{m} \delta t = \frac{ne^2 E \delta t}{m}$$

which responds to Ohm's law.

6.6.2 Electric field decay

From Ohm's law we have that

$$\begin{aligned}
J &= \frac{ne^2 E \delta t}{m} \Rightarrow \delta J = \frac{ne^2 \delta E \delta t}{m} \\
\Leftrightarrow \frac{\delta J}{\delta t} &= \frac{ne^2 \delta E}{m} = \frac{ne^2 (0 - E)}{m} = -\frac{ne^2 E}{m}
\end{aligned}$$

$$\Leftrightarrow J(0) \left(-\frac{1}{\tau} \right) e^{-t/\tau} = \frac{-J}{\tau} = \frac{-ne^2 E}{m}$$

We then obtain Drude's conductivity as

$$\sigma = \frac{J}{E} = \frac{ne^2 \tau}{m}$$

6.7 Brownian motion

It is the same solution as 6.4 ?

6.8 Verify the Kramers-Kronig relations

Suppose the susceptibility function for the 1D motion of a particle of mass m immersed in a fluid with a friction term of Γ is given by

$$\chi(\omega) = \frac{1}{\Gamma - i m \omega} \Rightarrow G_r(\tilde{\omega}) = \frac{-i}{2\pi m} \frac{1}{\tilde{\omega} + i\Gamma/m}$$

Where $\tilde{\omega} = \omega + i0^+$. Its useful to rewrite the GF as

$$\begin{aligned} G(\tilde{\omega}) &= \frac{-i}{2\pi m} \frac{1}{\tilde{\omega} + i\Gamma/m} \times \frac{\tilde{\omega} - i\Gamma/m}{\tilde{\omega} - i\Gamma/m} = \frac{-i}{2\pi m} \frac{\tilde{\omega} - i\Gamma/m}{\tilde{\omega}^2 + \Gamma/m^2} \\ &= \frac{-1}{2\pi m (\tilde{\omega}^2 + \Gamma/m^2)} (\Gamma/m + i\tilde{\omega}) \end{aligned}$$

The Kramers-Kronig relations show that the real and imaginary parts of the GF's are intricately related through integral expressions. In particular, for the retarded GF we have that

$$\Re G_r(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{+\infty} \Im G_r(\omega') \frac{d\omega'}{\omega - \omega'}$$

We want to prove that

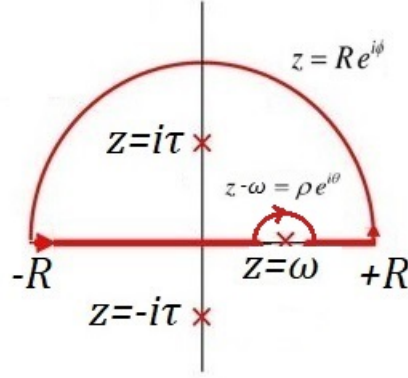
$$\frac{\cancel{1}}{2\pi m (\omega^2 + \Gamma/m^2)} \Gamma/m = \cancel{1} \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{+\infty} \frac{-\omega'}{2\pi m (\omega'^2 + \Gamma/m^2)} \frac{d\omega'}{\omega - \omega'}$$

This means we need to prove the following integral

$$\Leftrightarrow \mathcal{P} \left(\int_{-\infty}^{+\infty} \frac{z}{(z^2 + \tau^2)(z - \omega)} dz \right) = \pi \frac{\tau}{(\omega^2 + \tau^2)}$$

In general, the way to solve this type of integral is to consider instead a contour integral in the complex plane. First, let us define in the real line

$$I \equiv \mathcal{P} \left(\int_{-\infty}^{+\infty} \frac{z}{(z^2 + \tau^2)(z - \omega)} dz \right)$$



In the complex plane we have three poles, two at $z = \pm i\tau$ and another at $z = \omega$. If we choose a contour γ as shown in red in the figure below we obtain

$$\begin{aligned}
 I_\gamma &= \oint_\gamma \frac{z}{(z^2 + \tau^2)(z - \omega)} dz \\
 &= \lim_{\substack{R \rightarrow \infty \\ \rho \rightarrow 0}} \left(\int_{-R}^{\omega - \rho} + \int_{C_\rho} + \int_{\omega + \rho}^{+R} + \int_{C_R} \right) \frac{z}{(z^2 + \tau^2)(z - \omega)} dz
 \end{aligned}$$

where the contours C_R and C_ρ are the semi circular loops with respective radius R and ρ . Notice that, in the limit where $\rho \rightarrow 0$ we can go straight in the real line. Furthermore, in the limit where $R \rightarrow \infty$ we go across all real line meaning that

$$\lim_{\substack{R \rightarrow \infty \\ \rho \rightarrow 0}} \left(\int_{-R}^{\omega - \rho} + \int_{\omega + \rho}^{+R} \right) \frac{z}{(z^2 + \tau^2)(z - \omega)} dz = I$$

Substituting we solving the other integrals we get

$$\begin{aligned}
I_\gamma &= I + \lim_{\rho \rightarrow 0} \left(\int_{C_\rho} \frac{z}{(z^2 + \tau^2)(z - \omega)} dz \right) + \lim_{R \rightarrow \infty} \left(\int_{C_R} \frac{z}{(z^2 + \tau^2)(z - \omega)} dz \right) \\
&= I + \lim_{\rho \rightarrow 0} \left(\int_\pi^0 \frac{(\rho e^{i\theta} + \omega)}{((\rho e^{i\theta} + \omega)^2 + \tau^2) \cancel{(\rho e^{i\theta} + \omega) - \omega}} i \rho e^{i\theta} d\theta \right) + \lim_{R \rightarrow \infty} \left(\int_0^\pi \frac{(Re^{i\phi})}{((Re^{i\phi})^2 + \tau^2) ((Re^{i\phi}) - \omega)} i R d\phi \right) \\
&\approx I + i \lim_{\rho \rightarrow 0} \left(\int_\pi^0 \frac{(\rho e^{i\theta} + \omega)}{(\rho e^{i\theta} + \omega)^2 + \tau^2} d\theta \right) + i \lim_{R \rightarrow \infty} \left(\int_0^\pi \frac{(Re^{i\phi})^2}{(Re^{i\phi})^3} d\phi \right) \\
&\approx I + i \left(\frac{\omega}{\omega^2 + \tau^2} \int_\pi^0 d\theta \right) + i \lim_{R \rightarrow \infty} \frac{1}{R} \left(\int_0^\pi e^{-i\theta} d\phi \right) \\
&\approx I + i \left(\frac{\omega}{\omega^2 + \tau^2} (0 - \pi) \right) + 2i \lim_{R \rightarrow \infty} \frac{1}{R} \\
&\approx I - \pi i \frac{\omega}{\omega^2 + \tau^2} + 0
\end{aligned}$$

We finally obtain

$$I = I_\gamma + \pi i \frac{\omega}{\omega^2 + \tau^2}$$

We still need to solve for I_γ . This can be done relatively easy by using the residue theorem. Notice that given our choice of contour we only include the pole $z = i\tau$ of (multiplicity of one) inside our loop. By the residue theorem we obtain

$$\begin{aligned}
I_\gamma &= 2\pi i \lim_{z \rightarrow i\tau} \cancel{(z - i\tau)} \frac{z}{\cancel{(z - i\tau)} (z + i\tau) (z - \omega)} \\
&= 2\pi i \frac{\cancel{i\tau}}{\cancel{2i\tau} (i\tau - \omega)} \\
&= \pi i \frac{1}{i\tau - \omega}
\end{aligned}$$

Our final result is then

$$\begin{aligned}
I &= \pi i \frac{1}{i\tau - \omega} + \pi i \frac{\omega}{\omega^2 + \tau^2} \\
&= \pi i \left(\frac{(\cancel{\omega^2} + \tau^2) + \omega(i\tau - \cancel{\omega})}{(i\tau - \omega)(\omega^2 + \tau^2)} \right) \\
&= \pi i \left(\frac{(-i)\tau \cancel{(i\tau - \omega)}}{(i\tau - \cancel{\omega})(\omega^2 + \tau^2)} \right) \\
&= \pi \frac{\tau}{(\omega^2 + \tau^2)}
\end{aligned}$$

Which is exactly what we were looking for.

6.9 Macroscopic semiclassical torque equation of motion for the magnetization in a ferromagnet

In a macroscopic theory, the semiclassical torque equation of motion for the magnetization in a ferromagnet is

$$\frac{d\mathbf{M}}{dt} = \gamma(\mathbf{M} \times \mathbf{B})$$

where $\mathbf{M}$ and $\mathbf{B}$ denote the total effective magnetization and magnetic field and γ is the gyromagnetic ratio. In the dipole-dipole limit where exchange can be neglected, we can write both of these fields in terms of a static part and a small fluctuating part in the form

$$\mathbf{M} = M_0 \hat{\mathbf{z}} + \mathbf{m}(\mathbf{r})e^{-i\omega t} \quad \text{and} \quad \mathbf{B} = B_0 \hat{\mathbf{z}} + \mu_0 \mathbf{h}(\mathbf{r})e^{-i\omega t}$$

The first step is to make the cross-product explicit

$$\gamma(\mathbf{M} \times \mathbf{B}) = -\gamma \left((M_0 \hat{\mathbf{z}} + \mathbf{m}(\mathbf{r})e^{-i\omega t}) \times (B_0 \hat{\mathbf{z}} + \mu_0 \mathbf{h}(\mathbf{r})e^{-i\omega t}) \right)$$

We have a sum 4 cross products. The first one is trivial, giving

$$M_0 \hat{\mathbf{z}} \times B_0 \hat{\mathbf{z}} = 0$$

For the second one we have

$$M_0 \hat{\mathbf{z}} \times \mu_0 \mathbf{h}(\mathbf{r})e^{-i\omega t} = M_0 \mu_0 e^{-i\omega t} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ h_x & h_y & h_z \end{vmatrix} = M_0 \mu_0 e^{-i\omega t} (-h_y \hat{\mathbf{x}} + h_x \hat{\mathbf{y}}) = e^{-i\omega t} \begin{pmatrix} -M_0 \mu_0 h_y \\ M_0 \mu_0 h_x \\ 0 \end{pmatrix}$$

Analogously for the third one

$$\mathbf{m}(\mathbf{r})e^{-i\omega t} \times B_0 \hat{\mathbf{z}} = B_0 e^{-i\omega t} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ m_x & m_y & m_z \\ 0 & 0 & 1 \end{vmatrix} = B_0 e^{-i\omega t} (+m_y \hat{\mathbf{x}} - m_x \hat{\mathbf{y}}) = e^{-i\omega t} \begin{pmatrix} B_0 m_y \\ -B_0 m_x \\ 0 \end{pmatrix}$$

and finally

$$\begin{aligned} \mathbf{m}(\mathbf{r})e^{-i\omega t} \times \mu_0 \mathbf{h}(\mathbf{r})e^{-i\omega t} &= \mu_0 e^{-2i\omega t} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ m_x & m_y & m_z \\ h_x & h_y & h_z \end{vmatrix} \\ &= \mu_0 e^{-2i\omega t} \left((m_y h_z - m_z h_y) \hat{\mathbf{x}} - (m_x h_z - m_z h_x) \hat{\mathbf{y}} + (m_x h_y - m_y h_x) \hat{\mathbf{z}} \right) = \mu_0 e^{-2i\omega t} \begin{pmatrix} m_y h_z - m_z h_y \\ m_z h_x - m_x h_z \\ m_x h_y - m_y h_x \end{pmatrix} \end{aligned}$$

Because its said that $\mathbf{m}(\mathbf{r})$ and $\mathbf{h}(\mathbf{r})$ are small fluctuating part we can assume that, in second order, a product of their components is basicaly zero, this mean-

$$\mathbf{m}(\mathbf{r})e^{-i\omega t} \times \mu_0 \mathbf{h}(\mathbf{r})e^{-i\omega t} \approx 0$$

Adding them all up we obtain

$$\gamma(\mathbf{M} \times \mathbf{B}) = e^{-i\omega t} \begin{pmatrix} -\mu_0\omega_m h_y + \omega_0 m_y \\ \mu_0\omega_m h_x - \omega_0 m_x \\ 0 \end{pmatrix}$$

where the effective angular frequencies are given by $\omega_0 = \gamma B_0$ and $\omega_m = \gamma M_0$. The DE translates to

$$\begin{aligned} \frac{d}{dt} \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} e^{-i\omega t} &= (-i\omega) \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} e^{-i\omega t} = e^{-i\omega t} \begin{pmatrix} -\mu_0\omega_m h_y + \omega_0 m_y \\ \mu_0\omega_m h_x - \omega_0 m_x \\ 0 \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} &= \begin{pmatrix} 0 & +\frac{i\omega_0}{\omega} & 0 \\ -\frac{i\omega_0}{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} + \begin{pmatrix} 0 & -\mu_0 \frac{i\omega_m}{\omega} & 0 \\ \mu_0 \frac{i\omega_m}{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} 1 & -\frac{i\omega_0}{\omega} & 0 \\ +\frac{i\omega_0}{\omega} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} &= \begin{pmatrix} 0 & -\mu_0 \frac{i\omega_m}{\omega} & 0 \\ \mu_0 \frac{i\omega_m}{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ \Leftrightarrow \begin{pmatrix} m_x \\ m_y \\ m_z \end{pmatrix} &= \frac{1}{1 - (-\frac{i\omega_0}{\omega})(+\frac{i\omega_0}{\omega})} \begin{pmatrix} 1 & +\frac{i\omega_0}{\omega} & 0 \\ -\frac{i\omega_0}{\omega} & 1 & 0 \\ 0 & 0 & 1 - (-\frac{i\omega_0}{\omega})(+\frac{i\omega_0}{\omega}) \end{pmatrix} \begin{pmatrix} 0 & -\mu_0 \frac{i\omega_m}{\omega} & 0 \\ \mu_0 \frac{i\omega_m}{\omega} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ &= \frac{1}{1 - (\frac{\omega_0}{\omega})^2} \begin{pmatrix} +\frac{i\omega_0}{\omega} \mu_0 \frac{i\omega_m}{\omega} & -\mu_0 \frac{i\omega_m}{\omega} & 0 \\ \mu_0 \frac{i\omega_m}{\omega} & -\frac{i\omega_0}{\omega} (-\mu_0 \frac{i\omega_m}{\omega}) & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ &= \mu_0 \frac{1}{1 - (\frac{\omega_0}{\omega})^2} \begin{pmatrix} -\frac{\omega_0\omega_m}{\omega^2} & -i \frac{\omega_m\omega}{\omega^2} & 0 \\ i \frac{\omega_m}{\omega} & -\frac{\omega_0\omega_m}{\omega^2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ &= -\mu_0 \begin{pmatrix} \frac{\omega_0\omega_m}{\omega^2 - \omega_0^2} & i \frac{\omega_m\omega}{\omega^2 - \omega_0^2} & 0 \\ -i \frac{\omega_m\omega}{\omega^2 - \omega_0^2} & \frac{\omega_0\omega_m}{\omega^2 - \omega_0^2} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \\ &= \mu_0 \begin{pmatrix} \chi_a & i\chi_b & 0 \\ -i\chi_b & \chi_a & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} h_x \\ h_y \\ h_z \end{pmatrix} \end{aligned}$$

7 Linear Response Theory (Ficha 6 Bruno Amorim)

7.1 Matsubara charge susceptibility function

The Matsubara charge susceptibility function is given by

$$\chi(\tau) = -\langle T_\tau \rho(\tau) \rho(0) \rangle_0 + \langle \rho(0) \rangle_0^2$$

with

$$\rho(\tau) = \sum_{a,b} \rho_{a,b} c_a^\dagger(\tau) c_b(\tau)$$

being the density matrix and

$$H_0 = \sum_a \epsilon_a c_a^\dagger c_a$$

being the free Hamiltonian.

7.1.1 (a) Wick's Theorem

If we expand all the terms inside the thermal average we get

$$\chi(\tau) = - \sum_{a,b} \sum_{c,d} \rho_{a,b} \rho_{c,d} \langle T_\tau c_a^\dagger(\tau) c_b(\tau) c_c^\dagger c_d \rangle_0 + \left(\sum_{a,b} \rho_{a,b} \langle c_a^\dagger c_b \rangle_0 \right)^2$$

The last term is just

$$\left(\sum_{a,b} \rho_{a,b} \langle c_a^\dagger c_b \rangle_0 \right)^2 = \left(\sum_{a,b} \rho_{a,b} g_{a,b}(0^+) \right)^2 = \left(\sum_{a,b} \rho_{a,b} g_a(0^+) \delta_{a,b} \right)^2 = \left(\sum_a \rho_{a,a} g_a(0^+) \right)^2$$

By Wick's theorem we can express averages of an even number of operators as sum of the product of all the possible contractions. From the first term the possible contractions are

$$\left\langle T_\tau \overbrace{c_a^\dagger(\tau) c_b(\tau)}^{\text{contract}} \overbrace{c_c^\dagger c_d}^{\text{contract}} \right\rangle_0 \quad \text{and} \quad \left\langle T_\tau \overbrace{c_a^\dagger(\tau) c_b(\tau)}^{\text{contract}} \overbrace{c_c^\dagger c_d}^{\text{contract}} \right\rangle_0$$

and so

$$\begin{aligned} \langle T_\tau c_a^\dagger(\tau) c_b(\tau) c_c^\dagger c_d \rangle_0 &= \langle T_\tau c_a^\dagger(\tau) c_b(\tau) \rangle_0 \langle T_\tau c_c^\dagger c_d \rangle_0 + \langle T_\tau c_a^\dagger(\tau) c_d \rangle_0 \langle T_\tau c_b(\tau) c_c^\dagger \rangle_0 \\ &= \langle T_\tau c_a^\dagger(\tau) c_b(\tau) \rangle_0 \langle T_\tau c_c^\dagger c_d \rangle_0 + \langle T_\tau c_a^\dagger(\tau) c_d \rangle_0 \langle T_\tau c_b(\tau) c_c^\dagger \rangle_0 \end{aligned}$$

To make the propagator GF explicit we need to have the dagger operator for last. This way, we write

$$\langle T_\tau c_a^\dagger(\tau) c_d \rangle_0 = - \langle T_\tau c_d c_a^\dagger(\tau) \rangle_0 = - \langle T_\tau c_d(-\tau) c_a^\dagger \rangle_0 = -g_{d,a}(-\tau) = -g_a(-\tau) \delta_{a,d}$$

Substituting we obtain

$$\langle T_\tau c_a^\dagger(\tau) c_b(\tau) c_c^\dagger c_d \rangle_0 = g_a(0^+) \delta_{a,b} g_c(0^+) \delta_{c,d} - g_a(-\tau) \delta_{a,d} g_c(\tau) \delta_{c,b}$$

$$\begin{aligned}
&\Rightarrow - \sum_{a,b} \sum_{c,d} \rho_{a,b} \rho_{c,d} (g_a(0^+) \delta_{a,b} g_c(0^+) \delta_{c,d} - g_a(-\tau) \delta_{a,d} g_c(\tau) \delta_{c,b}) \\
&= - \sum_{a,c} \rho_{a,a} \rho_{c,c} g_a(0^+) g_c(0^+) + \sum_{a,c} \rho_{a,c} \rho_{c,a} g_a(-\tau) g_c(\tau) \\
&= - \left(\sum_a \rho_{a,a} g_a(0^+) \right)^2 + \sum_{a,c} \rho_{a,c} \rho_{c,a} g_a(-\tau) g_c(\tau)
\end{aligned}$$

The first term cancels out and we are left with

$$\chi(\tau) = \sum_{a,b} \rho_{a,b} \rho_{b,a} g_a(-\tau) g_b(\tau)$$

7.1.2 (b) GF in term of Matsubara frequencies

We can express the GF in term of Matsubara frequencies using a Fourier series

$$g_b(\tau) = \frac{1}{\beta} \sum_{ik_n} e^{-ik_n \tau} g_b(ik_n)$$

Where k_n are fermionic frequencies and ω_n are bosonic frequencies

$$\begin{cases} k_n = \frac{1}{\beta} 2\pi(n + \frac{1}{2}) & , \text{for fermions } \epsilon = -1 \\ \omega_n = \frac{1}{\beta} 2\pi n & , \text{for bosons } \epsilon = +1 \end{cases}$$

Substituting the expression for $g_a(-\tau)$ and $g_b(\tau)$ we obtain

$$\begin{aligned}
\chi(\tau) &= \sum_{a,b} \rho_{a,b} \rho_{b,a} g_a(-\tau) g_b(\tau) \\
&= \sum_{a,b} \rho_{a,b} \rho_{b,a} \left(\frac{1}{\beta} \sum_{ik_n} e^{+ik_n \tau} g_a(ik_n) \right) \left(\frac{1}{\beta} \sum_{ik_m} e^{-ik_m \tau} g_b(ik_m) \right) \\
&= \sum_{a,b} \rho_{a,b} \rho_{b,a} \frac{1}{\beta^2} \sum_{ik_n} \sum_{ik_m} e^{-i(k_m - k_n)\tau} g_a(ik_n) g_b(ik_m)
\end{aligned}$$

Notice that

$$k_m - k_n = \frac{1}{\beta} \frac{\pi}{2} (m + \frac{1}{2}) - \frac{1}{\beta} \frac{\pi}{2} (n + \frac{1}{2}) = \frac{1}{\beta} \frac{\pi}{2} (m - n) \equiv \omega_l$$

Because n and m are integers so is $m - n$. If we define $m - n \equiv 4l' \equiv l$ we now have a bosonic frequency ω_l . This way $k_m = \omega_l + k_n$ and we rather have

$$\begin{aligned}
\chi(\tau) &= \sum_{a,b} \rho_{a,b} \rho_{b,a} \frac{1}{\beta^2} \sum_{i\omega_l} \sum_{ik_n} e^{-i\omega_l \tau} g_a(ik_n) g_b(i\omega_l + ik_n) \\
&= \sum_{a,b} \rho_{a,b} \rho_{b,a} \frac{1}{\beta} \sum_{i\omega_l} e^{-i\omega_l \tau} \left\{ \frac{1}{\beta} \sum_{ik_n} g_a(ik_n) g_b(i\omega_l + ik_n) \right\} \\
&= \sum_{a,b} \rho_{a,b} \rho_{b,a} \frac{1}{\beta} \sum_{i\omega_l} e^{-i\omega_l \tau} \chi_{a,b}(i\omega_l)
\end{aligned}$$

7.1.3 (c) Matsubara sum

A Matsubara GF in the frequency space takes the form

$$g_a(ik_n) = \frac{1}{i\omega_n - \epsilon_a}$$

Substituting this result we get

$$\begin{aligned}\chi_{a,b}(i\omega_n) &= \frac{1}{\beta} \sum_{ik_n} g_a(ik_n) g_b(i\omega_n + ik_n) \\ &= \frac{1}{\beta} \sum_{ik_n} \frac{1}{ik_n - \epsilon_a} \frac{1}{i\omega_n + ik_n - \epsilon_b}\end{aligned}$$

If we extend the summation on ik_m to an integral on the complex plane i.e $ik_n \rightarrow z$ taking in account the fermi distribution $f(z)$ we get

$$\chi_{a,b}(i\omega_n) = \frac{1}{\beta} \oint_{\gamma_n} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)}$$

As seen in the figure, for a given ω_n we have one pole on $z = \epsilon_a$, another on $z = \epsilon_b - i\omega_n$ and n poles on $e^{\beta ik_n} = -1$ which is true for all fermionic frequencies k_n since

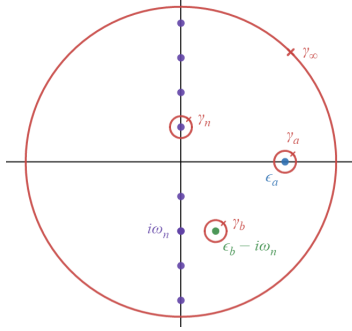
$$e^{\beta ik_n} = e^{i\frac{1}{\beta} 2\pi(n+\frac{1}{2})} = e^{i2\pi n} e^{i\pi} = 1 \times (-1) = -1$$

By the residue theorem, a contour integral around the path γ_∞ can be decomposed on the different paths arounds this poles, namely γ_a, γ_b and γ_n 's

$$\begin{aligned}I_\infty &= \frac{1}{\beta} \oint_{\gamma_\infty} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} \\ &= \frac{1}{\beta} \oint_{\gamma_a} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} \\ &\quad + \frac{1}{\beta} \oint_{\gamma_b} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} \\ &\quad + \sum_{ik_n} \frac{1}{\beta} \oint_{\gamma_n} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)}\end{aligned}$$

As proved in the (e) subsubsection $I_\infty = 0$ and so

$$\begin{aligned}\sum_{ik_n} \frac{1}{\beta} \oint_{\gamma_n} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} &= -\frac{1}{\beta} \oint_{\gamma_a} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} - \frac{1}{\beta} \oint_{\gamma_b} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} \\ \Leftrightarrow \chi_{a,b}(i\omega_n) &= -\frac{1}{\beta} \oint_{\gamma_a} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)} - \frac{1}{\beta} \oint_{\gamma_b} \frac{dz}{2\pi i} \frac{1}{e^{\beta z} + 1} \frac{1}{z - \epsilon_a} \frac{1}{z - (\epsilon_b - i\omega_n)}\end{aligned}$$



From the residues theorem, an integral around a pole is evaluated as

$$\oint_{\gamma} f(z) dz = 2\pi i \sum \text{Res}(f, p)$$

where the residue is given by

$$\text{Res}(f, c) = \frac{1}{(n-1)!} \lim_{z \rightarrow c} \frac{d^{n-1}}{dz^{n-1}} ((z-c)^n f(z))$$

with n being the multiplicity of the pole. Because all the poles have $n = 1$ the expression reduces to

$$\text{Res}(f, c) = \lim_{z \rightarrow c} (z-c) f(z)$$

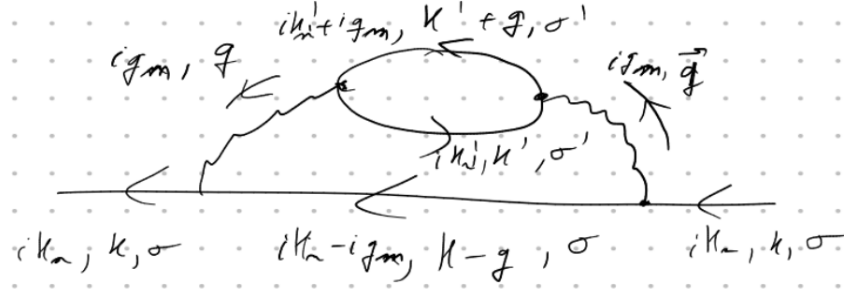
which gives

$$\begin{aligned} \chi_{a,b}(i\omega_n) &= -\frac{1}{\beta} \frac{1}{e^{\beta\epsilon_a} + 1} \frac{1}{\epsilon_a - (\epsilon_b - i\omega_n)} - \frac{1}{\beta} \frac{1}{e^{\beta(\epsilon_b - i\omega_n)} + 1} \frac{1}{(\epsilon_b - i\omega_n) - \epsilon_a} \\ &= -\frac{1}{\beta} f(\epsilon_a) \frac{1}{\epsilon_a - (\epsilon_b - i\omega_n)} - \frac{1}{\beta} \frac{1}{e^{\beta\epsilon_b} e^{-\beta i \frac{1}{2} 2\pi n} + 1} \frac{1}{(\epsilon_b - i\omega_n) - \epsilon_a} \\ &= -\frac{1}{\beta} f(\epsilon_a) \frac{1}{i\omega_n + \epsilon_a - \epsilon_b} - \frac{1}{\beta} f(\epsilon_b) \frac{-1}{i\omega_n + \epsilon_a - \epsilon_b} \\ &= -\frac{1}{\beta} \left(\frac{f(\epsilon_a) - f(\epsilon_b)}{i\omega_n + \epsilon_a - \epsilon_b} \right) \end{aligned}$$

7.1.4 (d) Analitical continuation

If if analitically continue $i\omega_n \rightarrow \omega + i0^+$ we obtain

$$\chi_{cs}^R(\omega + i0^+) \equiv \chi_{a,b}(\omega + i0^+) = \frac{f(\epsilon_a) - f(\epsilon_b)}{\omega + \epsilon_a - \epsilon_b + i0^+}$$



7.1.5 (e) Proof that $I_\infty = 0$

7.2 Self-energy of the bubble pair diagram

7.2.1 Self-energy from Feynman diagrams

The self-energy is given by

$$\sum_{\mathbf{k}, \sigma} (ik_n) = 8 \times \frac{(-1)^2}{2!} \times \frac{1}{4} \times (-1)^1 \times \left(-\frac{1}{\beta}\right)^2 \times \sum_{\mathbf{q}, iq_m, \mathbf{k}', ik'_n, \sigma'} g_{\mathbf{k}-\mathbf{q}, \sigma}(ik_n - iq_m) (-V_{\mathbf{q}})^2 g_{\mathbf{k}'+\mathbf{q}, \sigma'}(ik'_n - iq_m) g_{\mathbf{k}', \sigma'}(ik'_n)$$

where the factor of 8 comes from all the different ways to wire the the two vertex with the initial and final propagator to obtain this very same diagram, the factor of $(-1)^2/2!$ comes from the exponential expansion, the factor of 4 and $1/V^2$ comes simply from the hamiltonean and the factor of $(-1)^1$ is because of the one closed fermionic loop, the $(-1/\beta)^2$ comes from the Fourier transform of the two internal energies. The GF's pertains to the fermion propagators of a given wave vector, frequency and spin. Giving

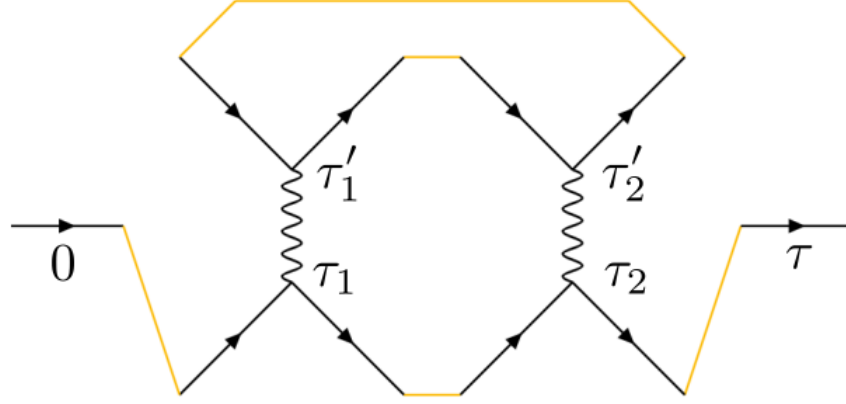
$$\begin{aligned} V_{\mathbf{q}}^{\text{eff}}(iq_m) &= V_{\mathbf{q}} \times \frac{1}{\beta V} \sum_{\mathbf{k}', ik'_n, \sigma'} g_{\mathbf{k}'+\mathbf{q}, \sigma'}(ik'_n - iq_m) g_{\mathbf{k}', \sigma'}(ik'_n) \times V_{\mathbf{q}} \\ &= V_{\mathbf{q}} \chi_{\mathbf{q}}(iq_m) V_{\mathbf{q}} \end{aligned}$$

7.2.2 Self-energy from Wick's Theorem in real space and imaginary time

The interaction Hamiltonean is given by

$$W = \frac{1}{2} \sum_{\sigma, \sigma'} \int d^3\mathbf{r} \int d^3\mathbf{r}' \psi_{\sigma}^{\dagger}(\mathbf{r}) \psi_{\sigma'}^{\dagger}(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \psi_{\sigma'}(\mathbf{r}') \psi_{\sigma}(\mathbf{r})$$

The auto-energy is, in a certain way, the energy that a particle due to interactions with its environment. Diagrammatically it can be expressed as a sum of all the diagrams that can't be further divided into smaller diagrams. It can be



Por bolinhas no diagrama no espaço das posições mas não no dos momentos
 Nos momentos tenho de por setinhas
 Por a tracejado inves de ondulado
 Ter cuidado que criar uma carga negativa é destruir uma positiva por isso tem
 um sinal de menos global, tinha de trocar as setinhas

found by calculating the total GF

$$g_M(\mathbf{r}|\tau) = \frac{\langle \hat{T}_D \{ \psi^\dagger_\sigma(\mathbf{r},0) \psi_\sigma(\mathbf{r},\tau) S(\beta) \} \rangle_0}{\langle S(\beta) \rangle_0}$$

where $S(\beta)$ is the S matrix given by

$$S(\beta) = 1 + \sum_{n=1} \frac{(-1)^n}{n!} \hat{T}_D \left\{ \int_0^\beta d\tau_1 \dots \int_0^\beta d\tau_n W(\tau_1) \dots W(\tau_n) \right\}.$$

Although, in general only the connected diagrams give contribution to the total GF meaning that the contribution of the terms with the contraction of the initial and final free propagator, $\psi^\dagger_{\mathbf{k},\sigma}(0) \psi_{\mathbf{k},\sigma}(\tau)$, is zero. We are up to second order so the S matrix is just

$$S(\beta) = 1 - \int_0^\beta d\tau_1 W(\tau_1) + \frac{1}{2} \int_0^\beta d\tau_1 \int_0^\beta d\tau_2 \hat{T}_D \{ W(\tau_1) W(\tau_2) \}$$

We are looking for the contraction (on second order) that represents the pair bubble diagram. For this we draw the diagram as shown below to help us find that contraction

Because the interaction is instantaneous i.e the interaction Hamiltonian is time-independent the times $\tau_1 = \tau'_1$ and $\tau_2 = \tau'_2$. We have for the contractions

$$\begin{aligned}
& \psi_{\sigma}^{\dagger}(\mathbf{r}, 0) \psi_{\sigma}(\mathbf{r}, \tau) \psi_{\sigma_1}^{\dagger}(\mathbf{r}_1, \tau_1) \psi_{\sigma_1'}^{\dagger}(\mathbf{r}_1', \tau_1) \psi_{\sigma_1'}(\mathbf{r}_1', \tau_1) \psi_{\sigma_1}(\mathbf{r}_1, \tau_1) \psi_{\sigma_2}^{\dagger}(\mathbf{r}_2, \tau_2) \psi_{\sigma_2'}^{\dagger}(\mathbf{r}_2', \tau_2) \psi_{\sigma_2'}(\mathbf{r}_2', \tau_2) \psi_{\sigma_2}(\mathbf{r}_2, \tau_2) \\
& = (-1)^{14} \left\langle \hat{T}_D \psi_{\sigma}^{\dagger}(\mathbf{r}, 0) \psi_{\sigma_1}(\mathbf{r}_1, \tau_1) \right\rangle \left\langle \hat{T}_D \psi_{\sigma_1}^{\dagger}(\mathbf{r}_1, \tau_1) \psi_{\sigma_2}(\mathbf{r}_2, \tau_2) \right\rangle_0 \left\langle \hat{T}_D \psi_{\sigma_2}^{\dagger}(\mathbf{r}_2, \tau_2) \psi_{\sigma}(\mathbf{r}, \tau) \right\rangle_0 \\
& \times \left\langle \hat{T}_D \psi_{\sigma_1'}^{\dagger}(\mathbf{r}_1', \tau_1) \psi_{\sigma_2'}(\mathbf{r}_2', \tau_2) \right\rangle_0 \left\langle \hat{T}_D \psi_{\sigma_1'}(\mathbf{r}_1', \tau_1) \psi_{\sigma_2'}^{\dagger}(\mathbf{r}_2', \tau_2) \right\rangle_0
\end{aligned}$$

With the GF being

$$g_{\sigma}(\mathbf{r}, \tau; \mathbf{r}', \tau') \delta_{\sigma, \sigma'} = - \left\langle \hat{T}_D \psi_{\sigma}(\mathbf{r}, \tau) \psi_{\sigma'}^{\dagger}(\mathbf{r}', \tau') \right\rangle_0$$

we obtain

$$\begin{aligned}
& = [-g_{\sigma}(\mathbf{r}, 0; \mathbf{r}_1, \tau_1)] \delta_{\sigma, \sigma_1} [-g_{\sigma_1}(\mathbf{r}_1, \tau_1; \mathbf{r}_2, \tau_2)] \delta_{\sigma_1, \sigma_2} [-g_{\sigma_2}(\mathbf{r}_2, \tau_2; \mathbf{r}, \tau)] \delta_{\sigma_2, \sigma} [-g_{\sigma_1'}(\mathbf{r}_1', \tau_1'; \mathbf{r}_2', \tau_2')] \delta_{\sigma_1', \sigma_2'} [g_{\sigma_1'}(\mathbf{r}_2', \tau_2'; \mathbf{r}_1', \tau_1')] \\
& = [-g_{\sigma}(0, 1)] [-g_{\sigma_1}(1, 2)] [-g_{\sigma_2}(2, \tau)] [-g_{\sigma_1'}(1', 2')] [g_{\sigma_1'}(2', 1')] \delta_{\sigma, \sigma_1} \delta_{\sigma_1, \sigma_2} \delta_{\sigma_2, \sigma} \delta_{\sigma_1', \sigma_2'} \delta_{\sigma_1', \sigma_2'}
\end{aligned}$$

We now need to come back and get the integral and pre-factors we left. We obtain

$$\begin{aligned}
g_M^{\text{bolha}}(\mathbf{r}|\tau) & = (-1)^2 \frac{8}{2! \times 4} \left(\frac{1}{2} \sum_{\sigma_1, \sigma_1'} \int d^3 \mathbf{r}_1 \int d^3 \mathbf{r}_1' \right) \left(\frac{1}{2} \sum_{\sigma_2, \sigma_2'} \int d^3 \mathbf{r}_2 \int d^3 \mathbf{r}_2' \right) \left(\int_0^{\beta} d\tau_1 \int_0^{\beta} d\tau_2 \right) V(\mathbf{r}_1 - \mathbf{r}_1') \\
& \times g_{\sigma}(0, 1) g_{\sigma_1}(1, 2) g_{\sigma_1'}(1', 2') g_{\sigma_1'}(2', 1') g_{\sigma_2}(2, \tau) \times \delta_{\sigma, \sigma_1} \delta_{\sigma_1, \sigma_2} \delta_{\sigma_2, \sigma} \delta_{\sigma_1', \sigma_2'} \delta_{\sigma_1', \sigma_2'}
\end{aligned}$$

The deltas kills all the sums as follows

$$\begin{aligned}
& = \sum_{\sigma_1, \sigma_1'} \sum_{\sigma_2, \sigma_2'} g_{\sigma}(0, 1) g_{\sigma_1}(1, 2) g_{\sigma_1'}(1', 2') g_{\sigma_1'}(2', 1') g_{\sigma_2}(2, \tau) \times \delta_{\sigma, \sigma_1} \delta_{\sigma_1, \sigma_2} \delta_{\sigma_2, \sigma} \delta_{\sigma_1', \sigma_2'} \delta_{\sigma_1', \sigma_2'} \\
& = \sum_{\sigma_1, \sigma_1'} \sum_{\sigma_2} g_{\sigma}(0, 1) g_{\sigma_1}(1, 2) g_{\sigma_1'}(1', 2') g_{\sigma_1'}(2', 1') g_{\sigma_2}(2, \tau) \times \delta_{\sigma, \sigma_1} \delta_{\sigma_1, \sigma_2} \delta_{\sigma_2, \sigma} \\
& = \sum_{\sigma_1'} g_{\sigma}(0, 1) g_{\sigma}(1, 2) g_{\sigma_1'}(1', 2') g_{\sigma_1'}(2', 1') g_{\sigma}(2, \tau) \times \delta_{\sigma, \sigma} \\
& = \sum_{\sigma_1'} g_{\sigma}(0, 1) g_{\sigma}(1, 2) g_{\sigma_1'}(1', 2') g_{\sigma_1'}(2', 1') g_{\sigma}(2, \tau)
\end{aligned}$$

We can also use the notation

$$\frac{1}{2} \int d^3 \mathbf{r}_1 \int d^3 \mathbf{r}_1' \int_0^{\beta} d\tau_1 \equiv \int d(1)$$

and and we are left with

$$g_M^{\text{bolha}}(\mathbf{r}|\tau) = \int d(1) \int d(2) \left(\int_0^\beta d\tau_1 \int_0^\beta d\tau_2 \right) V(1, 1') V(2, 2') \\ \times \sum_{\sigma'} g_\sigma(0, 1) g_\sigma(1, 2) g_{\sigma'}(1', 2') g_{\sigma'}(2', 1') g_\sigma(2, \tau)$$

By definition

$$g_M^{\text{bolha}}(\mathbf{r}|\tau) \stackrel{!}{=} \int_0^\beta d\tau_1 \int_0^\beta d\tau_2 \int d\mathbf{r}_1 \int d\mathbf{r}_2 g_\sigma(0, 1) \Sigma^{\text{bolha}}(1, 2) g_\sigma(2, \tau)$$

The self-energy is then

$$\Rightarrow \Sigma^{\text{bolha}} = \int d\mathbf{r}'_1 \int d\mathbf{r}'_2 V(1, 1') V(2, 2') \times \sum_{\sigma'} g_\sigma(1, 2) g_{\sigma'}(1', 2') g_{\sigma'}(2', 1') \\ = g_\sigma(1, 2) \int d\mathbf{r}'_1 \int d\mathbf{r}'_2 V(1, 1') V(2, 2') \times \left(\sum_{\sigma'} g_{\sigma'}(1', 2') g_{\sigma'}(2', 1') \right) \\ = g_\sigma(1, 2) \left(\int d\mathbf{r}'_1 \int d\mathbf{r}'_2 V(1, 1') V(2, 2') \times \chi_0 \right) \\ = g_\sigma(1, 2) V^{\text{eff}}$$

7.2.3 Verificação Diagramática

Using Fourier series we can go from the imaginary time space to the frequencies space and check our diagramatic result of $\sum_{\mathbf{k}, \sigma} (ik_n)$ from $\Sigma(\mathbf{r}'_1, \tau'_1; \mathbf{r}'_2, \tau'_2)$. We just need to substitute the Gf's as

$$g_\sigma(\tau_1, \mathbf{r}_1; \tau_2, \mathbf{r}_2) = \frac{1}{\beta V} \sum_{ik_n, \mathbf{k}} e^{ik_n(\tau_1 - \tau_2)} e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} g_{\mathbf{k}\sigma}(ik_n)$$

the interaction as

$$V(\mathbf{r}_1 - \mathbf{r}_2) = \frac{1}{V} \sum_{\mathbf{q}} e^{i\mathbf{q} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} V_{\mathbf{q}}$$

and the self energy as

$$\Sigma_\sigma(\tau, \mathbf{r}; \tau', \mathbf{r}') = \frac{1}{\beta V} \sum_{ik_n, \mathbf{k}} e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} e^{-ik_n(\tau - \tau')} \Sigma_{\mathbf{k}, \sigma}(ik_n).$$

We want to go from time-position space to frequency-momentum space

$$\Sigma = \int d\mathbf{r}'_1 \int d\mathbf{r}'_2 V(1, 1') V(2, 2') \times \sum_{\sigma'} g_\sigma(1, 2) g_{\sigma'}(1', 2') g_{\sigma'}(2', 1')$$

For this we just need to make the substitution above.

7.2.4 Soma de Matsubara

8 Interação Eletrão-Fonão (Ficha 7 Bruno Amorim)

The hamiltonean who describes eletron-phonon interactions is given by

$$H = H_e + H_f + H_{e-f}$$

where

$$\begin{cases} H_e = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}} \\ H_f = \sum_{\mathbf{q}, \lambda} \hbar \omega_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda}^{\dagger} a_{\mathbf{q}, \lambda}) \\ H_{e-p} = \int d\mathbf{r} D(\nabla \cdot \mathbf{u}(\mathbf{r})) \psi^{\dagger}(\mathbf{r}) \psi(\mathbf{r}) \end{cases}$$

where $\mathbf{u}(\mathbf{r})$ is the grid deformation, $\psi(\mathbf{r})$ is the eletronic anihilation field operator and D is the deformation potential

$$\begin{cases} \mathbf{u}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} e^{i\mathbf{q} \cdot \mathbf{r}} \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^{\dagger}) \\ \psi(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\mathbf{k}} \end{cases}$$

8.1 (a) Electron-phonon interactions hamiltonean

$$\begin{aligned}
H_{e-p} &= \int d\mathbf{r} D(\nabla \cdot \mathbf{u}(\mathbf{r})) \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}) \\
&= \int d\mathbf{r} D \left[\nabla \cdot \left(\frac{1}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} e^{i\mathbf{q} \cdot \mathbf{r}} \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) \right] \frac{1}{\sqrt{N}} \sum_{\mathbf{k}'} e^{-i\mathbf{k}' \cdot \mathbf{r}} c_{\mathbf{k}'}^\dagger \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\mathbf{k}} \\
&= \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \frac{D}{\sqrt{N}} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \int d\mathbf{r} \left[\nabla \cdot \left(e^{i\mathbf{q} \cdot \mathbf{r}} \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) \right] e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} \\
&= \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \frac{D}{N^{3/2}} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \int d\mathbf{r} \left[\left(e^{i\mathbf{q} \cdot \mathbf{r}} (i\mathbf{q}) \cdot \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) \right] e^{-i(\mathbf{k}' - \mathbf{k}) \cdot \mathbf{r}} \\
&= \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \frac{Di}{N^{3/2}} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \left(\mathbf{q} \cdot \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) \int d\mathbf{r} e^{-i(\mathbf{k}' - \mathbf{k} - \mathbf{q}) \cdot \mathbf{r}} \\
&= \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \sum_{\mathbf{k}'} \frac{Di}{N^{3/2}} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} c_{\mathbf{k}}^\dagger c_{\mathbf{k}'} \left(\mathbf{q} \cdot \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) V_{uc} \delta_{\mathbf{k}' - \mathbf{k} - \mathbf{q}} \\
&= \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \frac{V_{uc} Di}{N^{3/2}} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} c_{\mathbf{k}}^\dagger c_{\mathbf{k} + \mathbf{q}} \left(\mathbf{q} \cdot \mathbf{e}_{\mathbf{q}, \lambda} (a_{\mathbf{q}, \lambda} + a_{-\mathbf{q}, \lambda}^\dagger) \right) \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \sum_{\mathbf{k}} \frac{V_{uc} Di}{N} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} \mathbf{q} \cdot \mathbf{e}_{\mathbf{q}, \lambda} c_{\mathbf{k}}^\dagger c_{\mathbf{k} + \mathbf{q}} \phi_{\mathbf{q}, \lambda} \\
&= \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}, \lambda} M_{\mathbf{k} + \mathbf{q}, \mathbf{k}}^\lambda c_{\mathbf{k}}^\dagger c_{\mathbf{k} + \mathbf{q}} \phi_{\mathbf{q}, \lambda} \\
&\Rightarrow M_{\mathbf{k} + \mathbf{q}, \mathbf{k}}^\lambda = \frac{V_{uc} Di}{N} \sqrt{\frac{\hbar}{2m\omega_{\mathbf{q}, \lambda}}} \mathbf{q} \cdot \mathbf{e}_{\mathbf{q}, \lambda}
\end{aligned}$$

8.2 (b) Auto-energy

To determine the auto-energy we need to recall its definition via the Dyson equation

$$G = G_0 + G_0 \sum G$$

The auto-energy is, in a certain way, the energy that a particle due to interactions with its environment. Diagrammatically it can be expressed as a sum of all the diagrams that can't be further divided into smaller diagrams. It can be found by calculating the total GF

$$g_M(\mathbf{k}|\tau_1 - \tau_2) = \frac{\langle \hat{T}_D S(\beta) c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau) \rangle_0}{\langle S(\beta) \rangle_0}$$

where $S(\beta)$ is the S matrix given by

$$S(\beta) = 1 + \sum_{n=1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \dots \int_0^\beta d\tau_n \hat{T}_D \{H_1(\tau_1) \dots H_1(\tau_n)\}.$$

Although, in general only the connected diagrams give contribution to the total GF and so, the expression can be though as

$$g_M(\mathbf{k}|\tau_1 - \tau_2) = g_M^0(\mathbf{k}|\tau_1 - \tau_2) + \frac{\text{“connected”}}{\langle S(\beta) \rangle_0}$$

Because the vertex of the interaction has only 3 propagators, at first order the Wick's contraction of the 3 operators results always in zero. In second order of perturbation theory we start by making the thermal average

$$\begin{aligned} & \left\langle \hat{T}_D S(\beta) a_{\mathbf{k}}^\dagger(\tau_1) a_{\mathbf{k}}(\tau_2) \right\rangle_0 \sim \left\langle \hat{T}_D H_1(\tau_1) H_1(\tau_2) c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau) \right\rangle_0 \\ &= \left\langle \hat{T}_D \frac{1}{N} \sum_{\mathbf{k}', \mathbf{q}', \lambda'} M_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'}^\lambda c_{\mathbf{k}'}^\dagger(\tau_1) c_{\mathbf{k}'+\mathbf{q}'}(\tau_1) \phi_{\mathbf{q}', \lambda'}(\tau_1) \sum_{\mathbf{k}'', \mathbf{q}'', \lambda''} M_{\mathbf{k}''+\mathbf{q}'', \mathbf{k}'}^\lambda c_{\mathbf{k}''}^\dagger(\tau_2) c_{\mathbf{k}''+\mathbf{q}''}(\tau_2) \phi_{\mathbf{q}'', \lambda''}(\tau_2) c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau) \right\rangle_0 \\ &= \frac{1}{N} \sum_{\mathbf{k}', \mathbf{q}', \lambda'} \sum_{\mathbf{k}'', \mathbf{q}'', \lambda''} M_{\mathbf{k}'+\mathbf{q}', \mathbf{k}'}^\lambda M_{\mathbf{k}''+\mathbf{q}'', \mathbf{k}'}^\lambda \left\langle \hat{T}_D c_{\mathbf{k}'}^\dagger(\tau_1) c_{\mathbf{k}'+\mathbf{q}'}(\tau_1) \phi_{\mathbf{q}', \lambda'}(\tau_1) c_{\mathbf{k}''}^\dagger(\tau_2) c_{\mathbf{k}''+\mathbf{q}''}(\tau_2) \phi_{\mathbf{q}'', \lambda''}(\tau_2) c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau) \right\rangle_0 \end{aligned}$$

By Wick's theorem, the possible contractions are

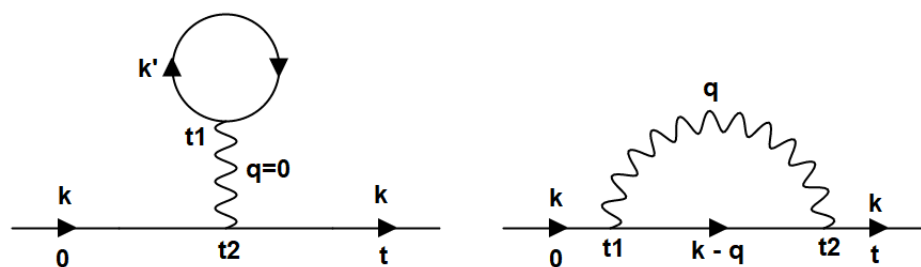
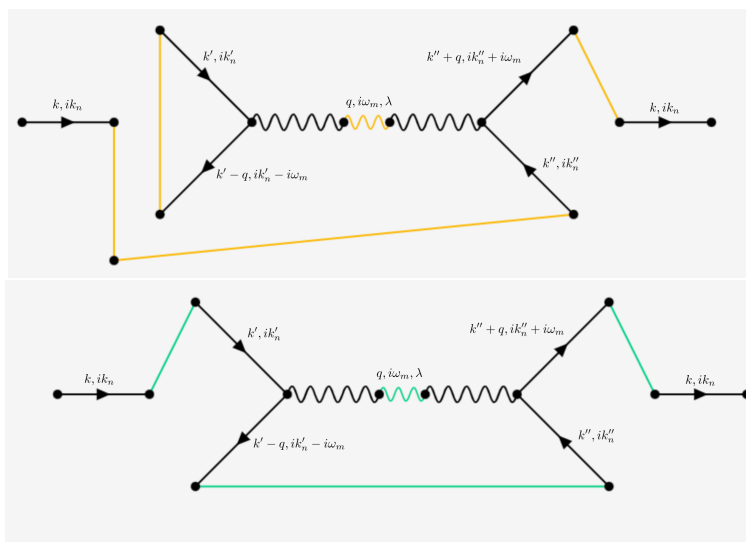
$$\begin{aligned} & (-1)^1 \left\langle \hat{T}_D \overbrace{c_{\mathbf{k}'}^\dagger(\tau_1) c_{\mathbf{k}'+\mathbf{q}'}(\tau_1)}^{\text{contract}} \overbrace{c_{\mathbf{q}', \lambda'}(\tau_1) c_{\mathbf{k}''}^\dagger(\tau_2)}^{\text{contract}} \overbrace{c_{\mathbf{k}''+\mathbf{q}''}(\tau_2) \phi_{\mathbf{q}'', \lambda''}(\tau_2)}^{\text{contract}} \overbrace{c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau)}^{\text{contract}} \right\rangle_0 \\ & (-1)^0 \left\langle \hat{T}_D \overbrace{c_{\mathbf{k}'}^\dagger(\tau_1) c_{\mathbf{k}'+\mathbf{q}'}(\tau_1)}^{\text{contract}} \overbrace{c_{\mathbf{q}', \lambda'}(\tau_1) c_{\mathbf{k}''}^\dagger(\tau_2)}^{\text{contract}} \overbrace{c_{\mathbf{k}''+\mathbf{q}''}(\tau_2) \phi_{\mathbf{q}'', \lambda''}(\tau_2)}^{\text{contract}} \overbrace{c_{\mathbf{k}}^\dagger(0) c_{\mathbf{k}}(\tau)}^{\text{contract}} \right\rangle_0 \end{aligned}$$

where we ignored the terms that make the contraction $\left\langle \overbrace{c_{\mathbf{k}''}^\dagger(\tau_1) c_{\mathbf{k}''}(\tau_2)}^{\text{contract}} \right\rangle_0$ because it would result in disconnected diagrams since the initial and final free propagators would connect leaving the rest to connect with itself. The $(-1)^n$ where n is the number of closed fermionic loops is to account for the anticommutator minus signs of the fermionic operators. Diagrammatically, given the initial vertex we could connect them respectively as

or, in a more easy way to read them as

Notice, because its said that $M_{\mathbf{k}, \mathbf{k}}^\lambda = 0$, the first diagram, the one with the tadpole, has zero contribution because the contraction give a $\delta_{\mathbf{k}, \mathbf{k}+\mathbf{q}}$. The only contribution for the self-energy is then self-interaction diagram (without the initial and final propagator). Using the Feynman rules it translates to

$$\sum(\mathbf{k}|ik_n) = \frac{1}{2!} \times 2 \times (-1)^2 \times \left(-\frac{1}{\beta}\right) \times \frac{1}{N} \sum_{\mathbf{q}, i\omega_n, \lambda} g(\mathbf{k} - \mathbf{q}|ik_n - i\omega_n) d(\mathbf{q}|i\omega_n) M_{\mathbf{k}, \mathbf{k}-\mathbf{q}} M_{\mathbf{k}-\mathbf{q}, \mathbf{k}}$$



where $g(\mathbf{k} - \mathbf{q}|ik_n - i\omega_n)$ is the electrons GF

$$g(\mathbf{k}|ik_n) = \frac{1}{(ik_n - i\omega_n) - \epsilon_{\mathbf{k}}},$$

and $d(\mathbf{q}|\tau_1 - \tau_2)$ is the phonons GF given by

$$\begin{aligned} d(\mathbf{q}|i\omega_n) &= G(a + a^\dagger, a + a^\dagger|\omega) \\ &= \cancel{G(a, a^\dagger|\omega)} + G(a, a^\dagger|\omega) + G(a^\dagger, a|\omega) + \cancel{G(a^\dagger, a^\dagger|\omega)} \end{aligned}$$

In section 3 we proved the properties $G^*(A, B|\omega) = \epsilon G(A^\dagger, B^\dagger|\omega)$ (in this case, for the phonon operators whom are bosonic $\epsilon = 1$) and so

$$\begin{aligned} &= G(a, a^\dagger|\omega) + G^*(a, a^\dagger|\omega) \\ &= \frac{1}{i\omega_n - \omega_{q,\lambda}} + \frac{1}{-i\omega_n - \omega_{q,\lambda}} \\ &= \frac{\cancel{i\omega_n} - \omega_{q,\lambda} + \cancel{i\omega_n} - \omega_{q,\lambda}}{(i\omega_n - \omega_{q,\lambda})(-i\omega_n - \omega_{q,\lambda})} \\ &= \frac{-2\omega_{q,\lambda}}{-(i\omega_n)^2 + \omega_{q,\lambda}^2} \\ &\Rightarrow d(\mathbf{q}|i\omega_n) = \frac{2\omega_{q,\lambda}}{(i\omega_n)^2 - \omega_{q,\lambda}^2} \end{aligned}$$

The factors $1/N, M_{\mathbf{k}, \mathbf{k}-\mathbf{q}}$ and $M_{\mathbf{k}-\mathbf{q}, \mathbf{k}}$ are simply from the Hamiltonian expression, the $(-1/\beta)$ factor comes from the Fourier transform of the internal frequencies, the factors $1/2!$ and $(-1)^2$ comes from the exponential expansion and the 2 factor comes from the two different ways to wire the diagram. The summations are done only on the internal variable because of the δ 's that come from the commutators. We can further simplify the expression if we use the property that $M_{\mathbf{k}, \mathbf{k}+\mathbf{q}}^\lambda = (M_{\mathbf{k}+\mathbf{q}, \mathbf{k}}^\lambda)^*$. We get

$$\Leftrightarrow \sum(\mathbf{k}|ik_n) = -\frac{1}{N} \sum_{\mathbf{q}, ik_n, \lambda} g(\mathbf{k} - \mathbf{q}|ik_n - i\omega_n) d(\mathbf{q}|i\omega_n) |M_{\mathbf{k}, \mathbf{k}-\mathbf{q}}|^2$$

To bring to the light the definition of self-energy we can define the effective potential of the electron-phonon interaction mediated by the phonons as

$$V_q^{\text{eff}}(i\omega_n) = |M_{\mathbf{k}, \mathbf{k}-\mathbf{q}}|^2 d(\mathbf{q}|i\omega_n)$$

Our final result if then

$$\sum(\mathbf{k}|ik_n) = -\frac{1}{N} \sum_{\mathbf{q}, ik_n, \lambda} g(\mathbf{k} - \mathbf{q}|ik_n - i\omega_n) d(\mathbf{q}|i\omega_n) |M_{\mathbf{k}, \mathbf{k}-\mathbf{q}}|^2$$

8.3 Life time of an electron